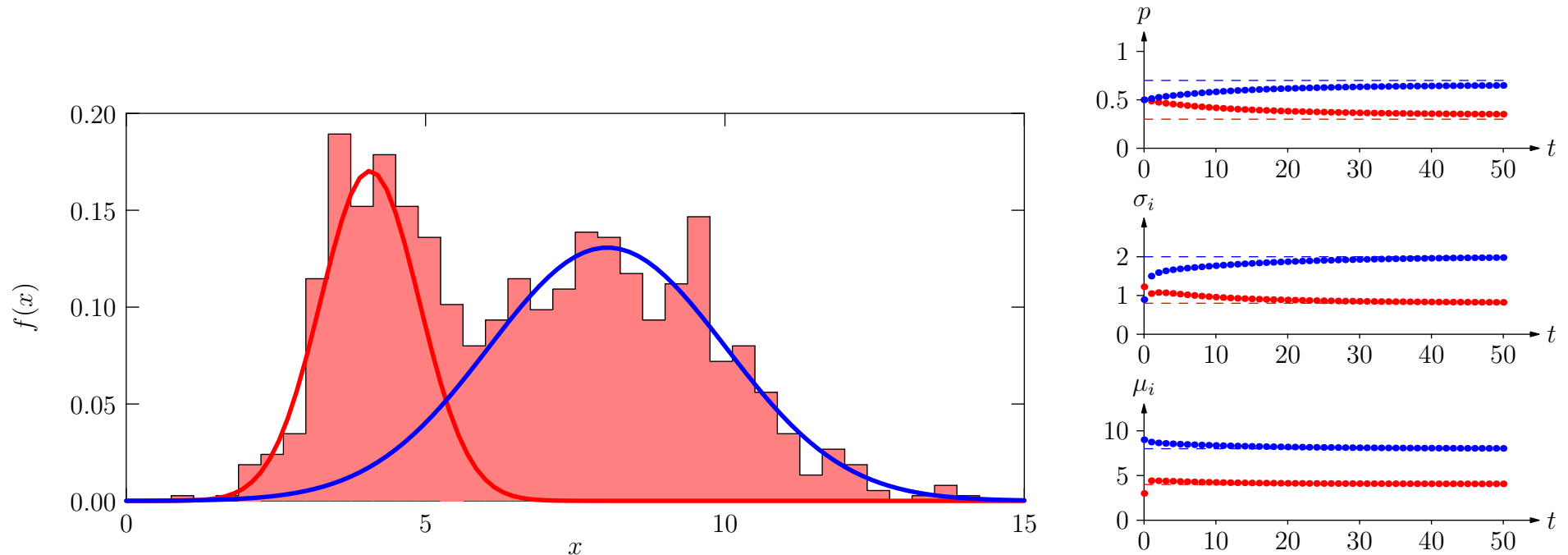


Mathematics of Machine Learning

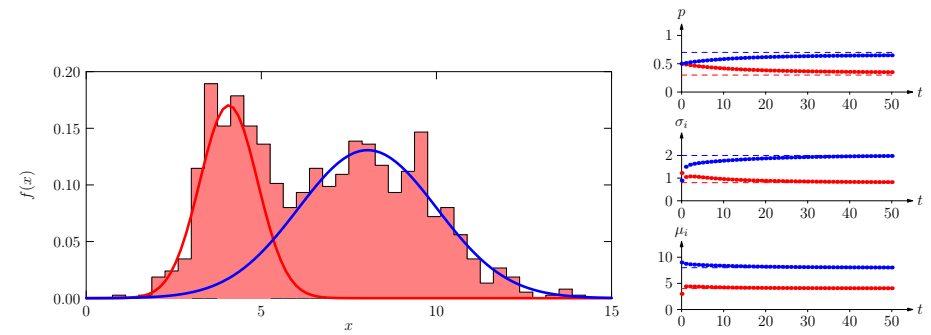
Probabilistic Inference



*Hierarchical Models, Mixture of Gaussians, Expectation
Maximisation*

Outline

1. **Building Probabilistic Models**
2. Mixture of Gaussians
3. Expectation Maximisation



Building Probabilistic Models

- To describe a system with uncertainty we use random variables, X, Y, Z , etc.
- We use the convention of writing random variables in capitals (this is sometimes confusing as when you observe a random variables it is no longer random)
- The variables are described by probability mass function $\mathbb{P}(X, Y, Z)$ or if our variables are continuous, but probability densities $f_{X, Y, Z}(x, y, z)$
- A major rule of probability is

$$\sum_X \mathbb{P}(X, Y, Z) = \mathbb{P}(Y, Z)$$

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Conditional Probabilities

- When developing models it is often useful to consider conditional probabilities e.g. $\mathbb{P}(X,Y|Z)$ or $f_{X|Y,Z}(x|y,z)$
- A second major rule in probabilistic modelling is

$$\mathbb{P}(X,Y) = \mathbb{P}(X|Y)\mathbb{P}(Y) = \mathbb{P}(Y|X)\mathbb{P}(X)$$

- This is a mathematical identity that does not imply causality (it defines conditional probability)
- It is the origins of Bayes' rule: $\mathbb{P}(X|Y) = \frac{\mathbb{P}(Y|X)\mathbb{P}(X)}{\mathbb{P}(Y)}$

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Discriminative Models

- We often think of our observations as given and the predictions as random variables
- For example we might be given some features $\mathbf{x}$ and we wish to predict a class $C \in \mathcal{C}$
- Our objective is then to find the probability $\mathbb{P}(C|\mathbf{x})$
- This is known as a **discriminative model**
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Generative Models

- Sometimes it is easy to think about the joint process of generating the features and outputs together
- This leads to a joint distribution $\mathbb{P}(\mathbf{X}, Y)$ where $\mathbf{X}$ are your features and Y is your output you are trying to predict
- This is known as a **generative model**
- Generative models are often more natural to think about
- We can use them to do discrimination using

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Modelling Virus

- Suppose we want to estimate the number of hospitalisation from Corona virus in the next month
- Our observable is the number of reported cases
- In our model we might want to estimate the number of actual cases
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- This will be a random variable (it is uncertain, but we can build a probabilistic model giving a distribution of number of actual cases)

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Hierarchical Models

- Of course, if I was really modelling the spread of a disease I would care about the probability, $f(C|A,V)$, of catching the disease, C , given the persons age A and the variant of the disease V
- I would want to know the distribution of ages $f(A)$ and try to infer the probability of different variants $\mathbb{P}(V)$
- I would care about the probability, $f(R|A,V)$, of cases being reported given age and variant
- And the probability, $f(H|A,V)$, of hospitalisation given A and V
- This would involve an elaborate (hierarchical) model with a large number of latent variables

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Probabilistic Inference

- We can use Bayes' rules to learn a set of parameter Θ that occur in our likelihood function

$$\mathbb{P}(\Theta|\mathcal{D}) = \frac{\mathbb{P}(\mathcal{D}|\Theta)\mathbb{P}(\Theta)}{\mathbb{P}(\mathcal{D})}$$

- This provides us a full probabilistic description of the parameters
- It doesn't overfit (we are not choosing the best)
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- Bayes is problematic because it is often hard
- The posterior is often not expressible as a nice probability function
- We need to compute the *evidence* or *margin likelihood* we use

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- Nevertheless we can usually do this using Monte Carlo techniques

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- When we assume a uniform prior then the MAP solution is just maximising the likelihood
- Weirdly this hack was accepted as part of mainstream statistics even when Bayesian statistics was considered unscientific
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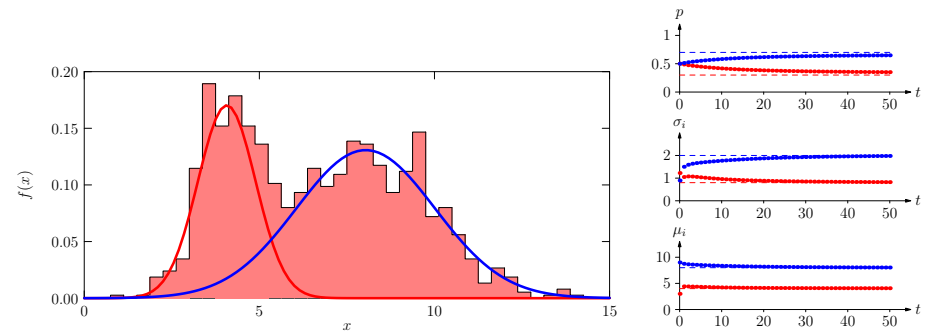
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Outline

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2. **Mixture of Gaussians**
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Mixture of Gaussians

- Suppose we were observing the decays from two types of short-lived particle, A or B
- We observe the half life, X_i , but not the particle type
- We assume X_i is normally distributed with unknown means and variances: $\Theta = \{\mu_A, \sigma_A^2, \mu_B, \sigma_B^2\}$
- Let $Z_i \in \{0,1\}$ be an indicator that particle i is of type A
- The probability of X_i is given by

$$f(X_i|Z_i, \Theta) = Z_i \mathcal{N}(X_i|\mu_A, \sigma_A^2) + (1 - Z_i) \mathcal{N}(X_i|\mu_B, \sigma_B^2)$$

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$$f(X_i|Z_i, \Theta) = Z_i \mathcal{N}(X_i|\mu_A, \sigma_A^2) + (1 - Z_i) \mathcal{N}(X_i|\mu_B, \sigma_B^2)$$

Mixture of Gaussians

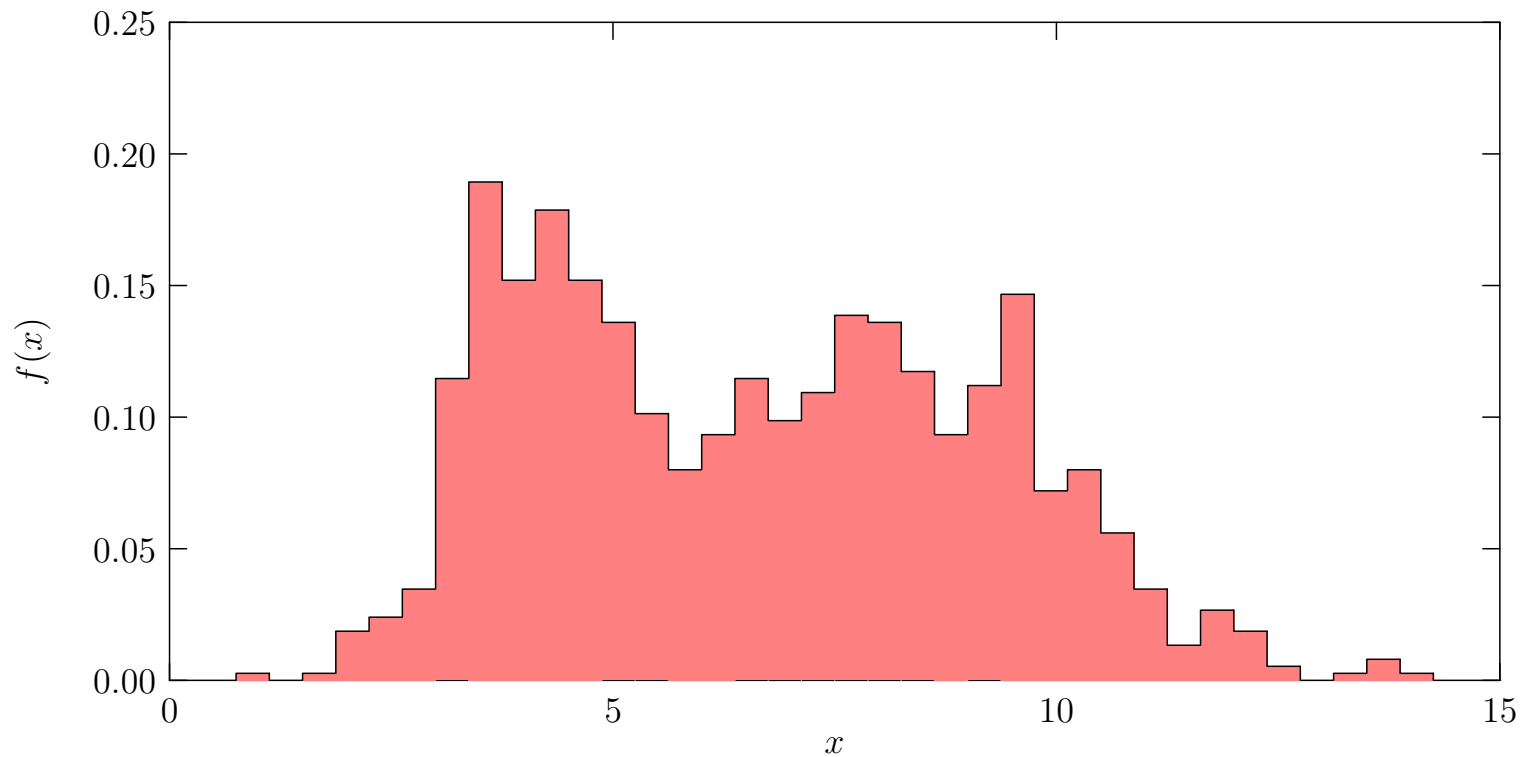
- Suppose we were observing the decays from two types of short-lived particle, A or B
- We observe the half life, X_i , but not the particle type
- We assume X_i is normally distributed with unknown means and variances: $\Theta = \{\mu_A, \sigma_A^2, \mu_B, \sigma_B^2\}$
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- Note that

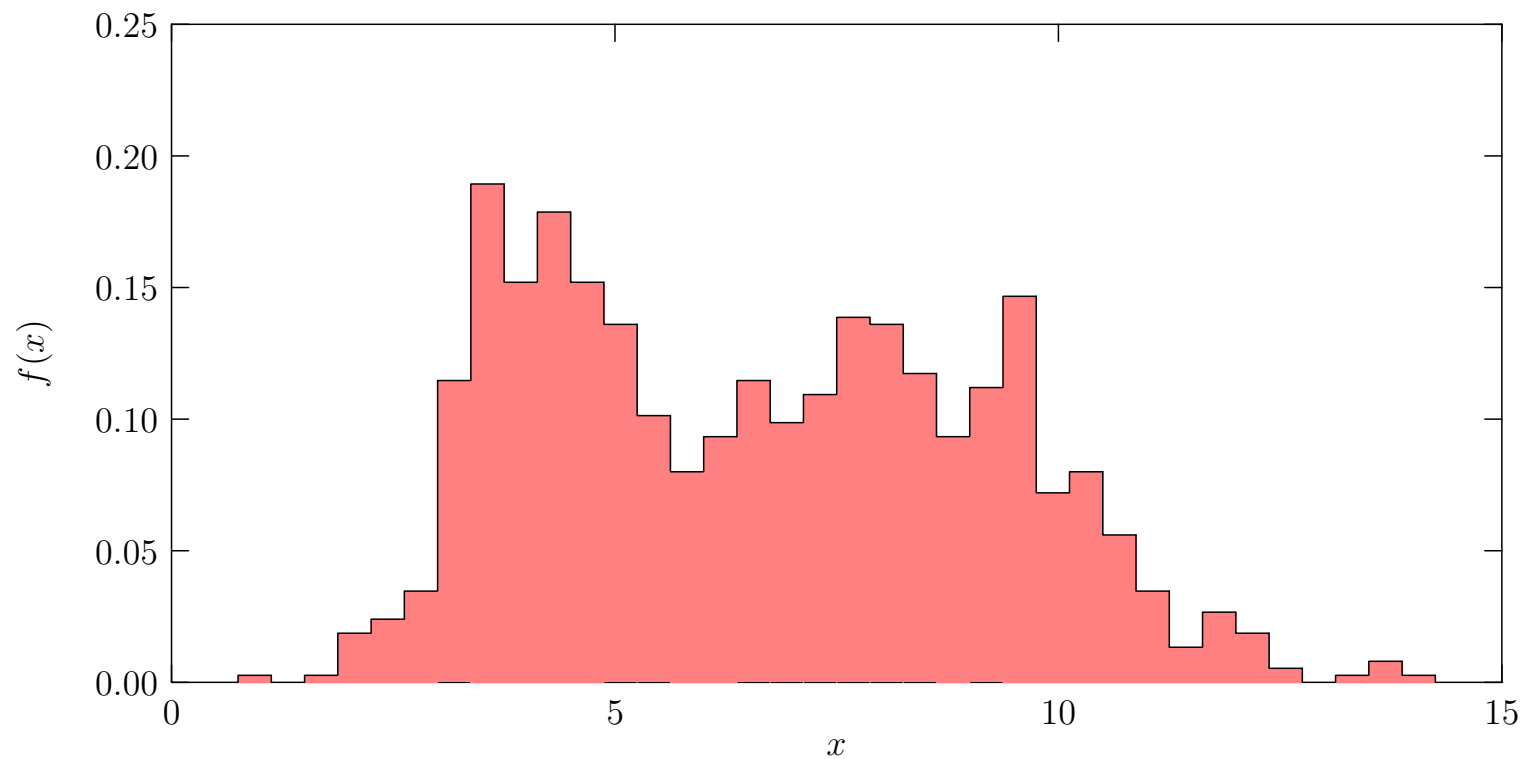
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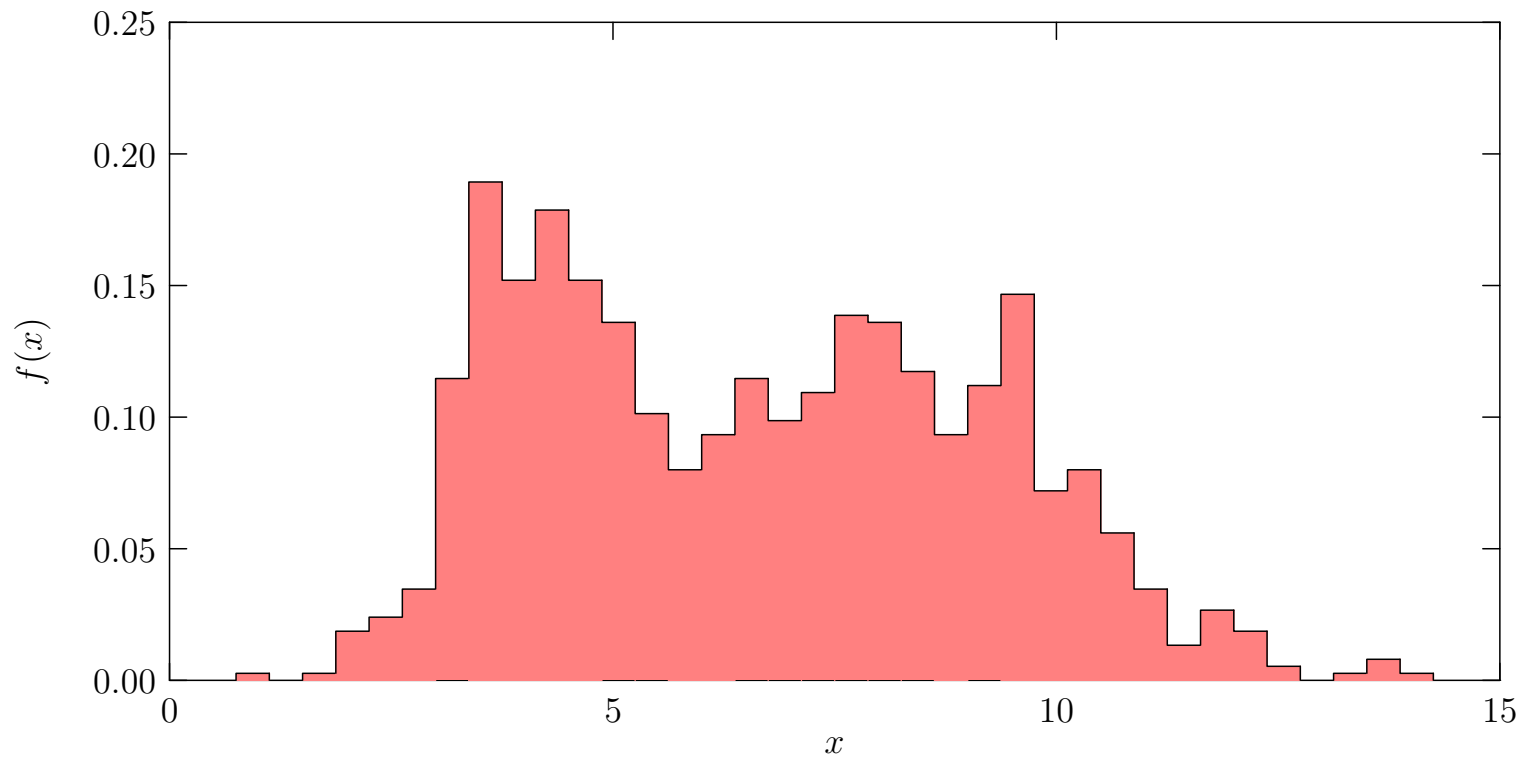
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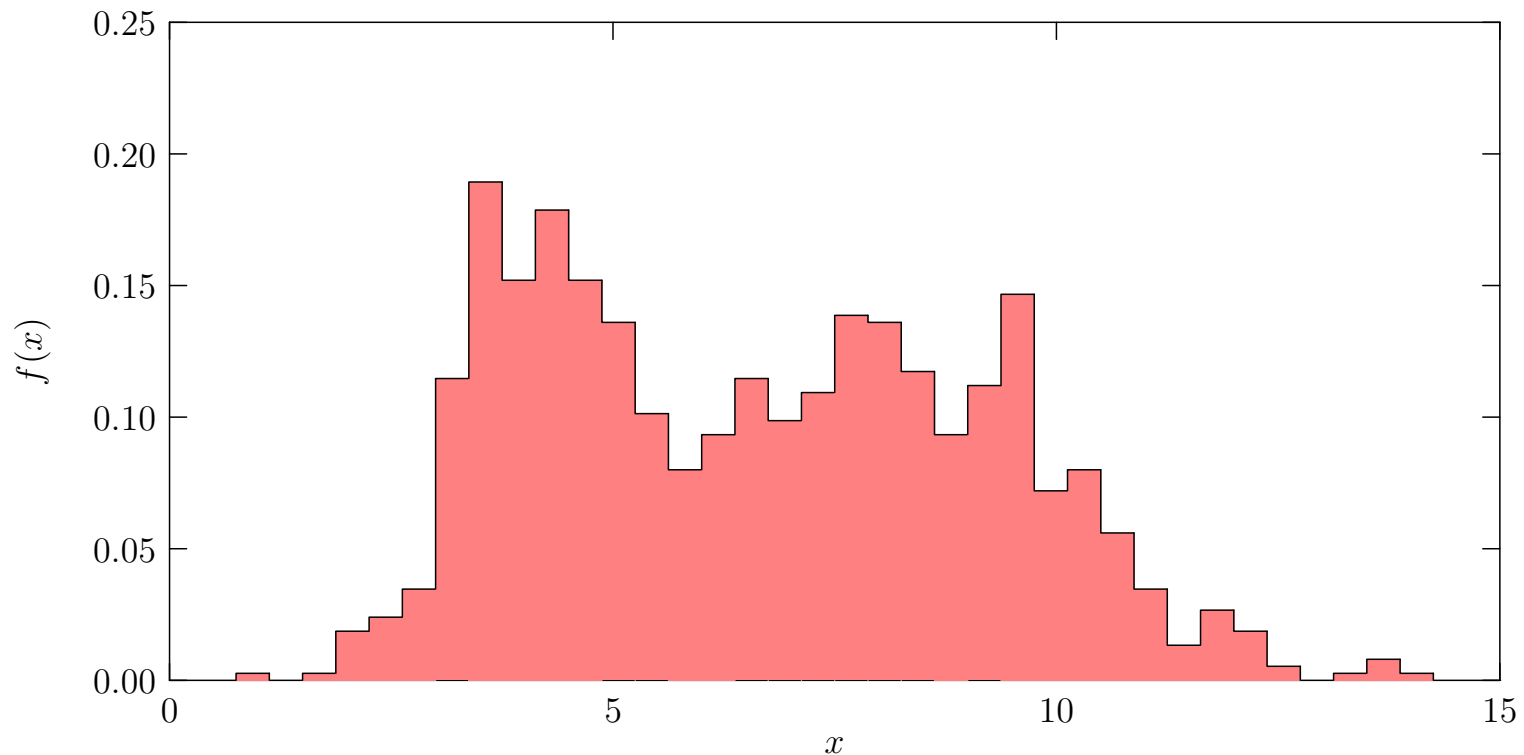
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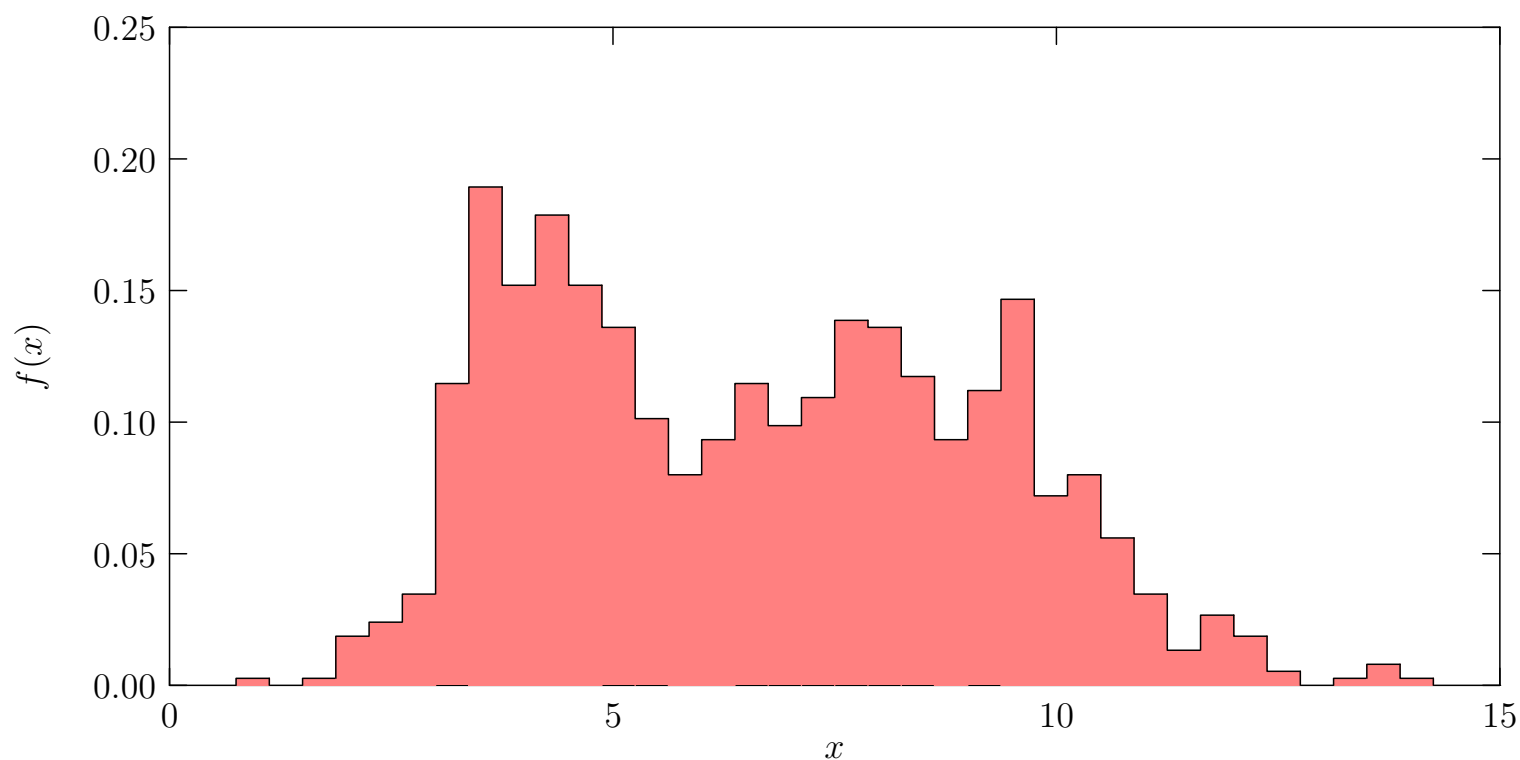
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Maximum Likelihood

- To solve the model as a Bayesian we would have to assign priors to our parameters $\Theta = (\mu_A, \sigma_A, \mu_B, \sigma_B, p)$
- This is doable, but complicated (we would also end up with a distribution for our parameters)
- Often we only want a reasonable estimate for some of our parameters (e.g. the half-lives μ_A and μ_B)
- A reasonable approach is to seek those parameters that maximise the likelihood of our observed data

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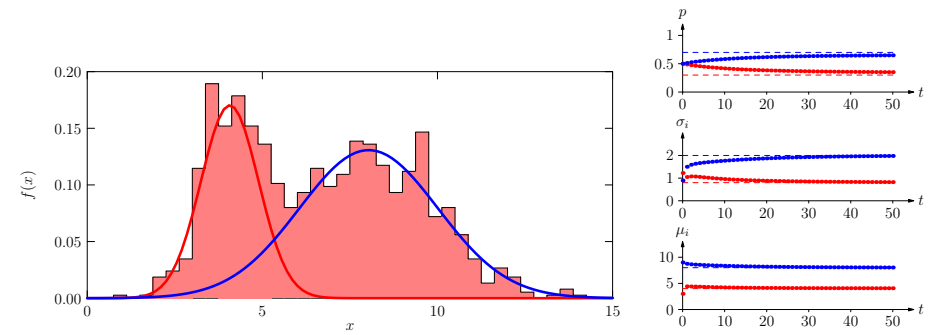
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Outline

1. Building Probabilistic Models
2. Mixture of Gaussians
3. **Expectation Maximisation**



Maximum Likelihood with Latent Variables

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- If we knew which type of particle a data-point belongs to (Z_i) then it would be straightforward to maximise the likelihood
- As we don't we need to estimate $\mathbb{P}(Z_i = 1)$, but this depends on $\mu_A, \sigma_A^2, \mu_B, \sigma_B^2$ and p
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- Instead we can use an **expectation-maximisation algorithm** usually known as an **EM algorithm**
- We proceed iteratively by maximising the expected log-likelihood with respect to the current set of parameters

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- The argument around why this works is quite involved
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$$Q(\Theta|\Theta^{(t)}) = \sum_{\mathbf{Z} \in \{0,1\}^m} \mathbb{P}(\mathbf{Z}|\mathcal{D}, \Theta^{(t)}) \log(f(\mathcal{D}|\mathbf{Z}, \Theta))$$

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$$\log(f(\mathcal{D}|\Theta^{(t+1)})) - \log(f(\mathcal{D}|\Theta^{(t)})) \geq Q(\Theta^{(t+1)}|\Theta^{(t)}) - Q(\Theta^{(t)}|\Theta^{(t)}) \geq 0$$

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Conditional Latent Variables

- We need to compute the distribution of latent variables conditioned on the data and current estimated parameters
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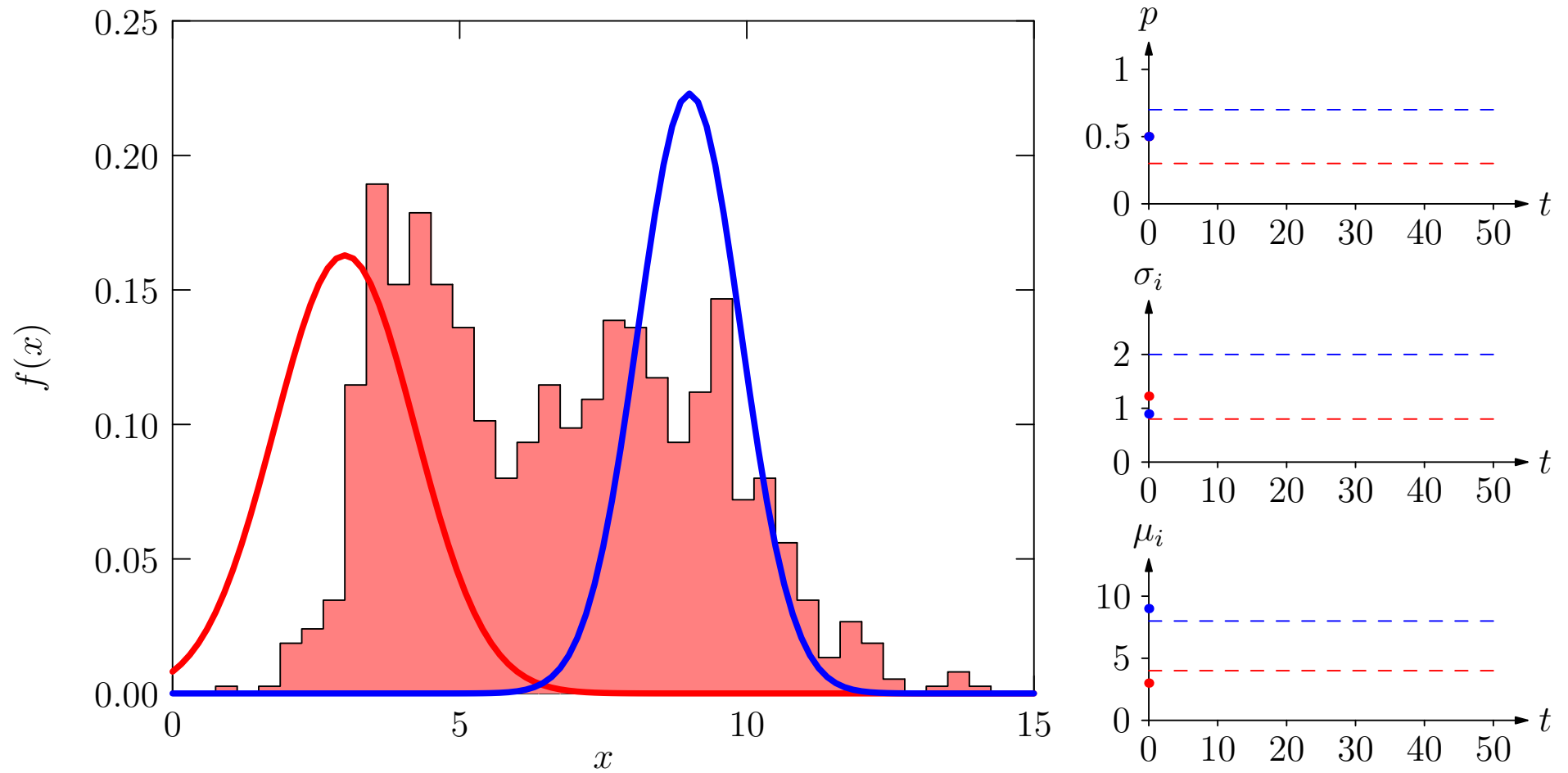
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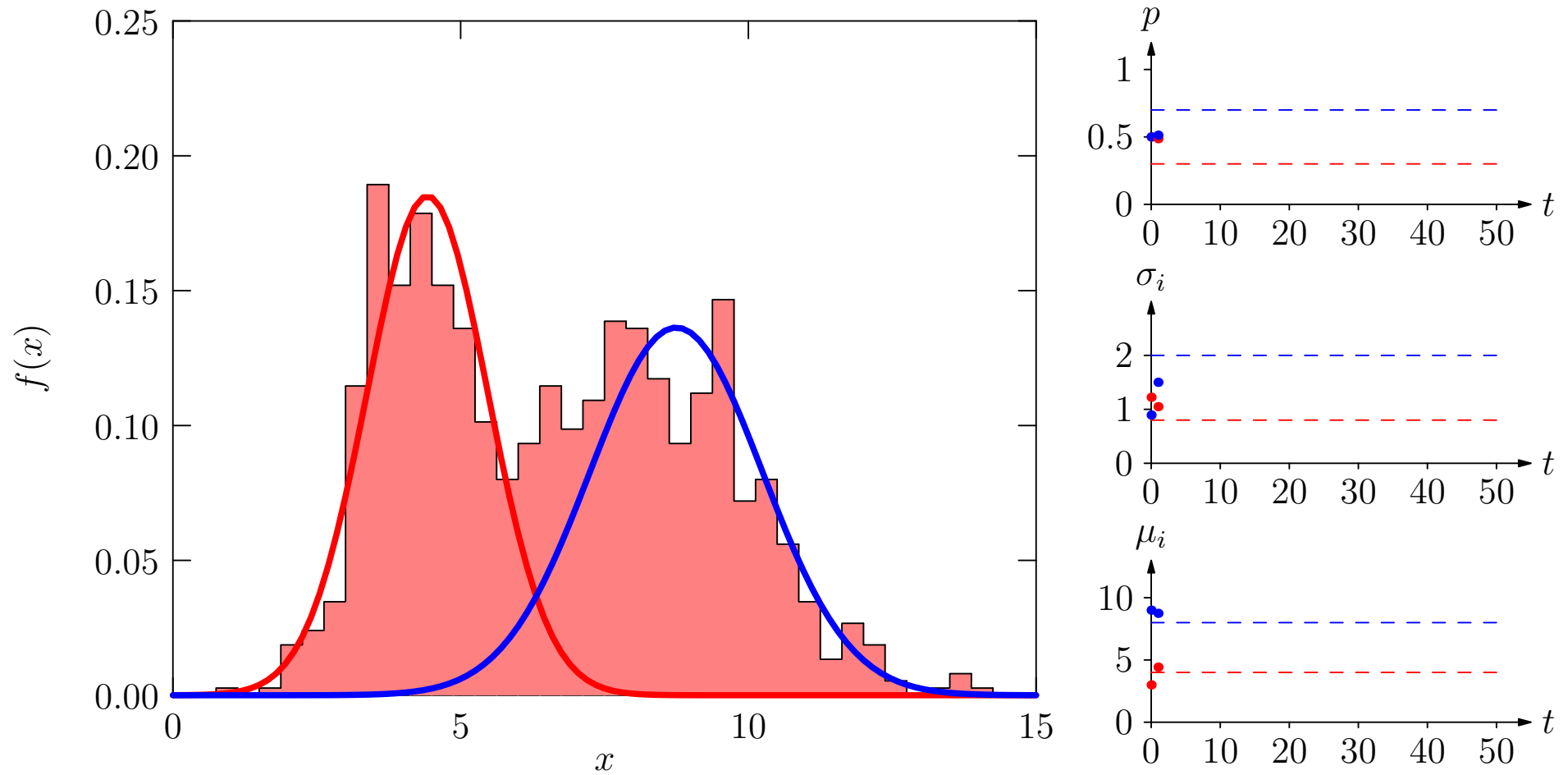
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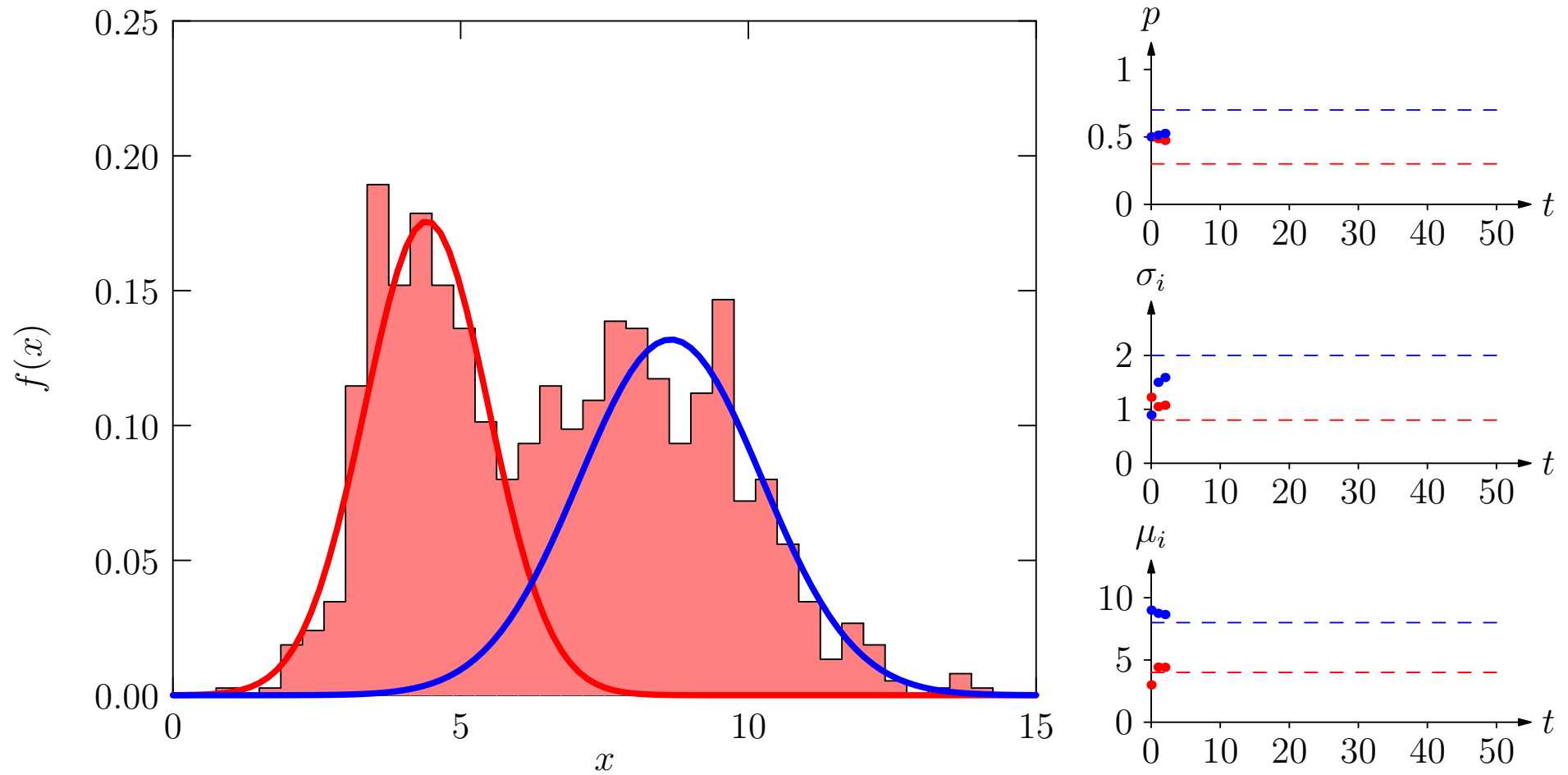
Example



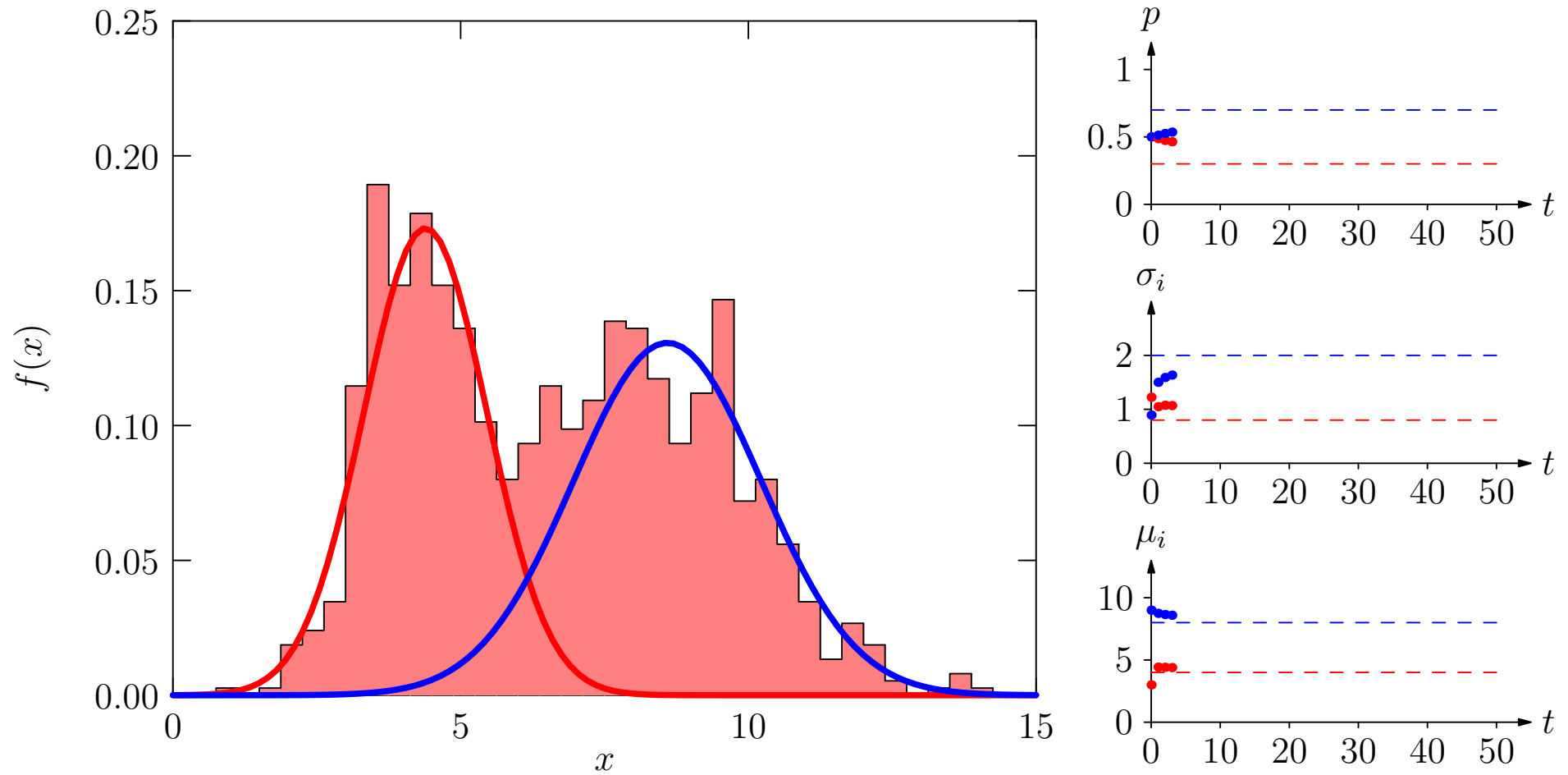
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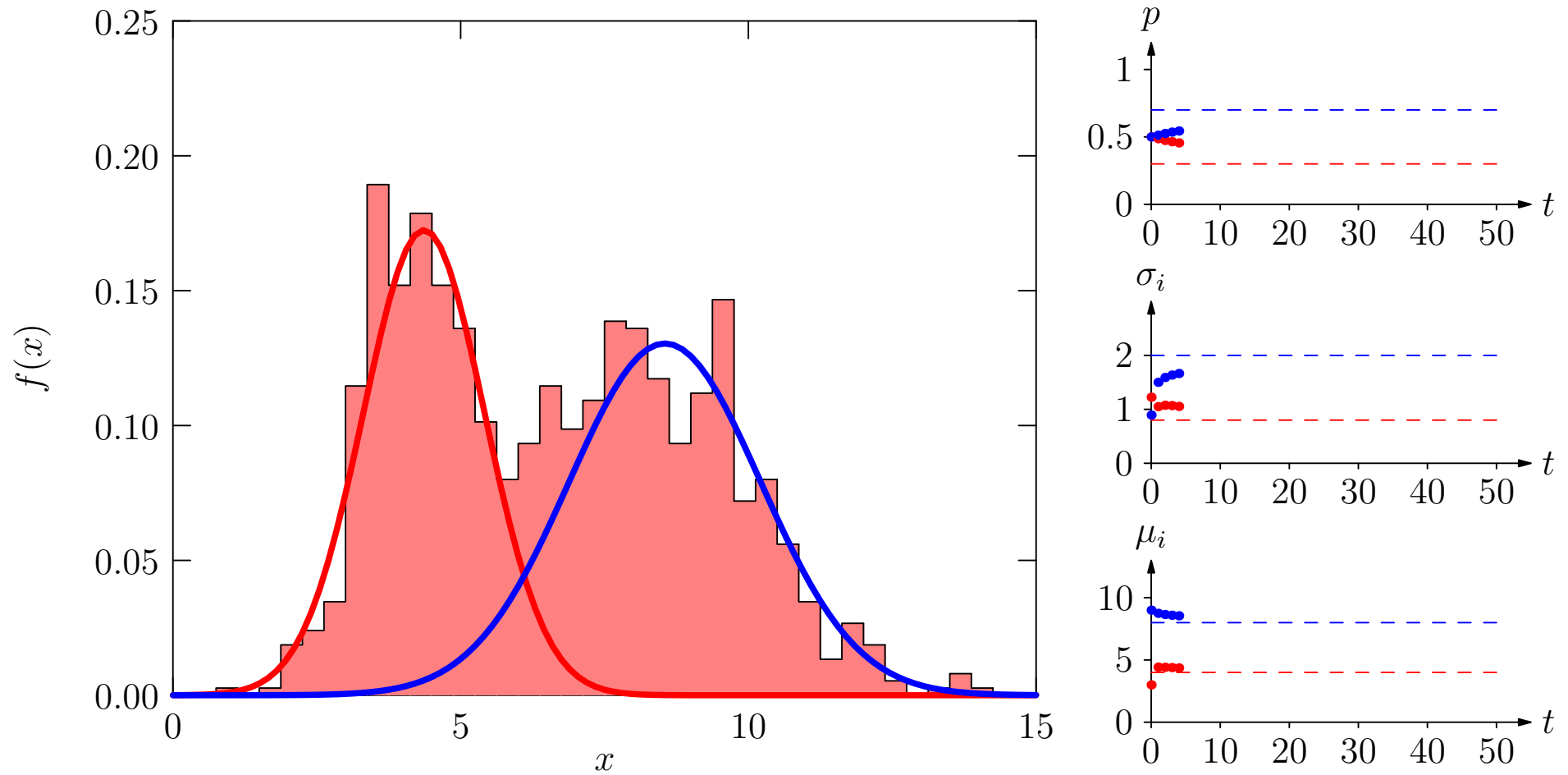
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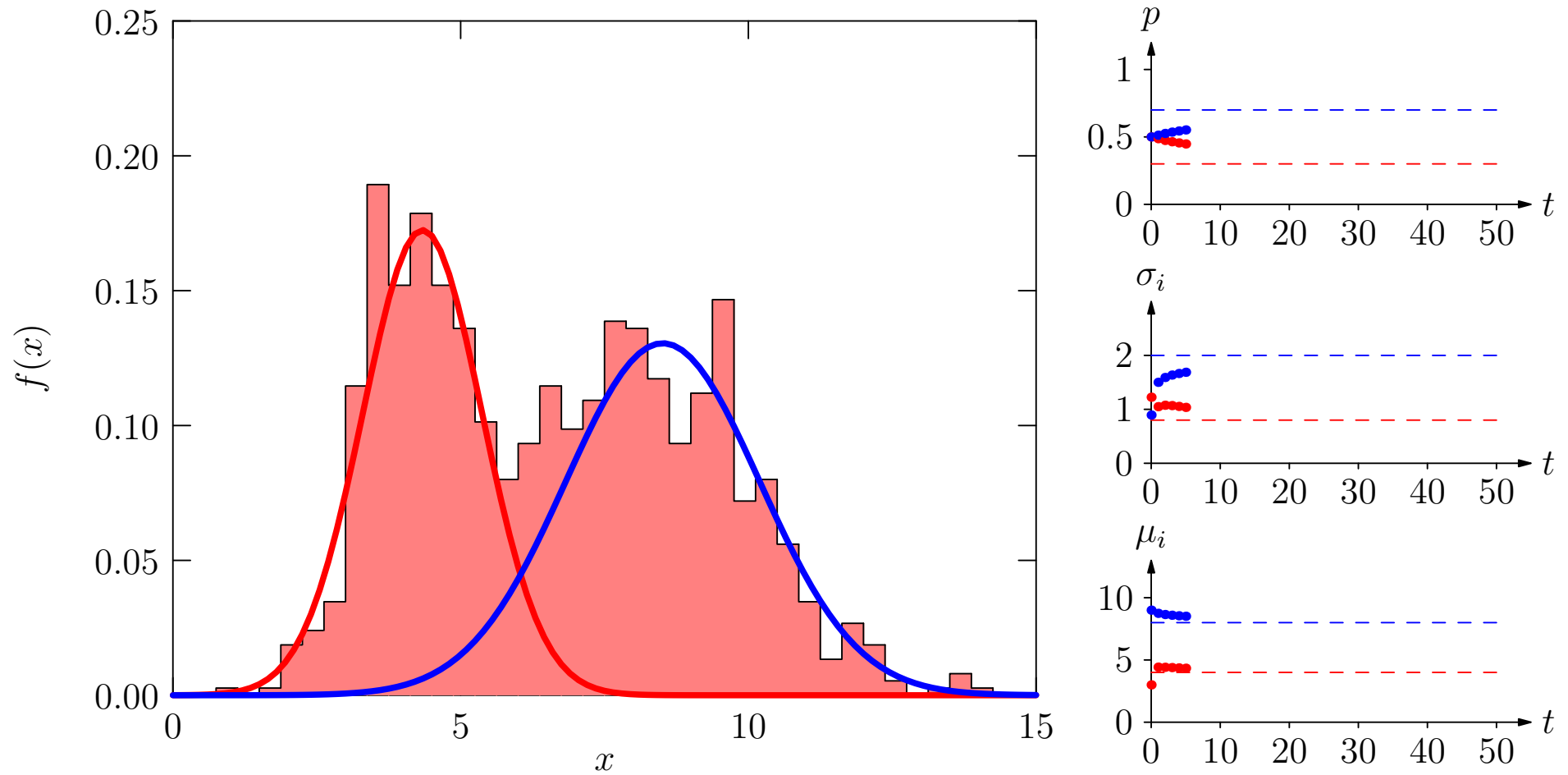
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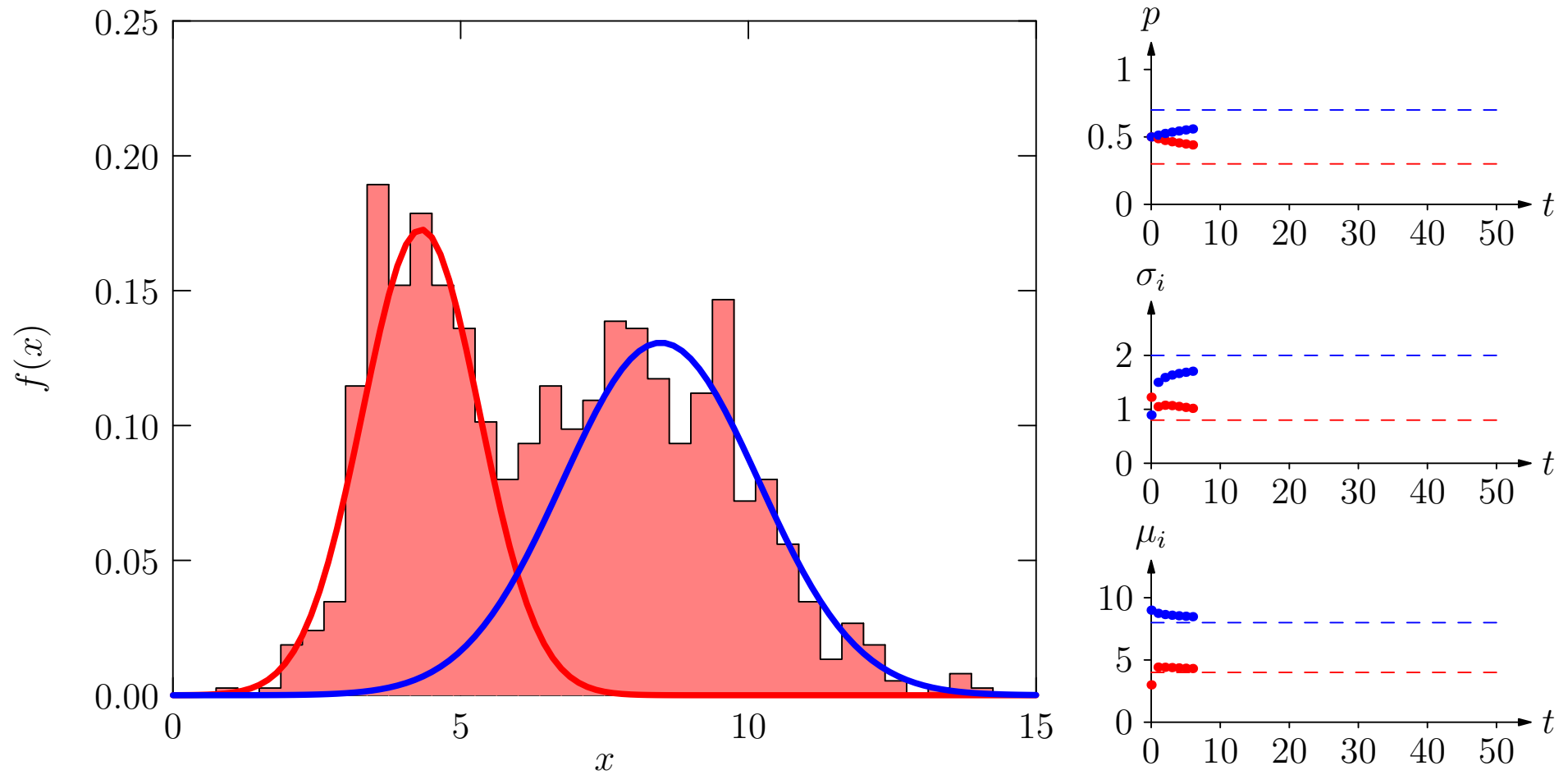
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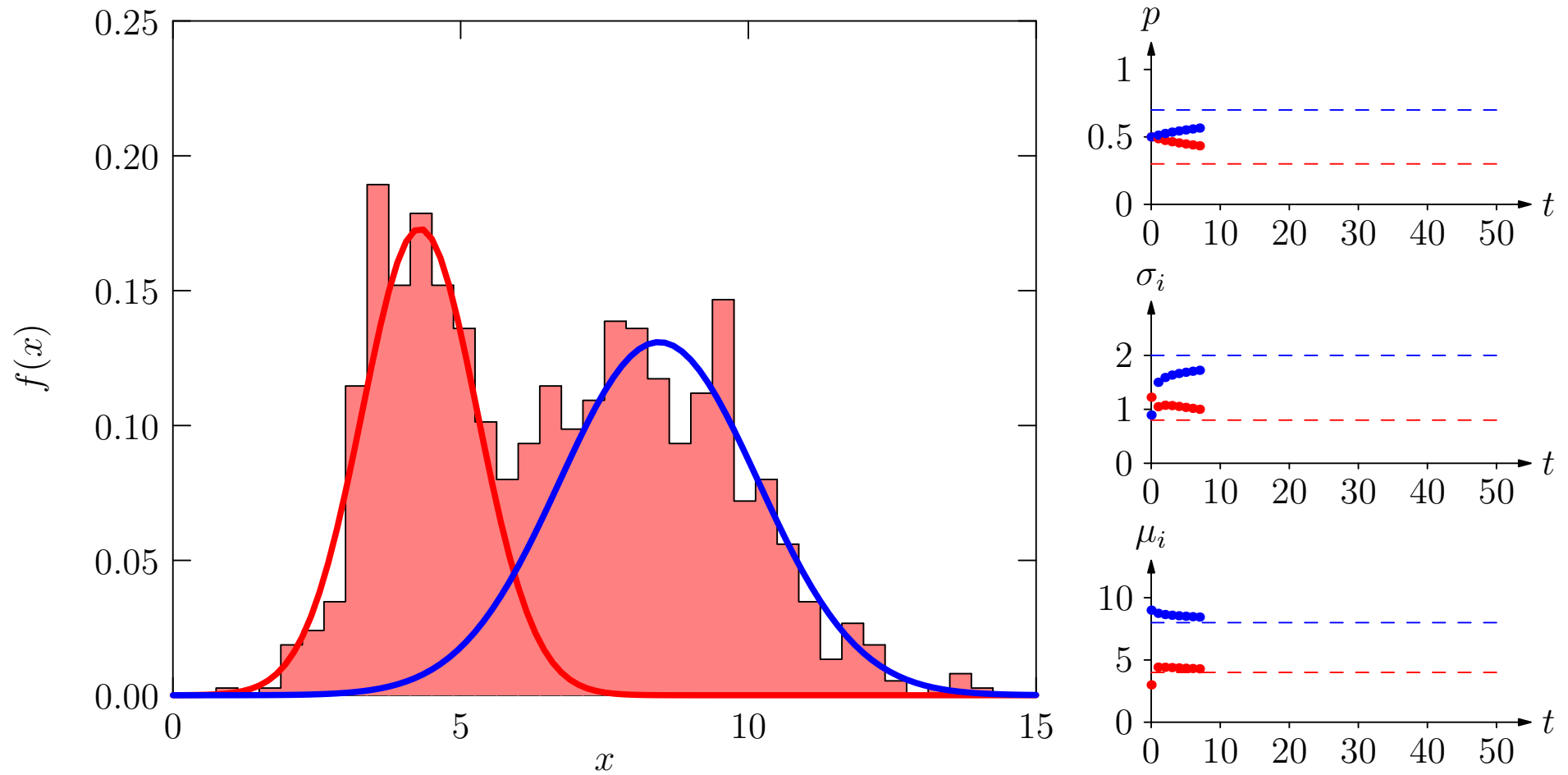
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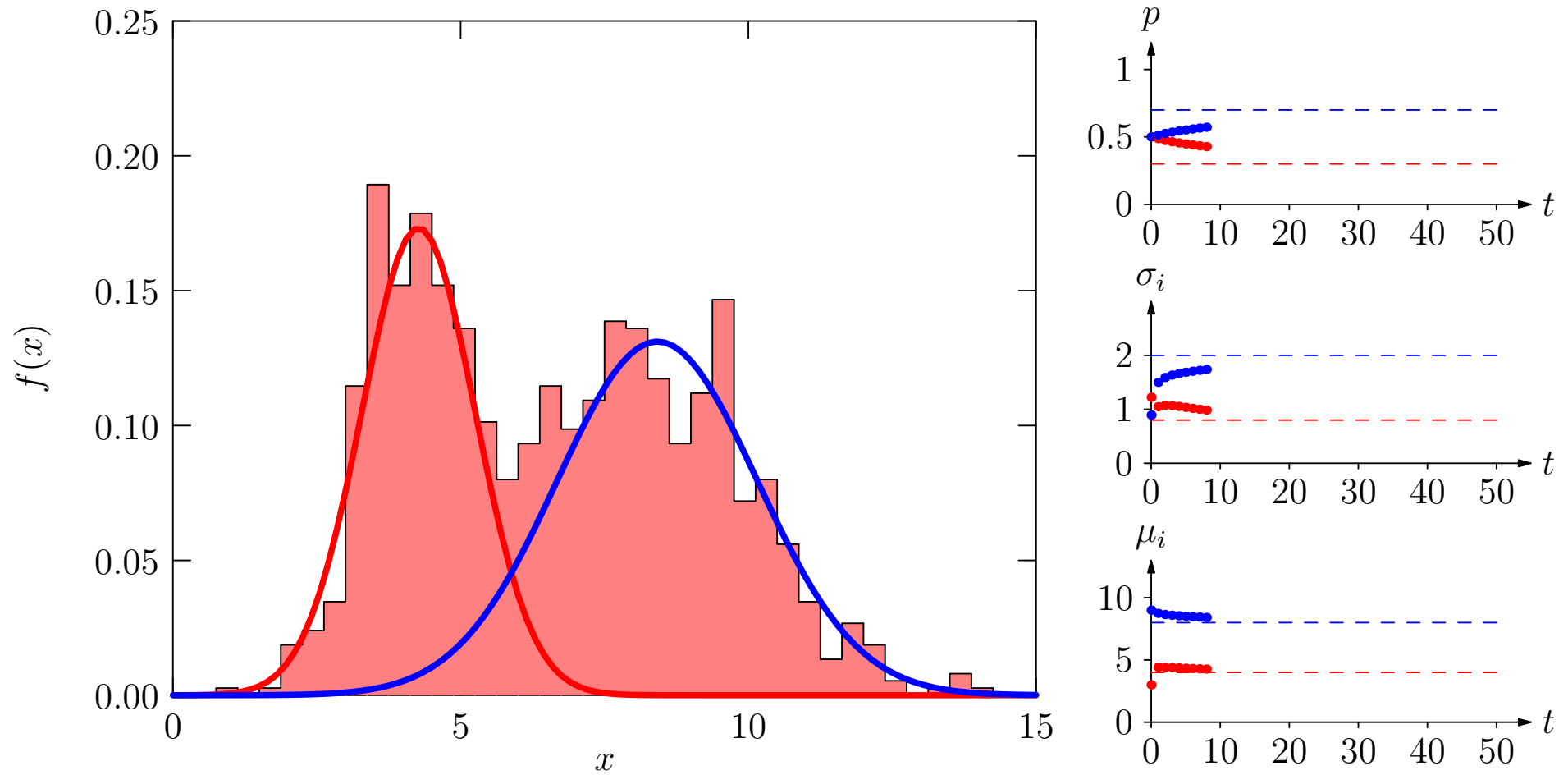
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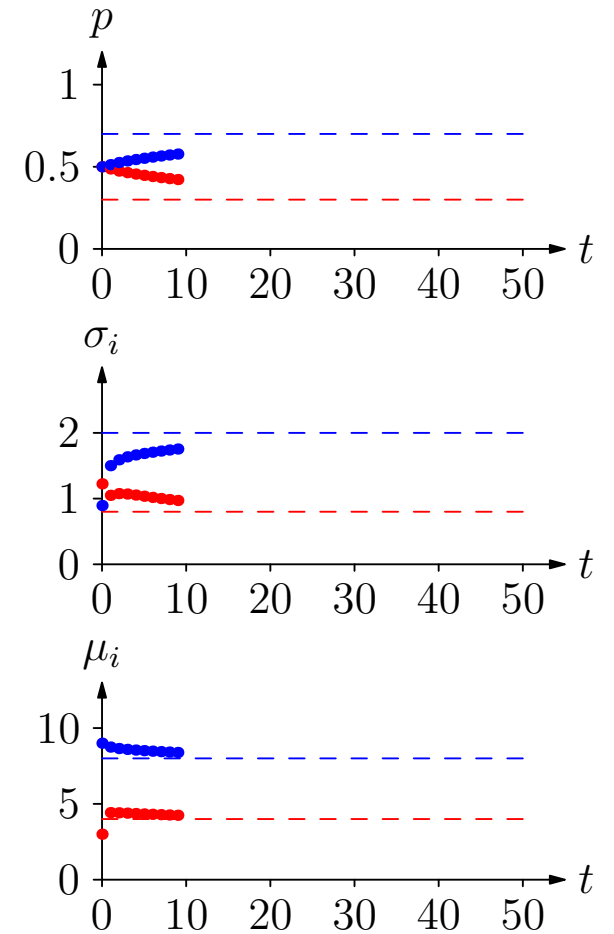
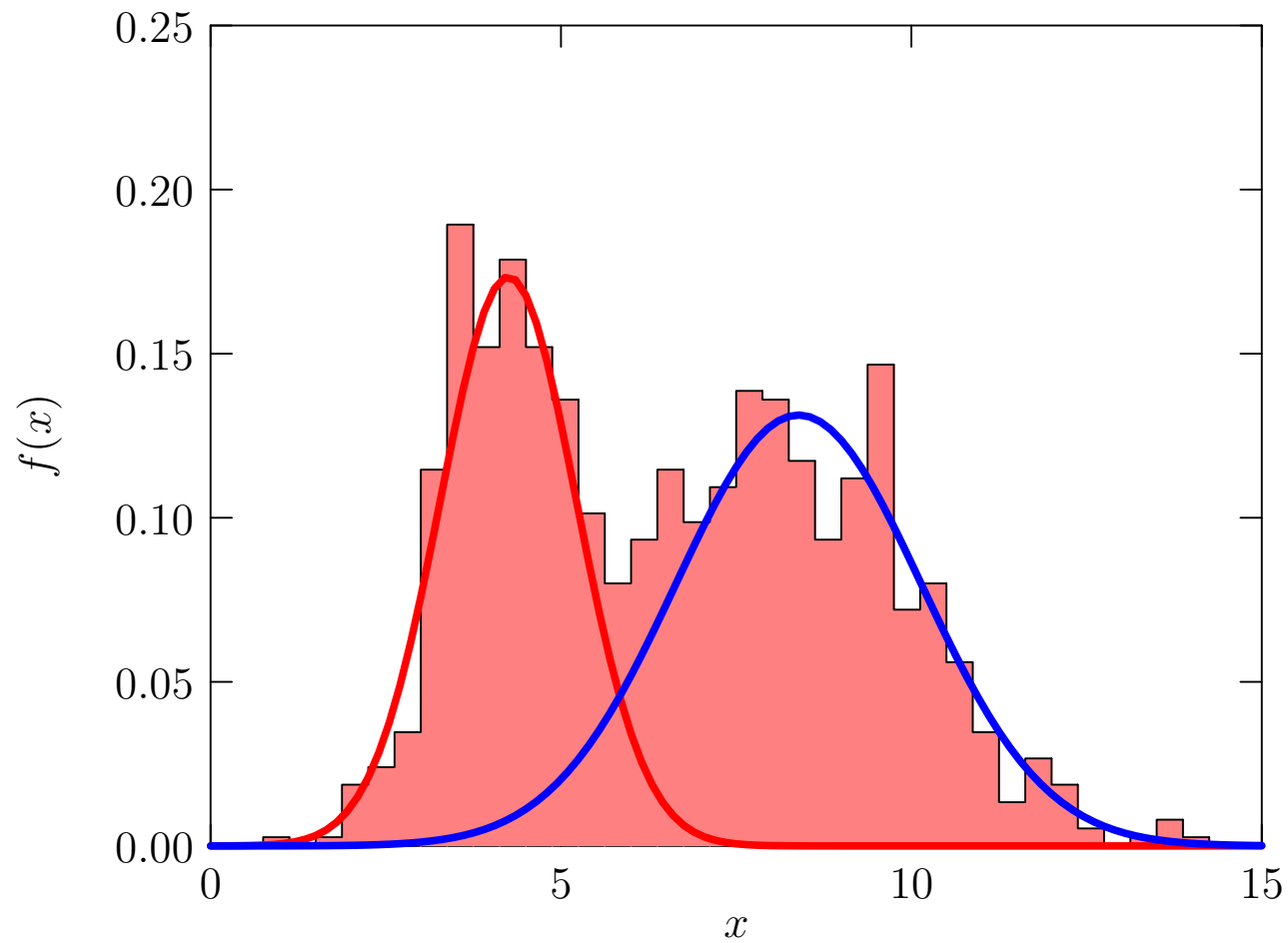
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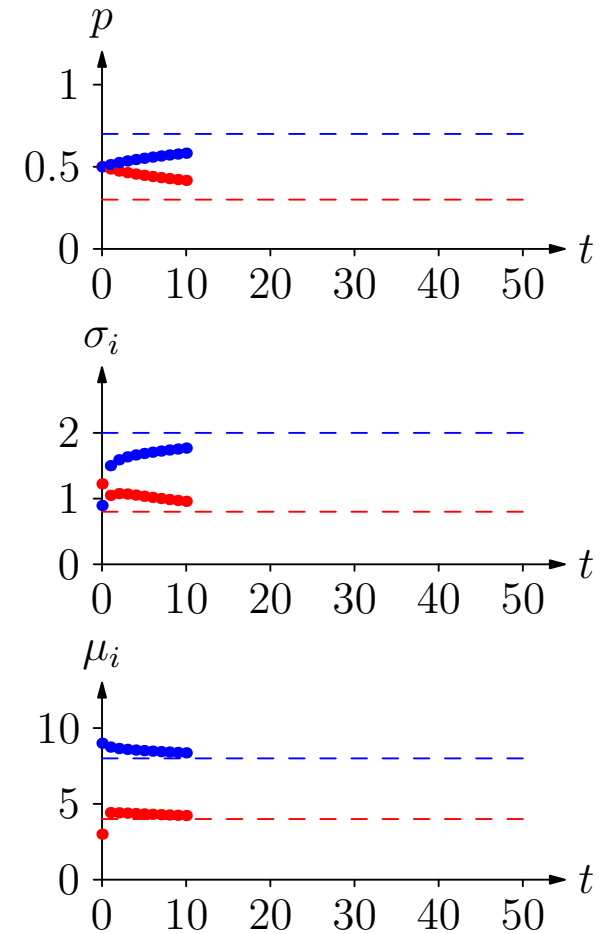
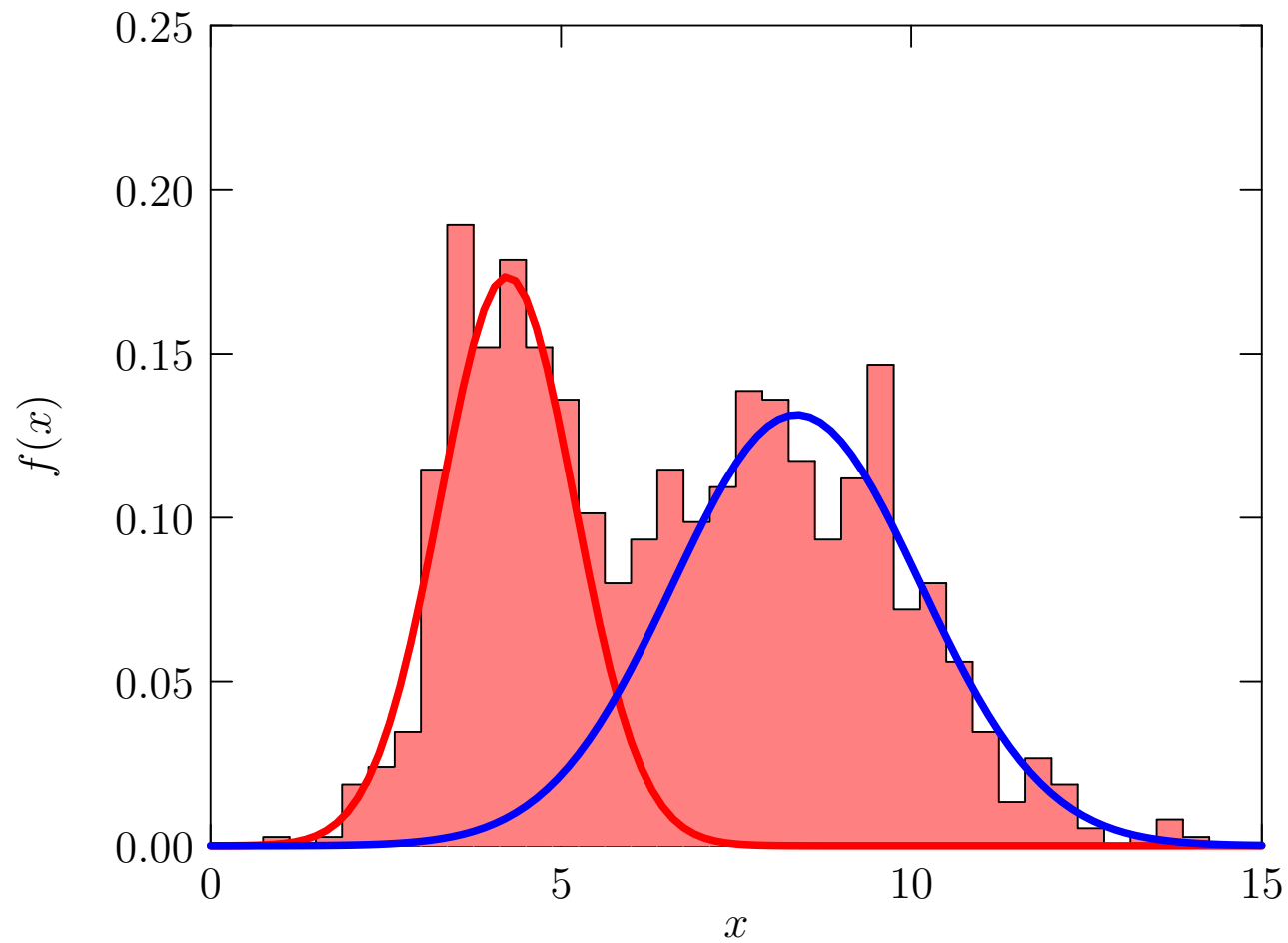
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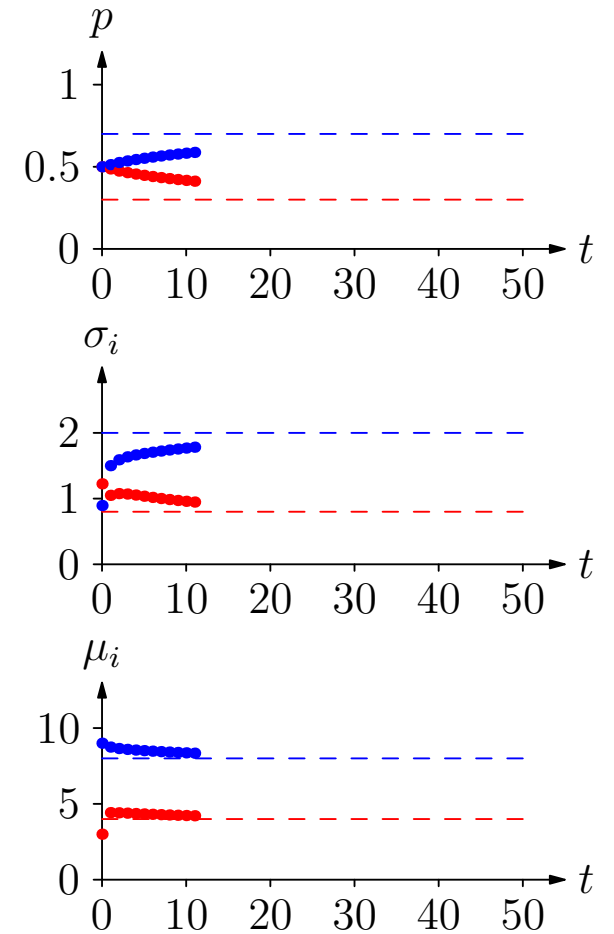
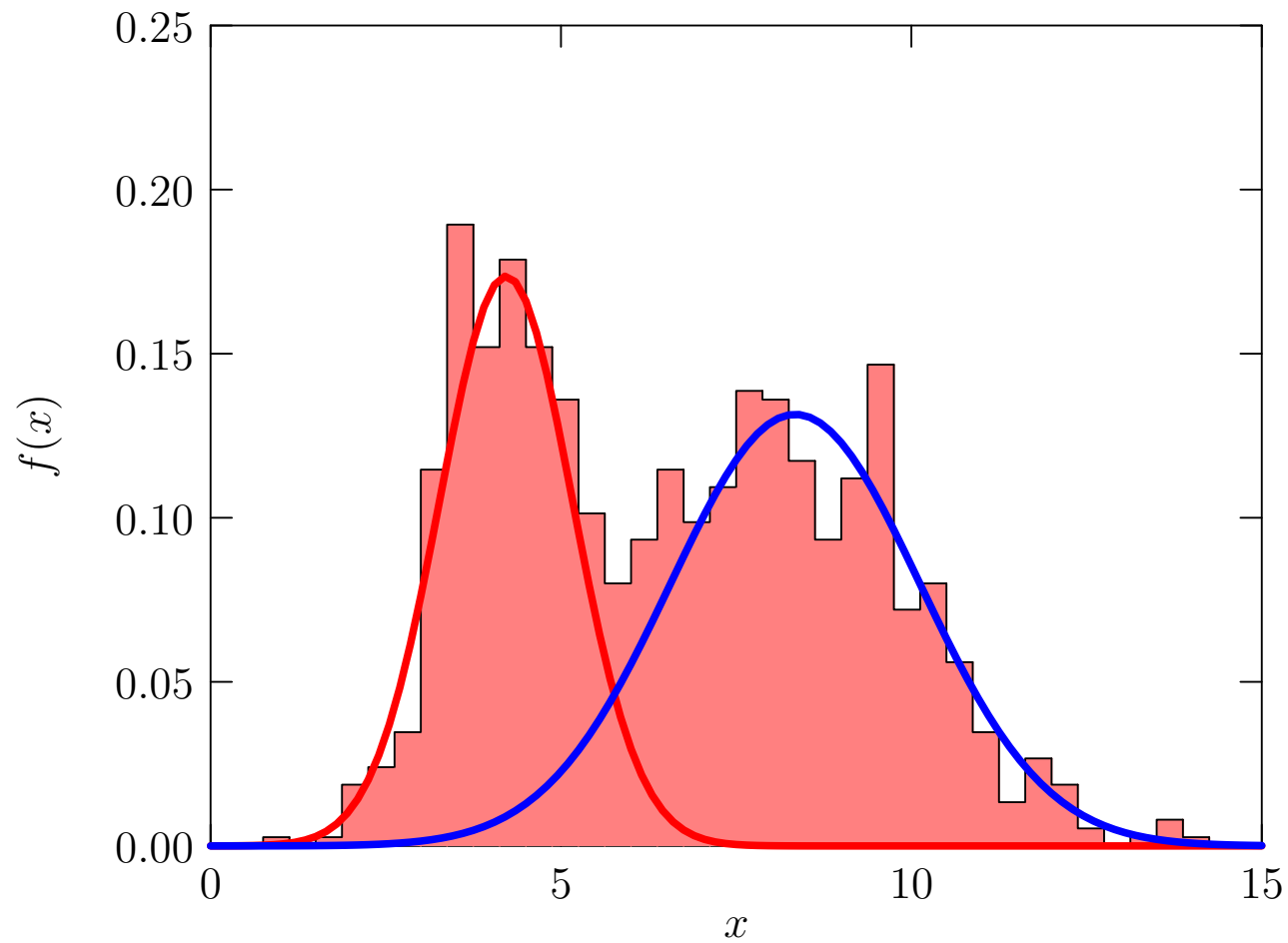
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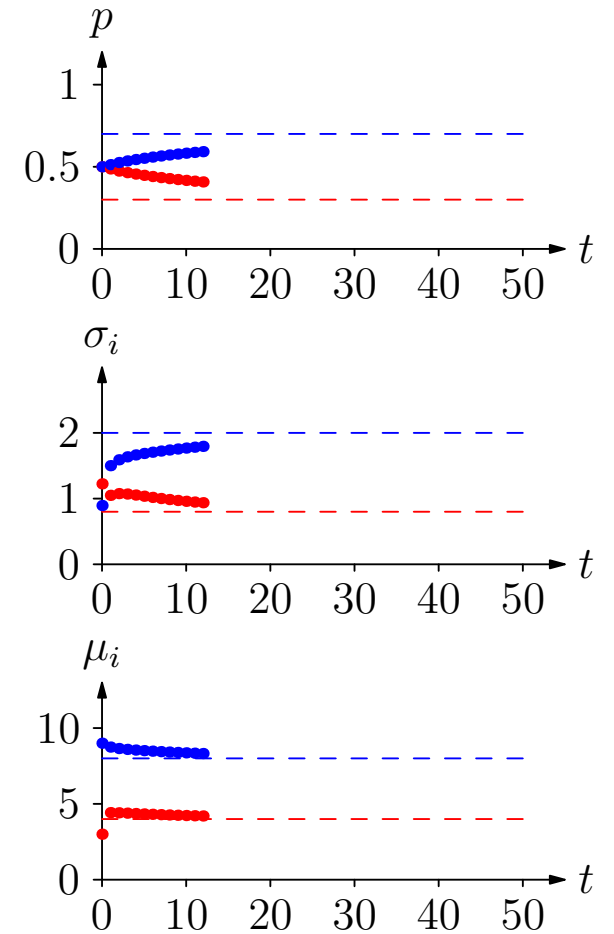
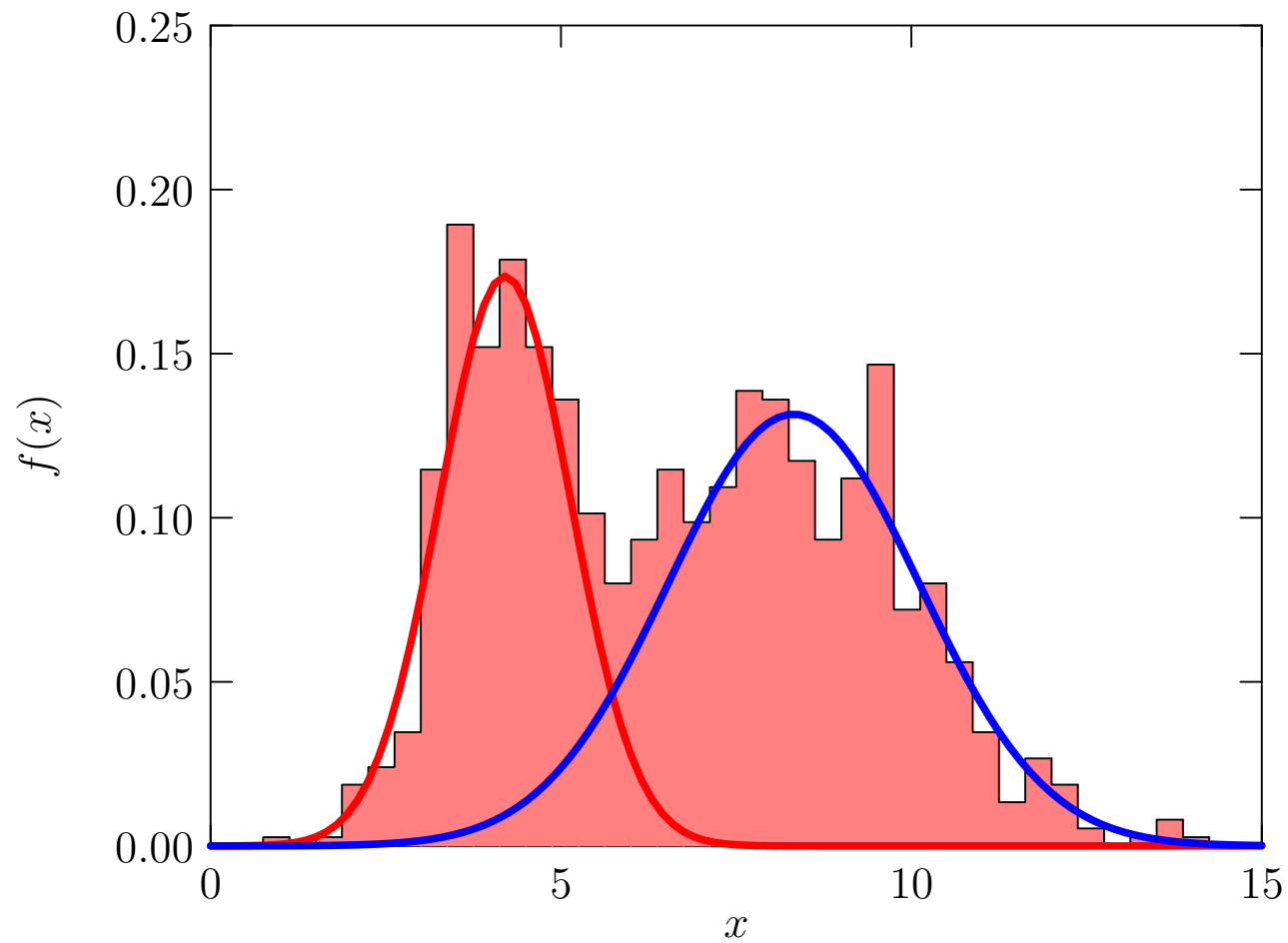
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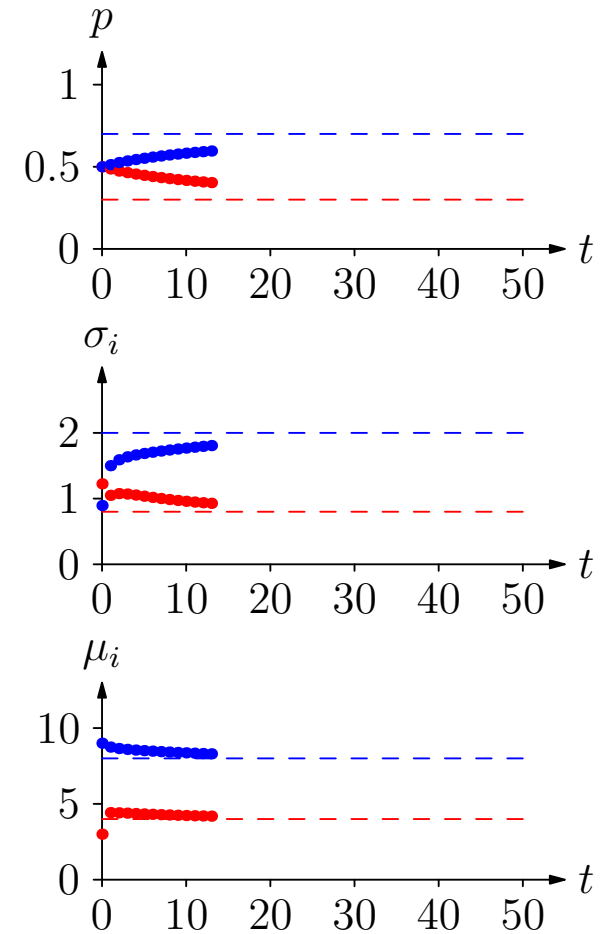
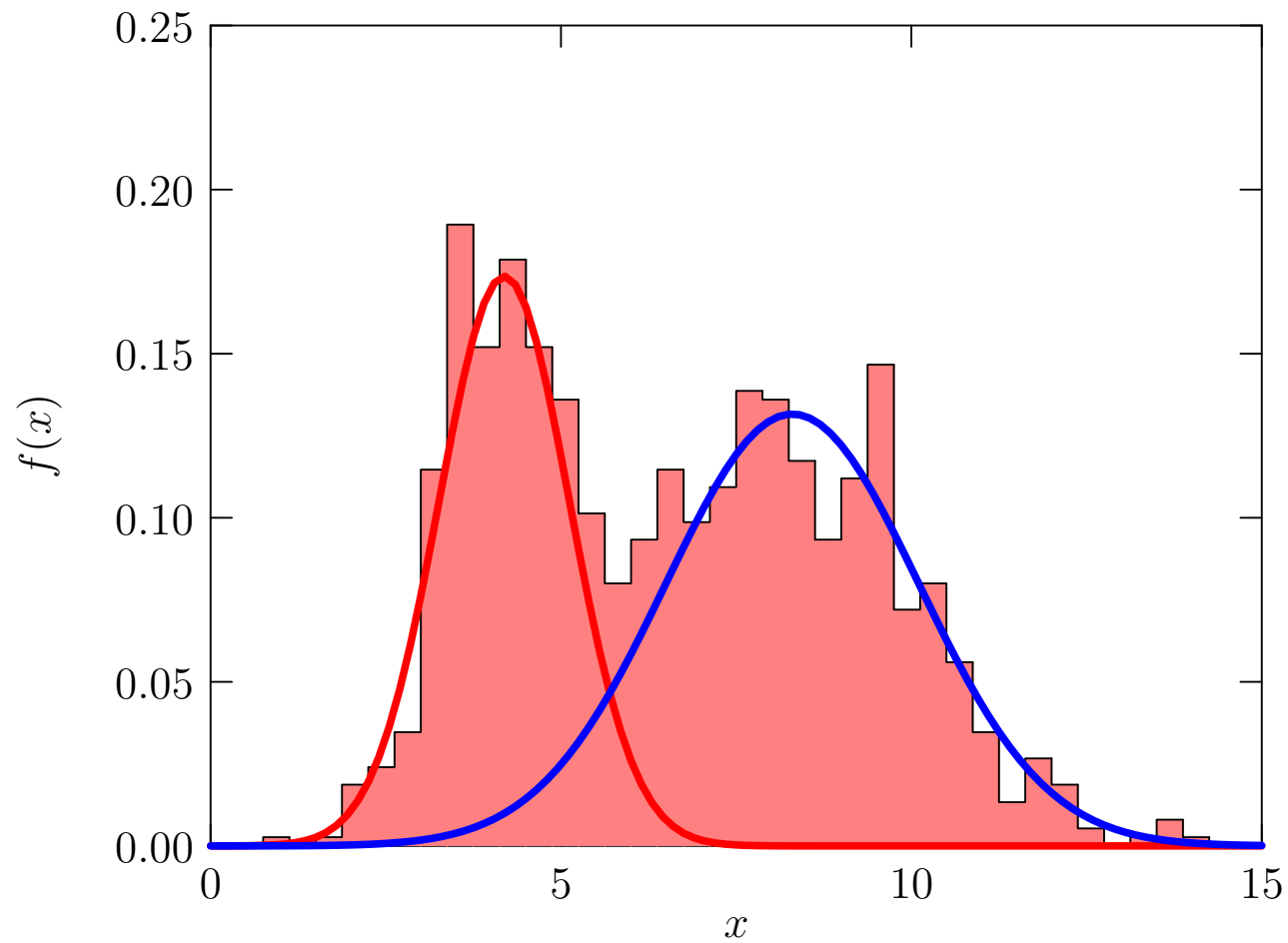
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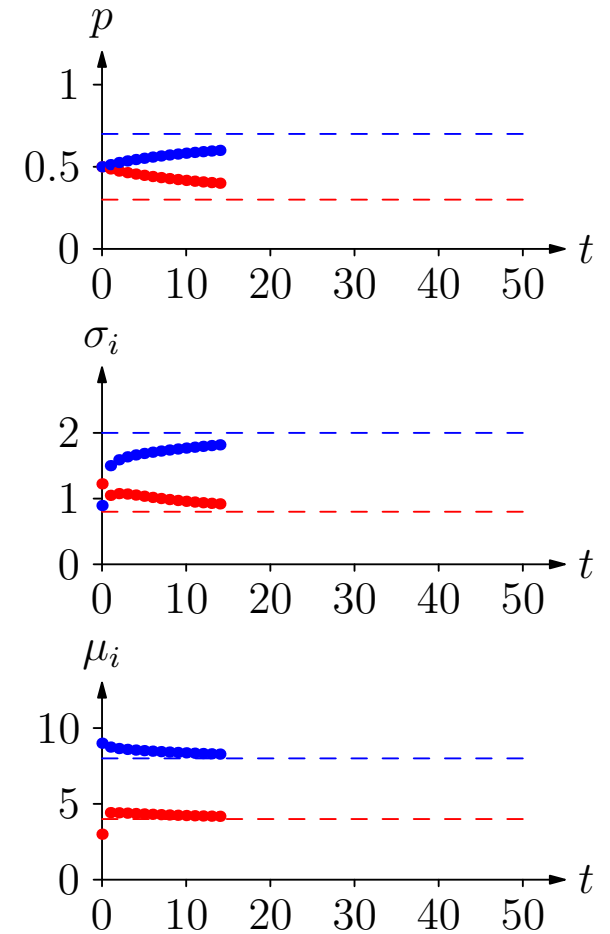
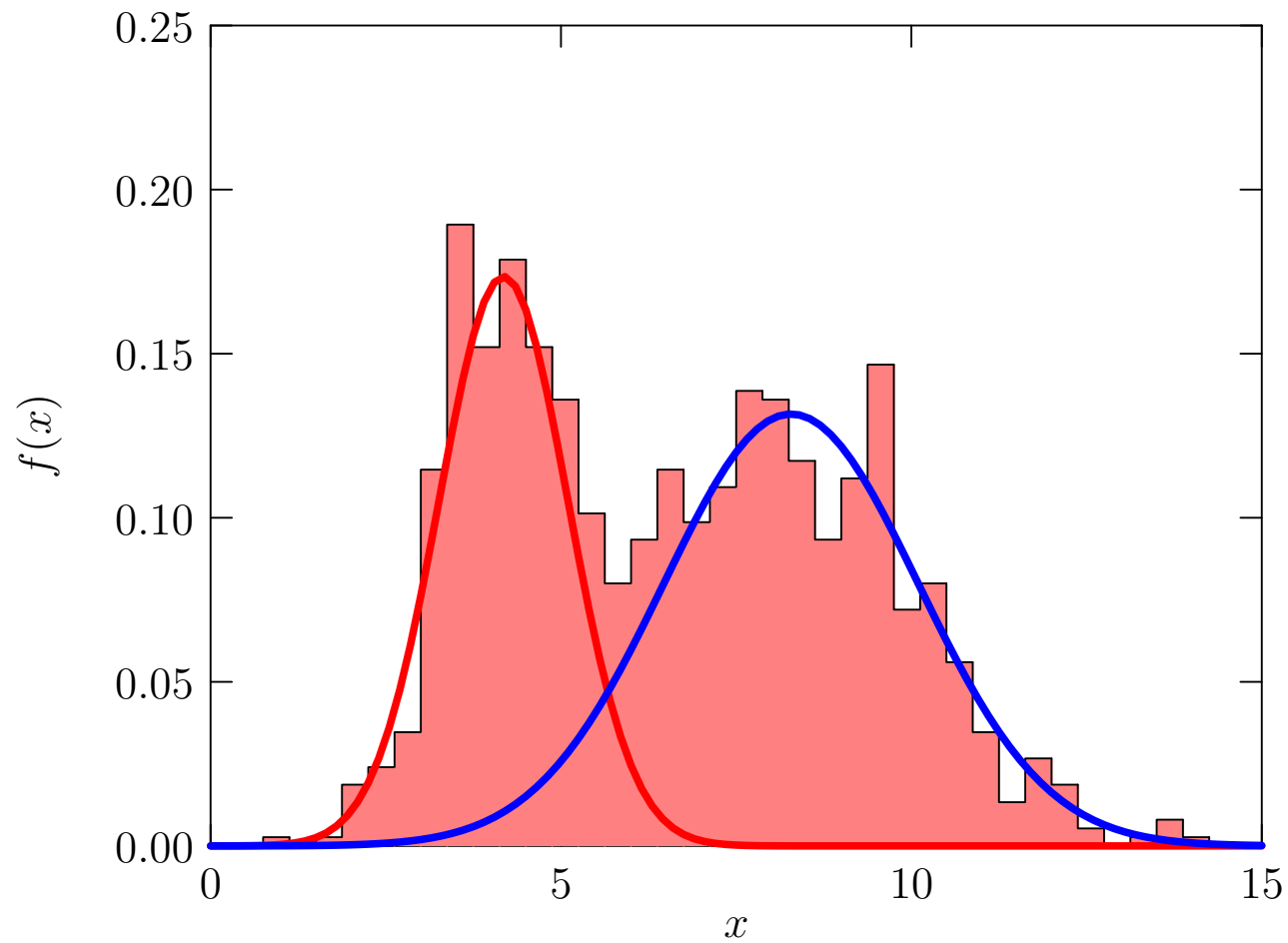
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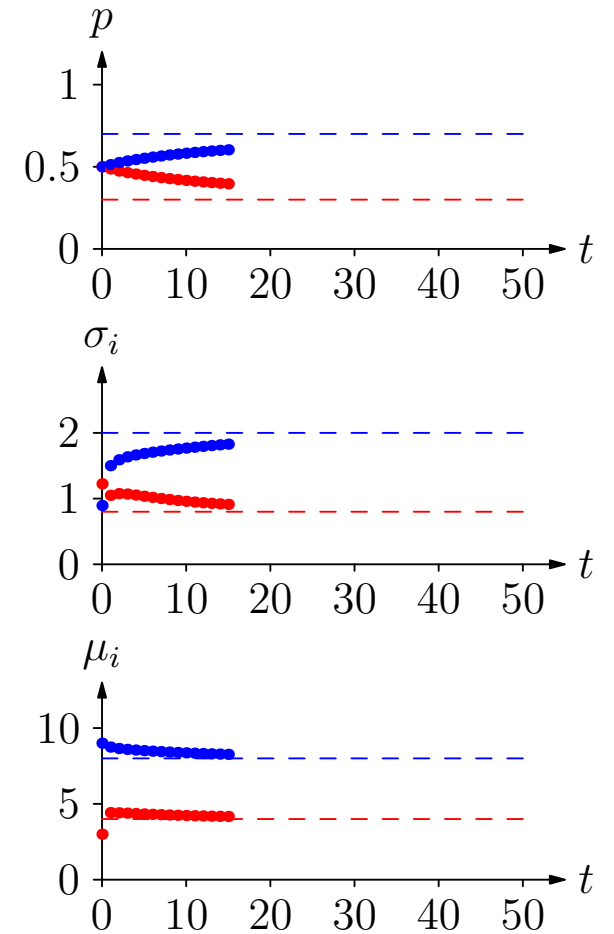
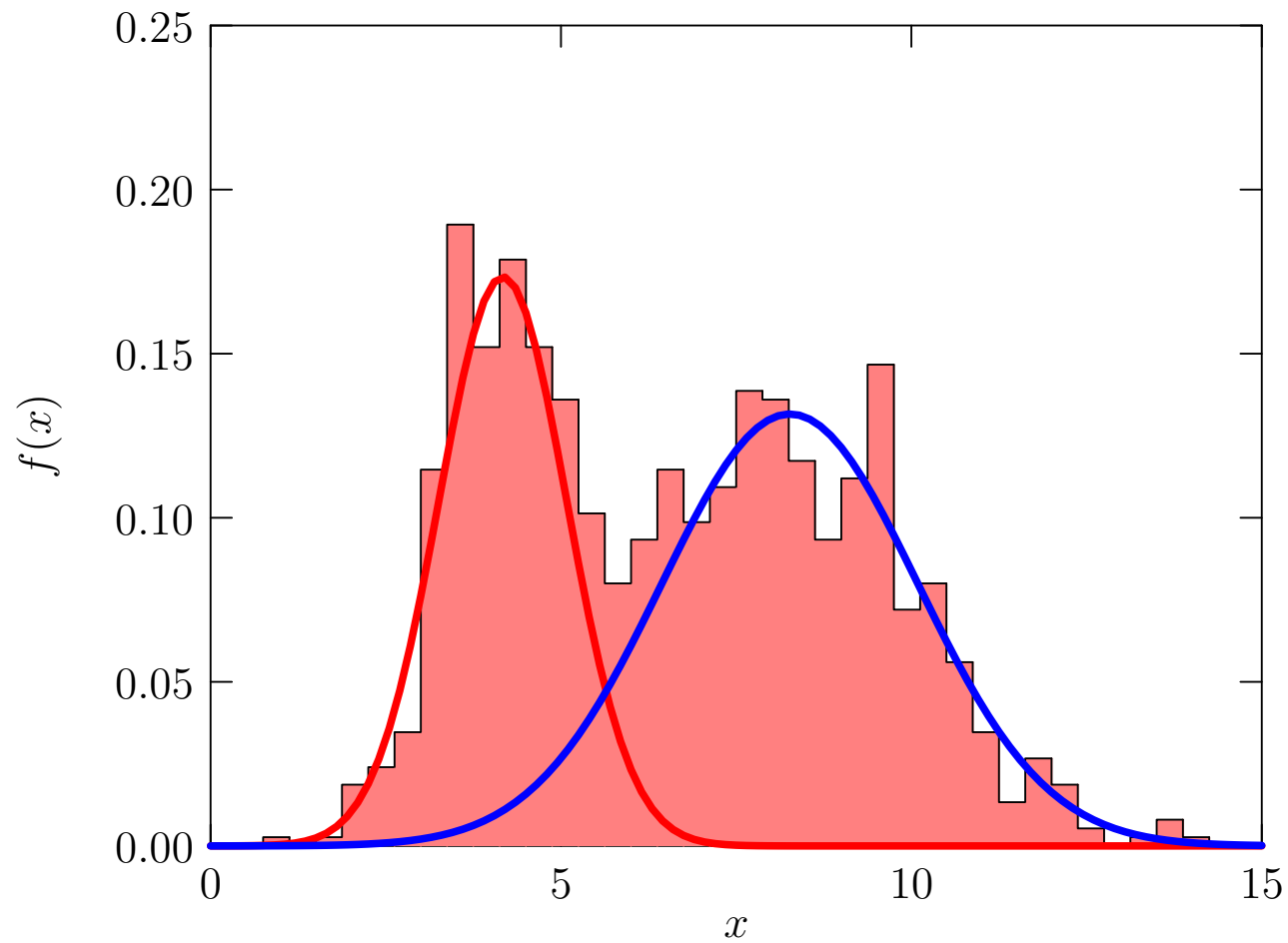
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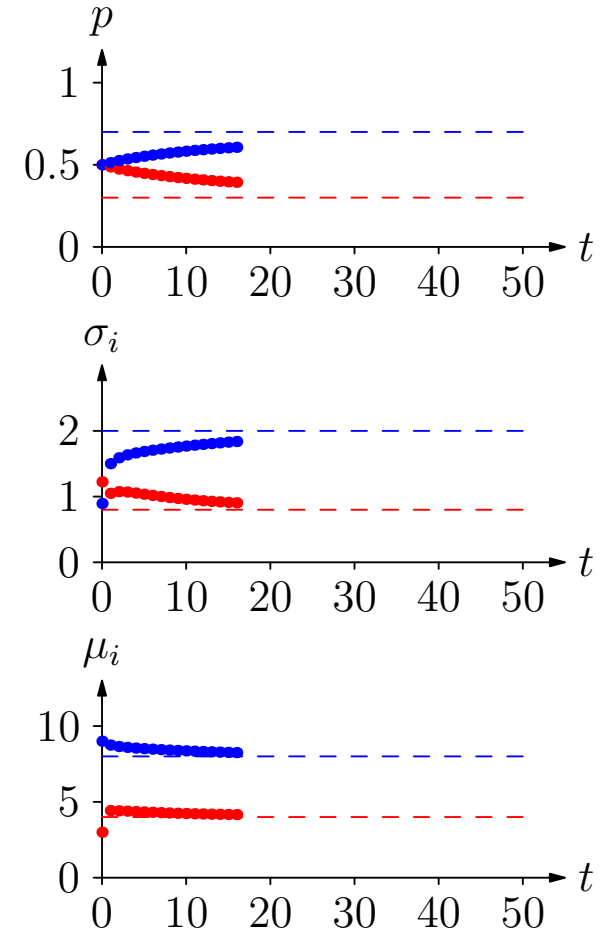
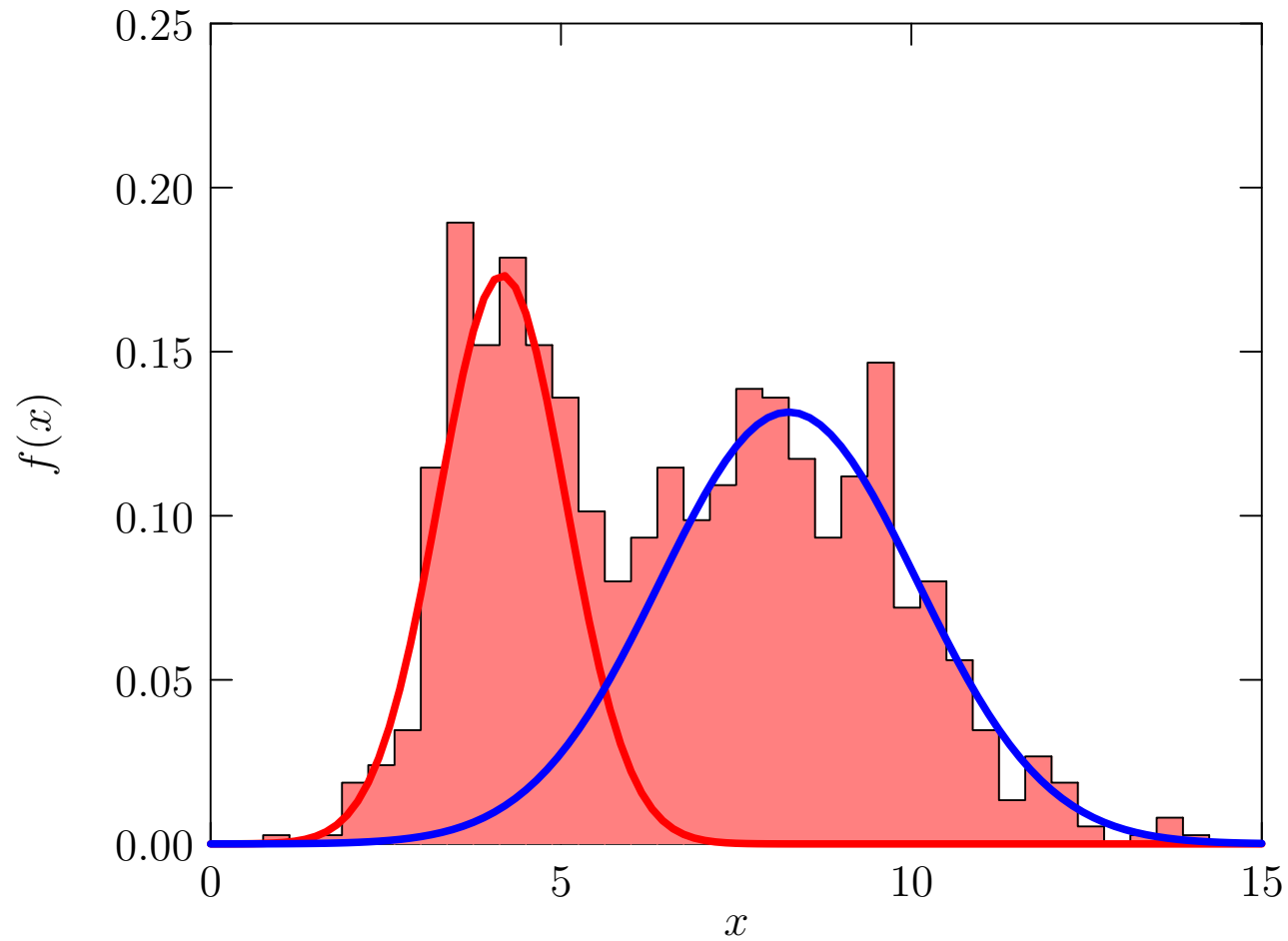
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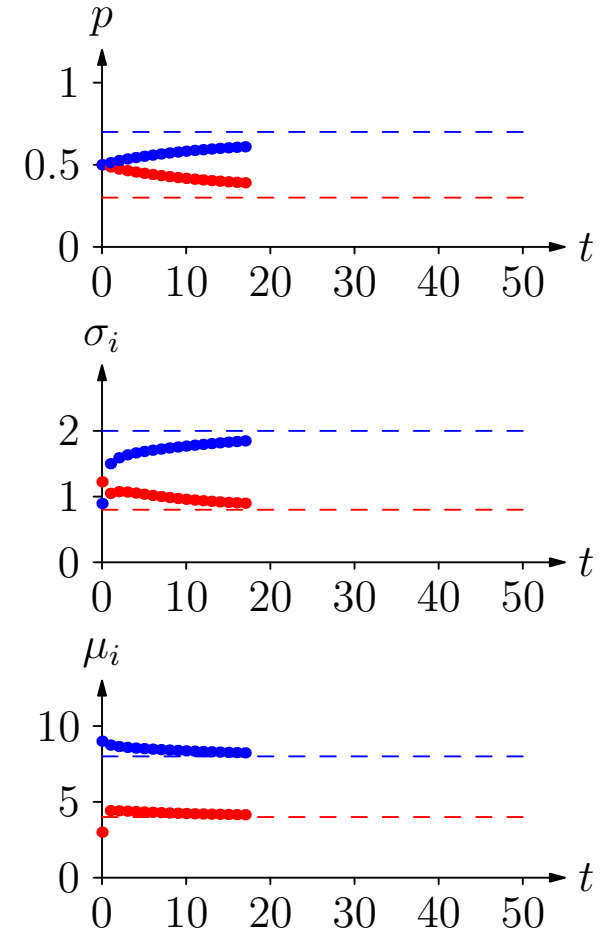
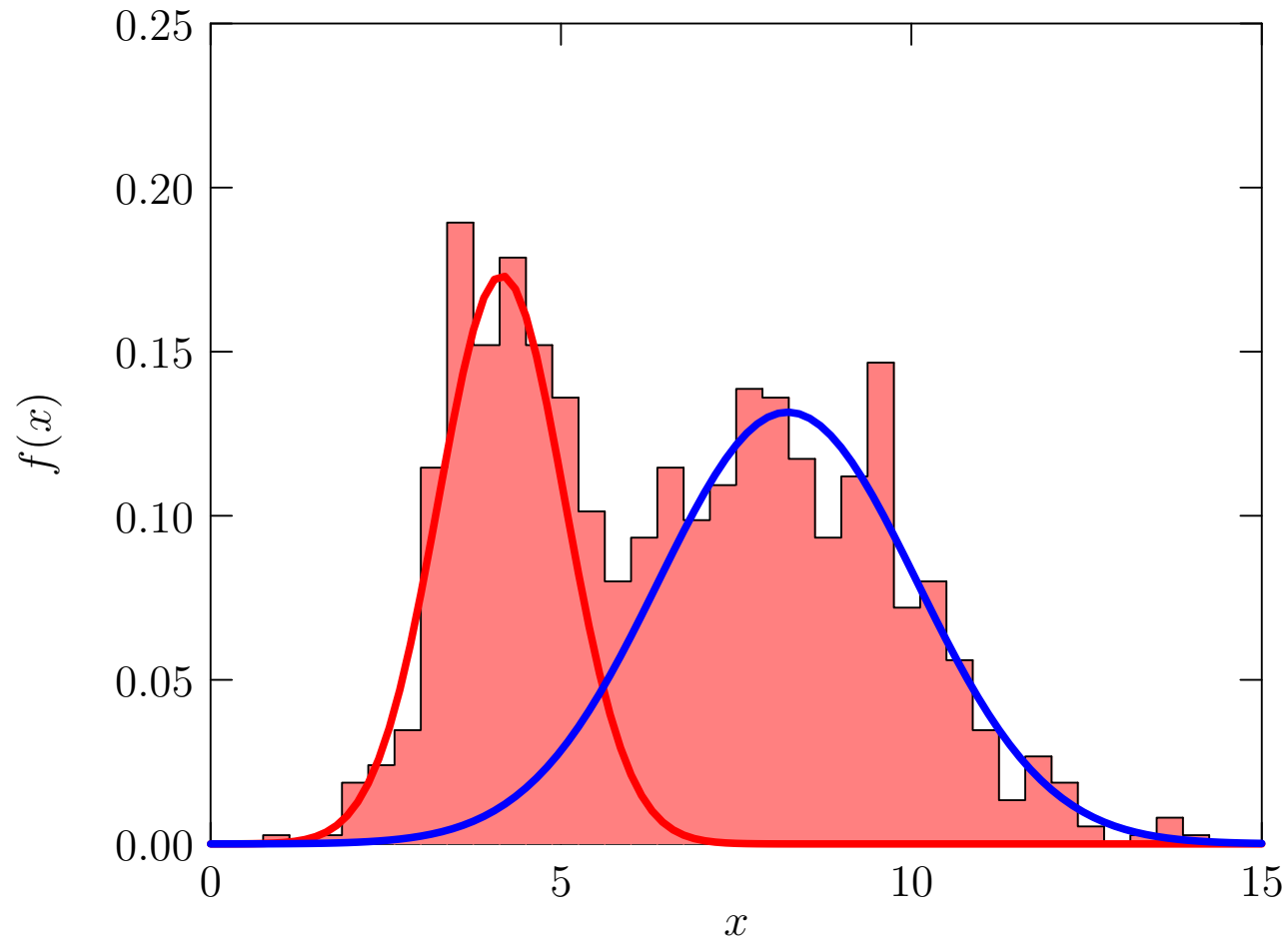
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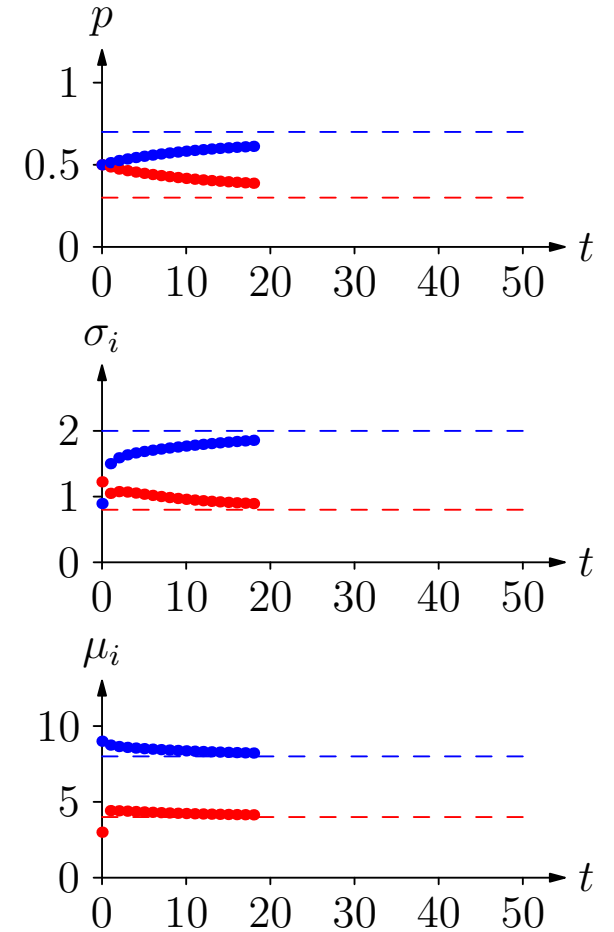
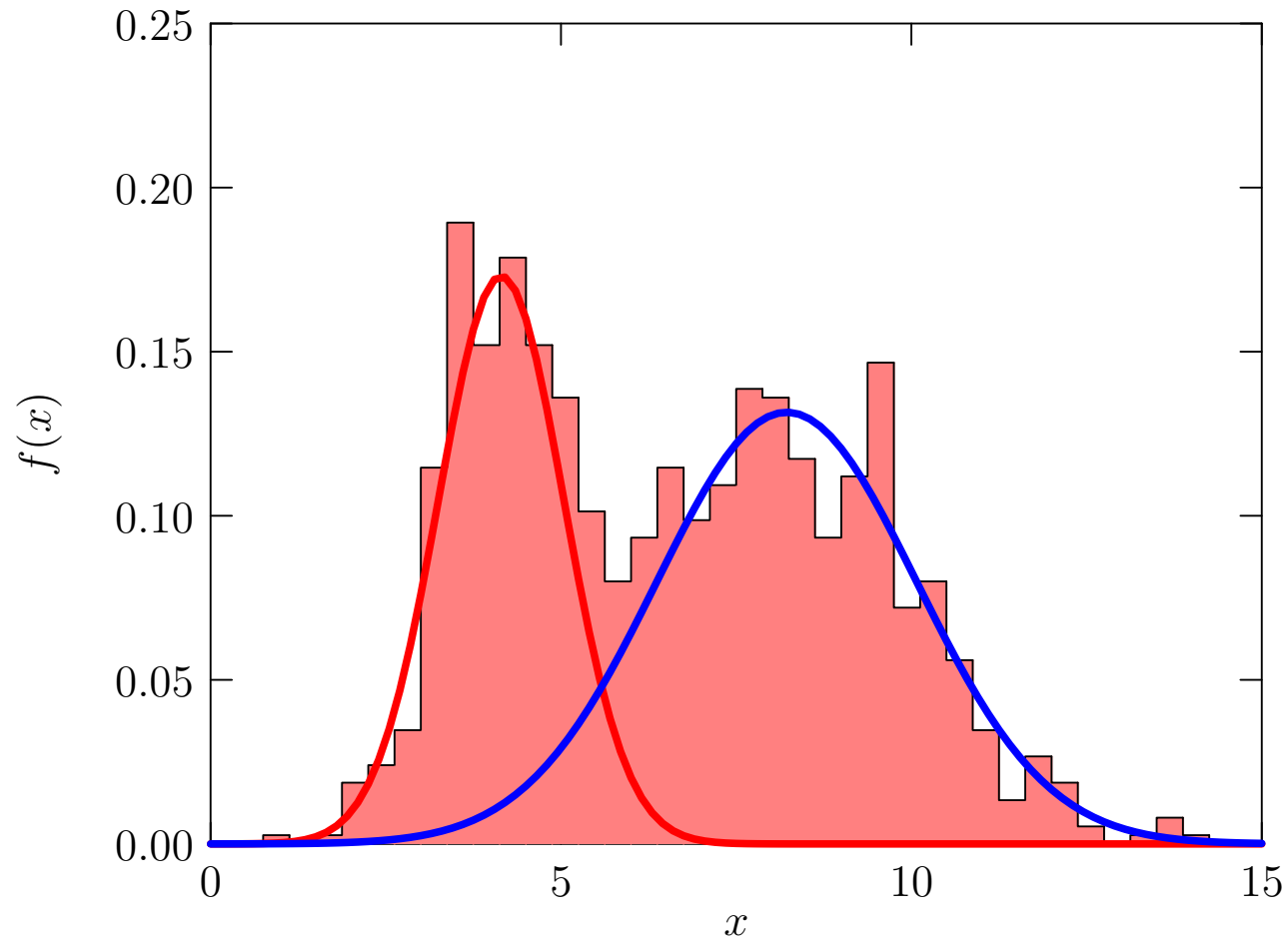
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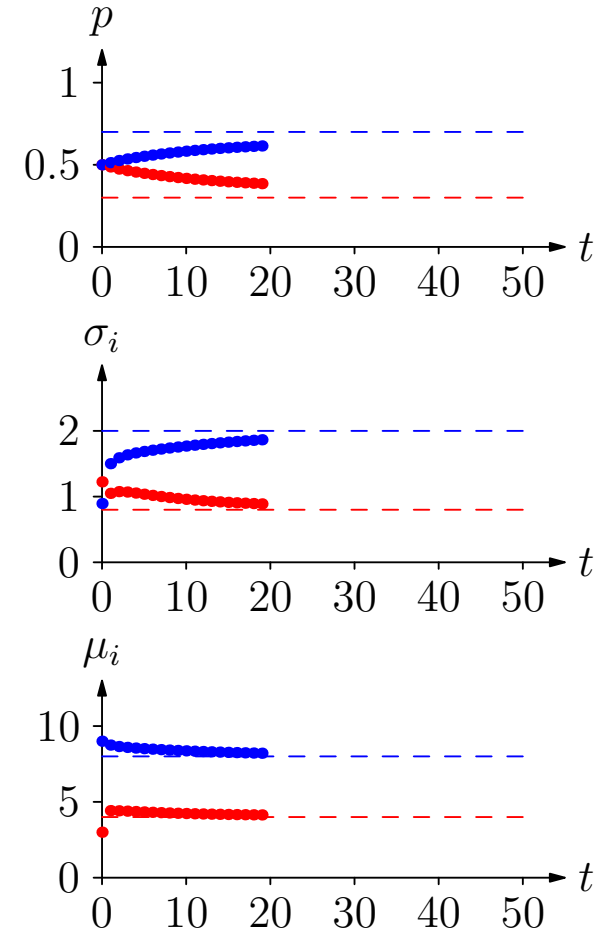
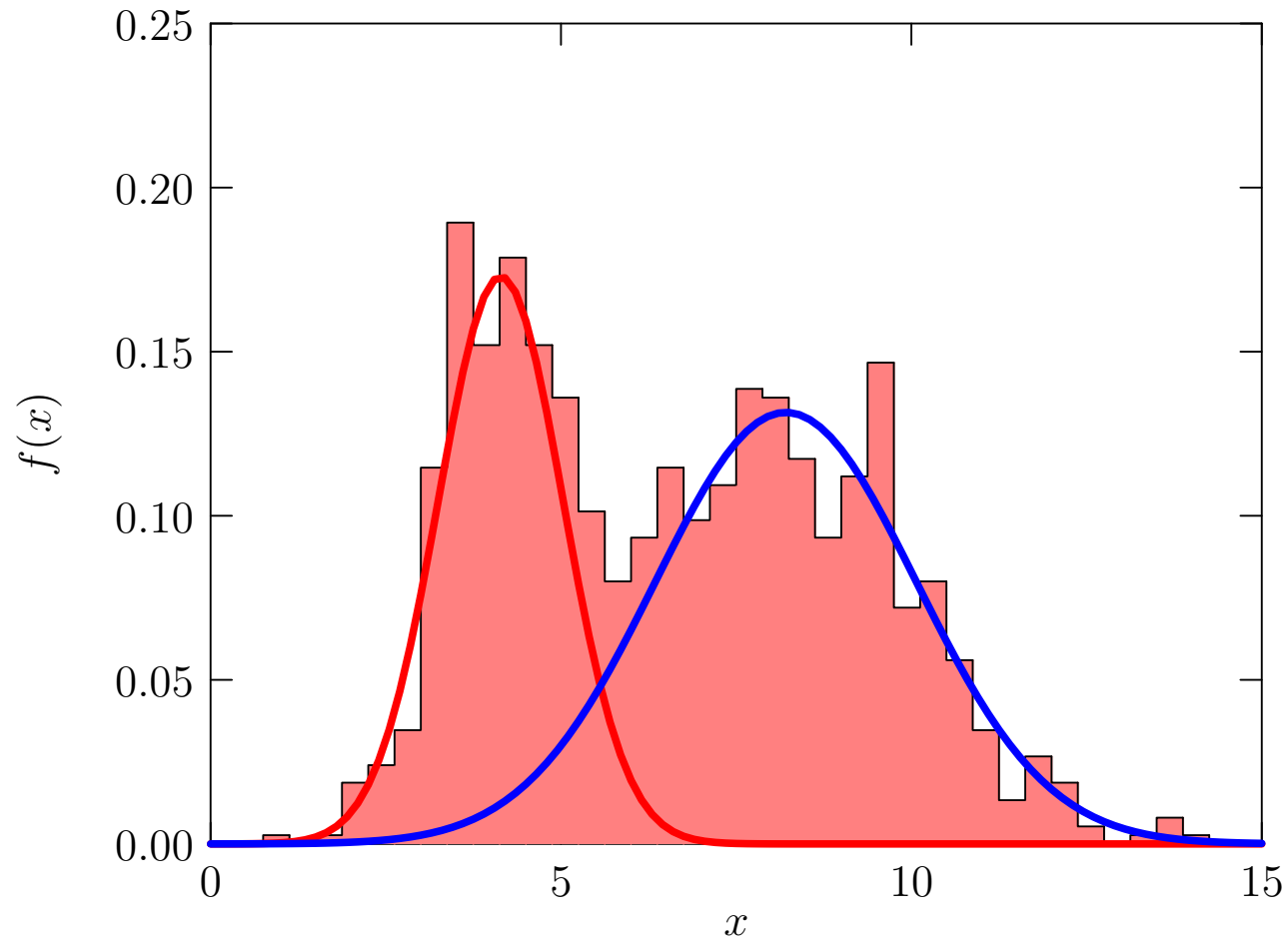
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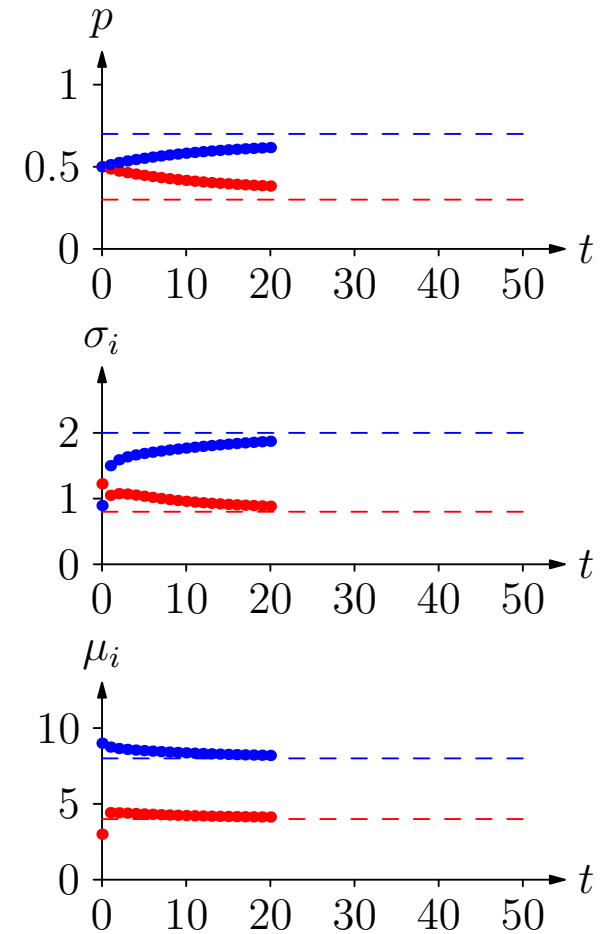
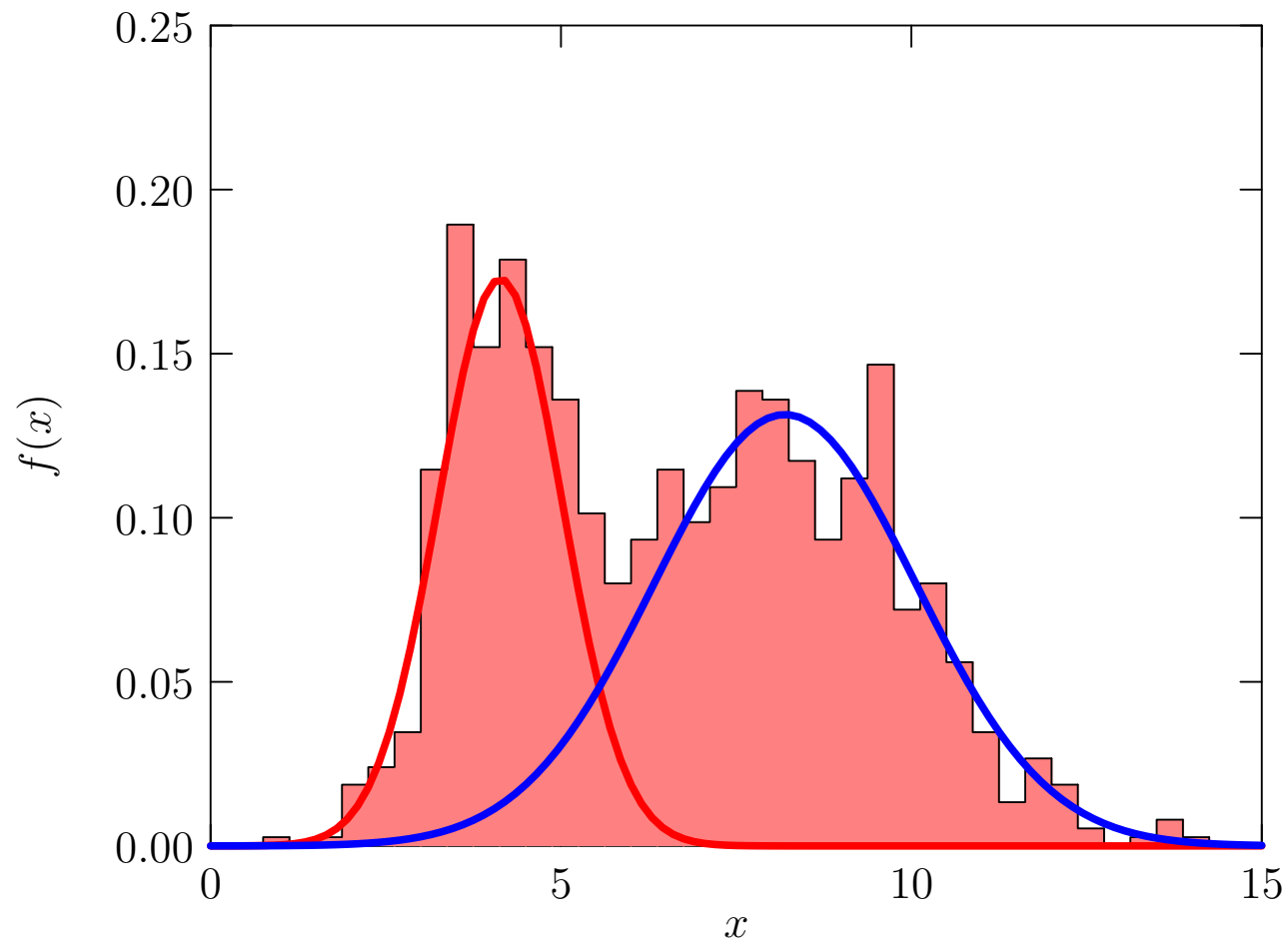
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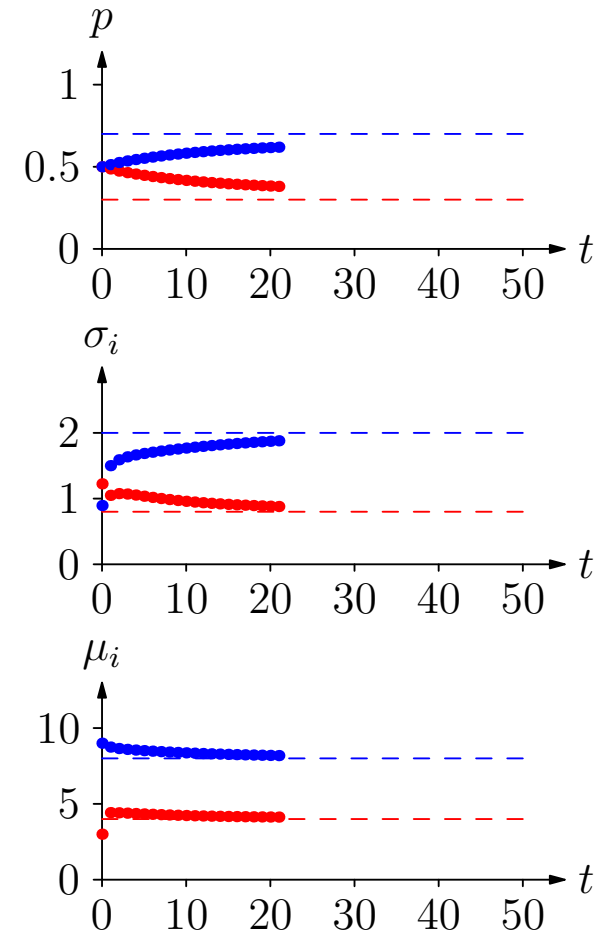
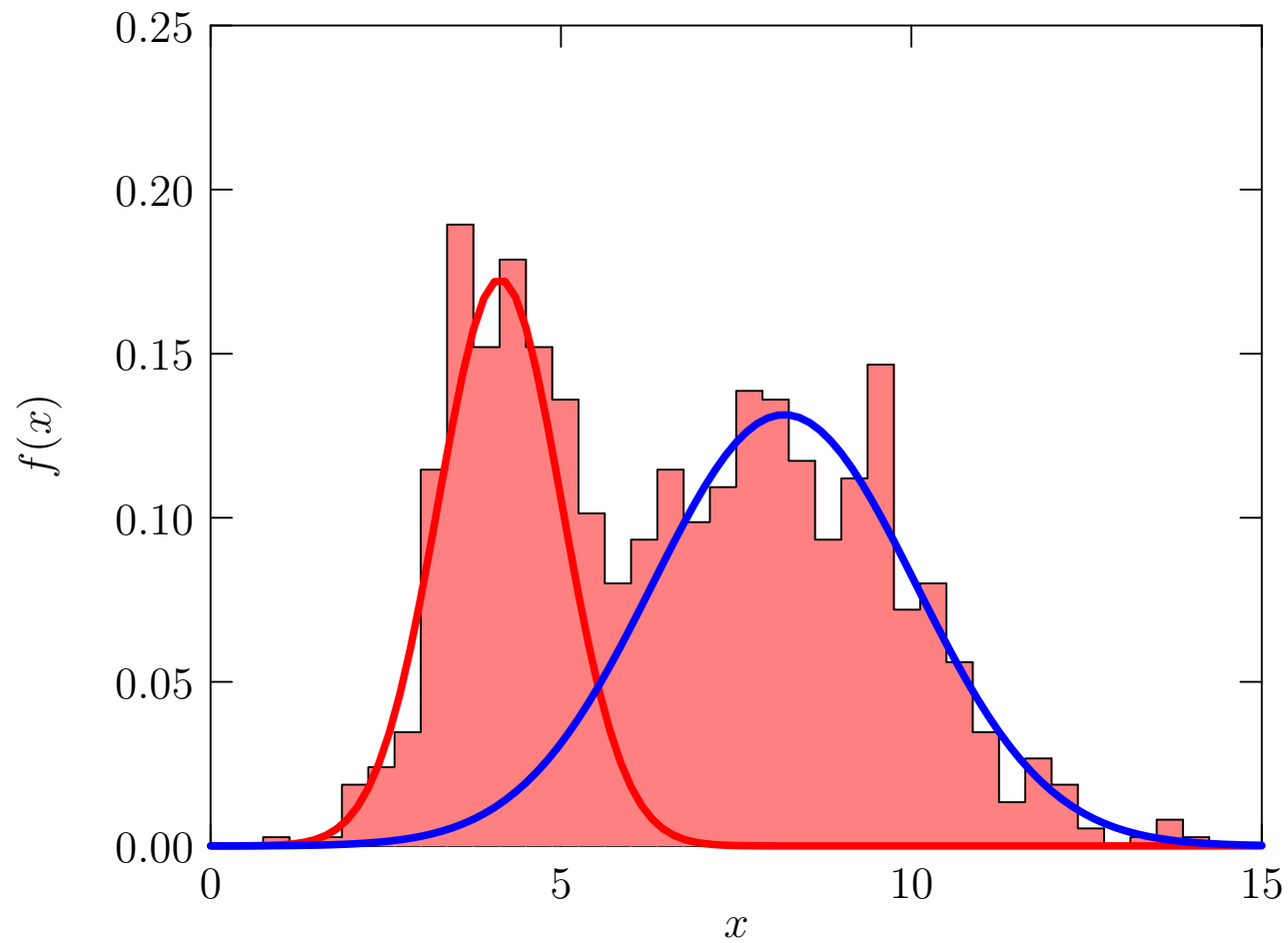
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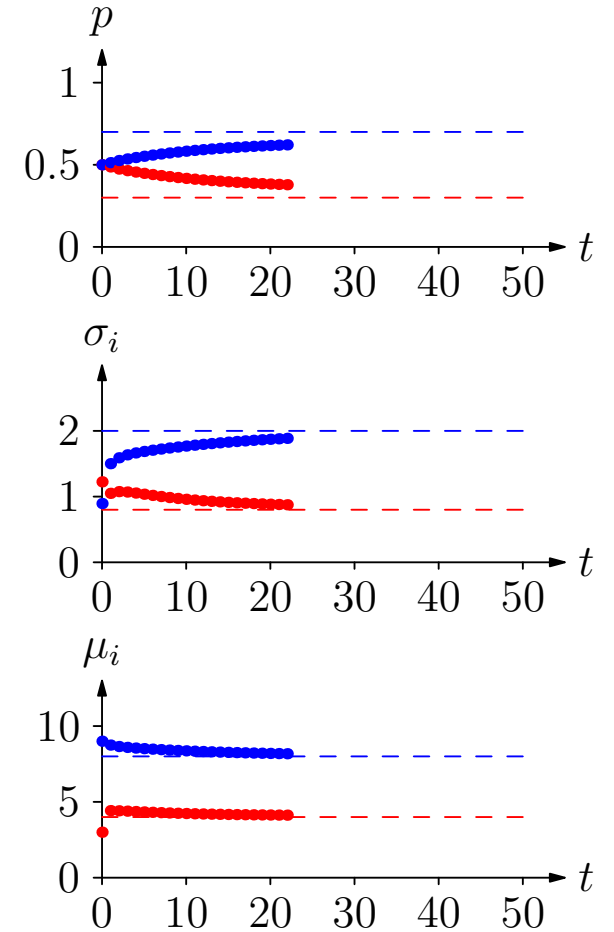
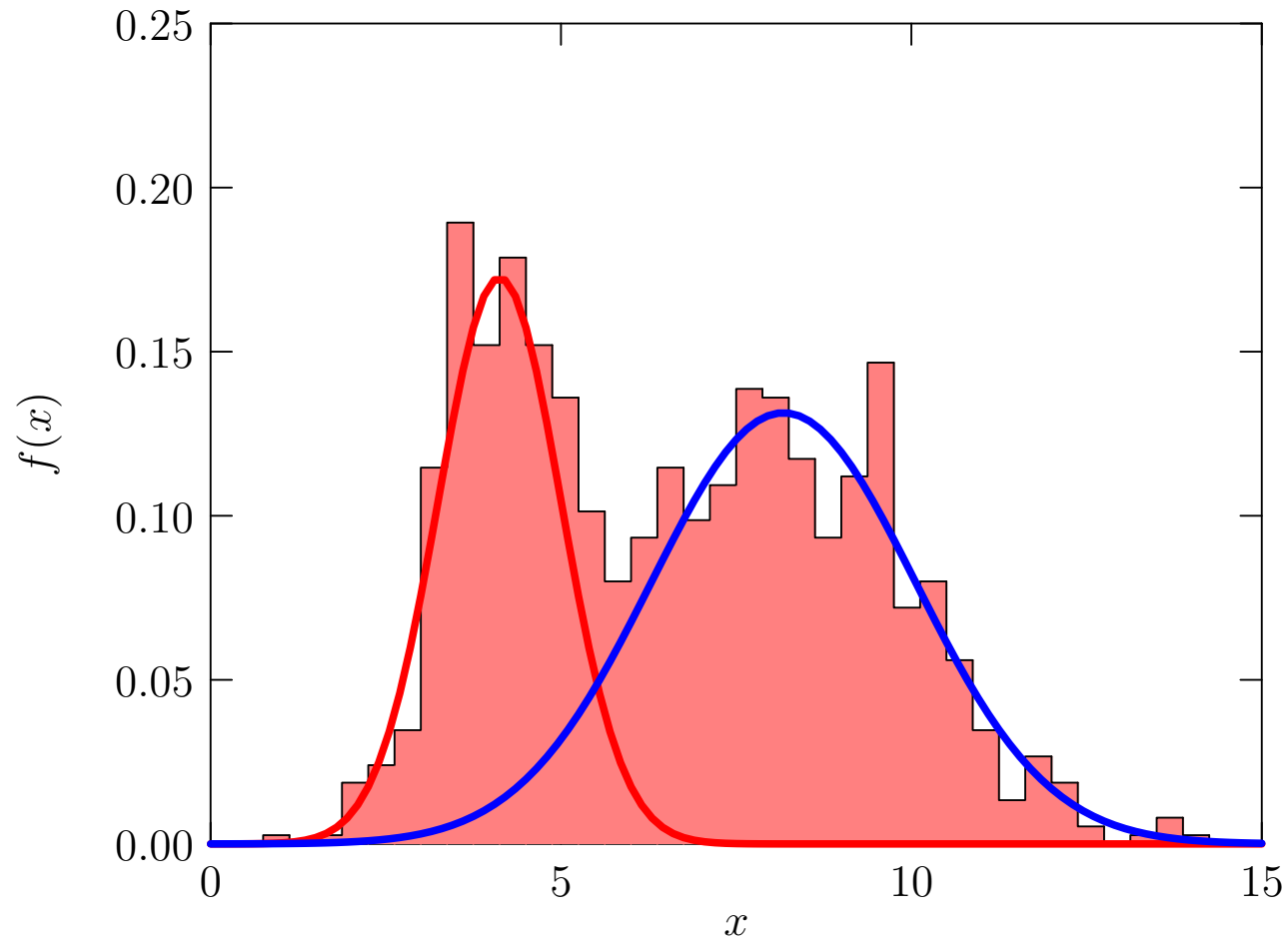
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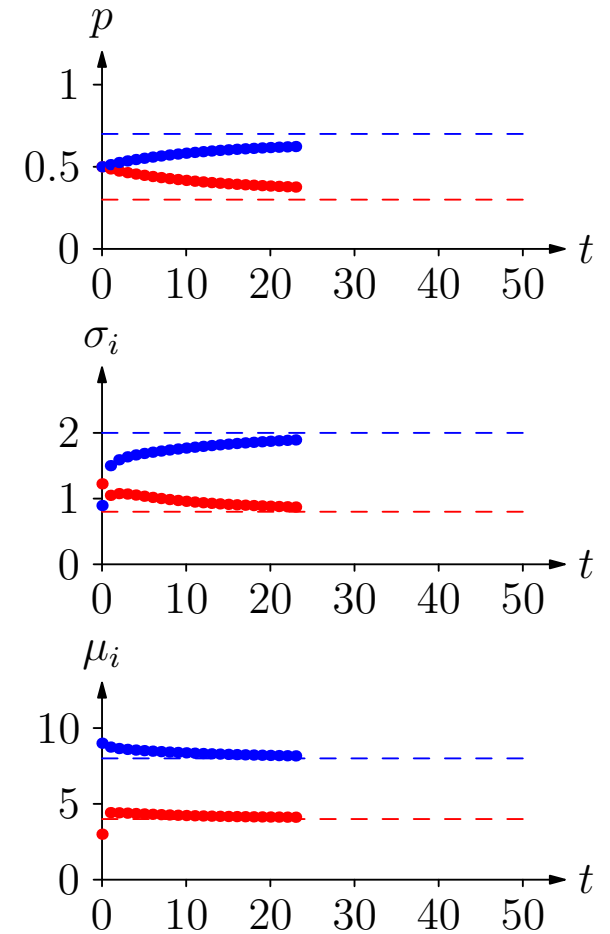
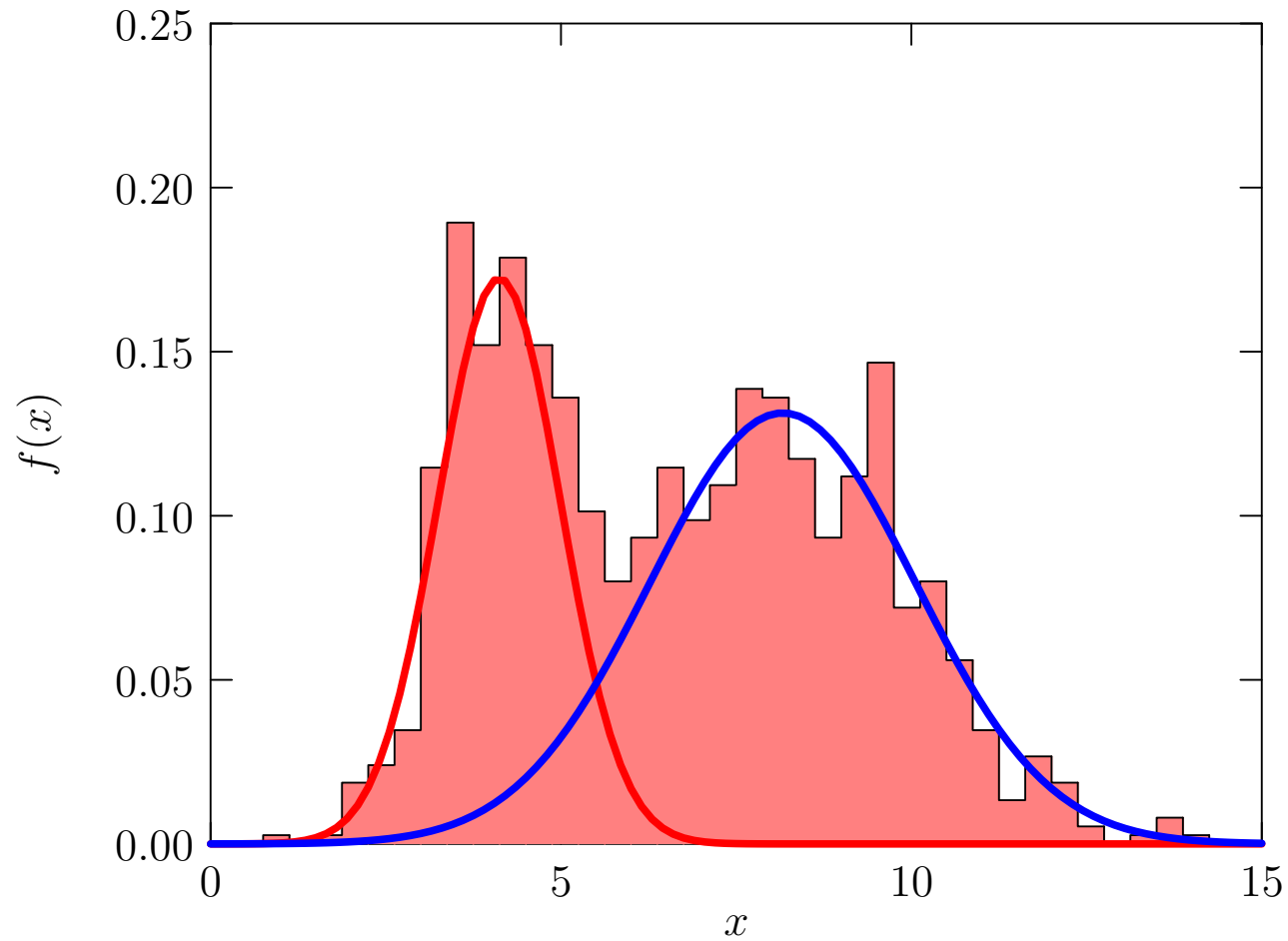
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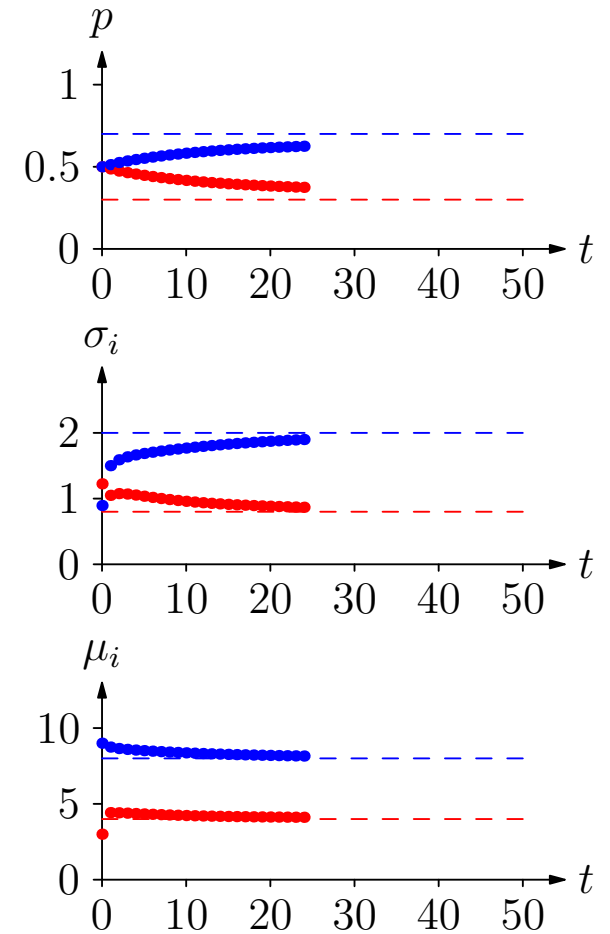
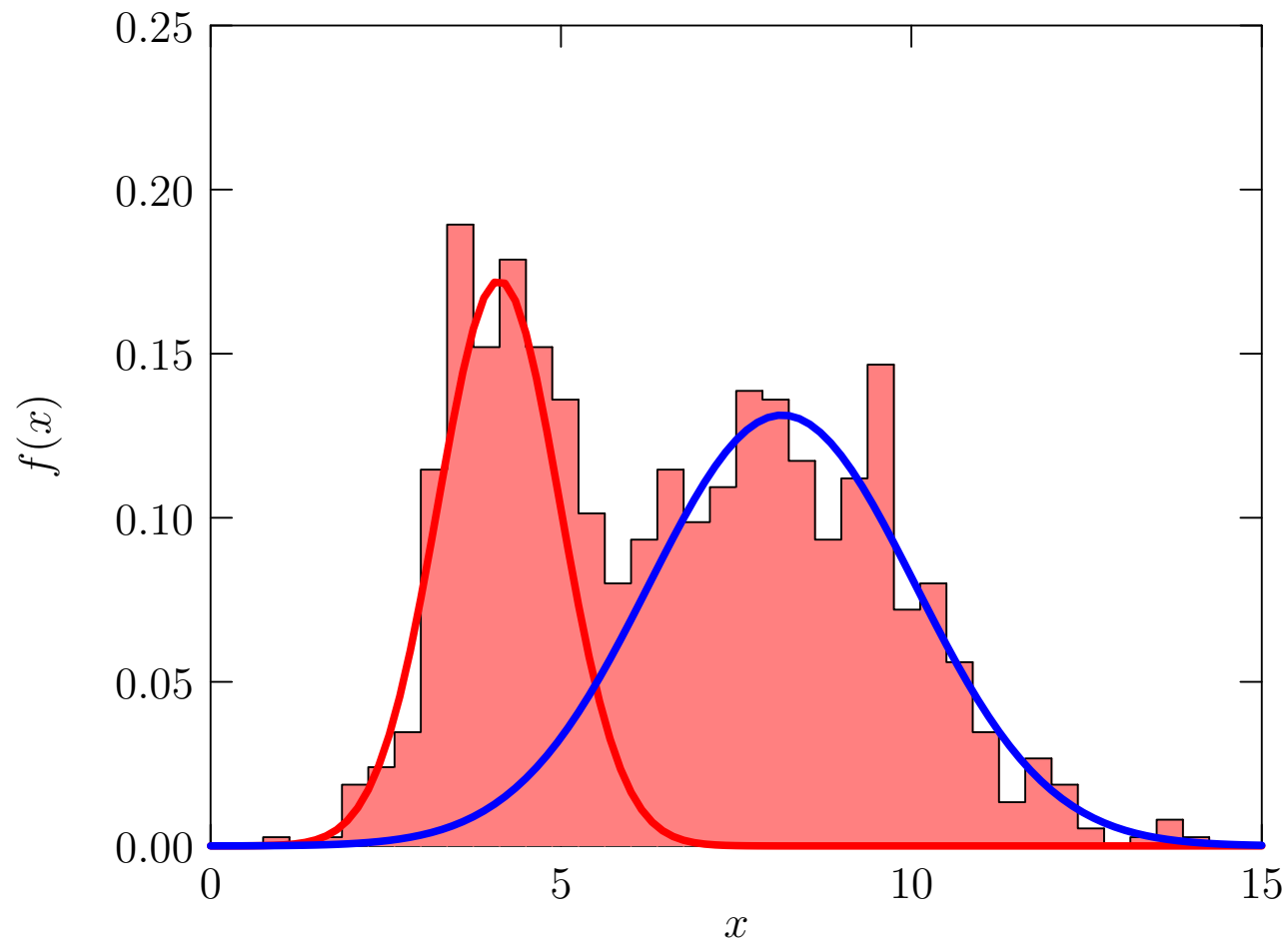
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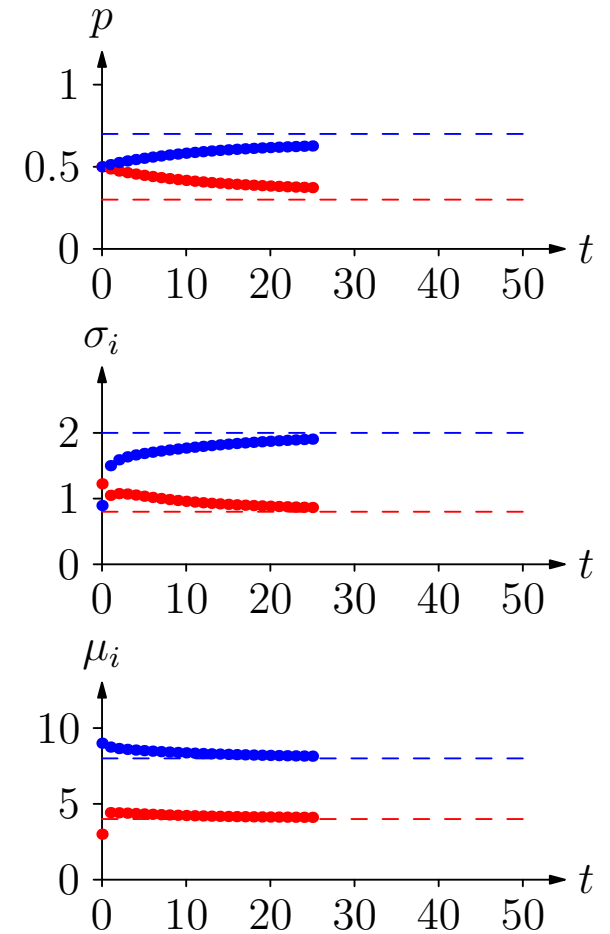
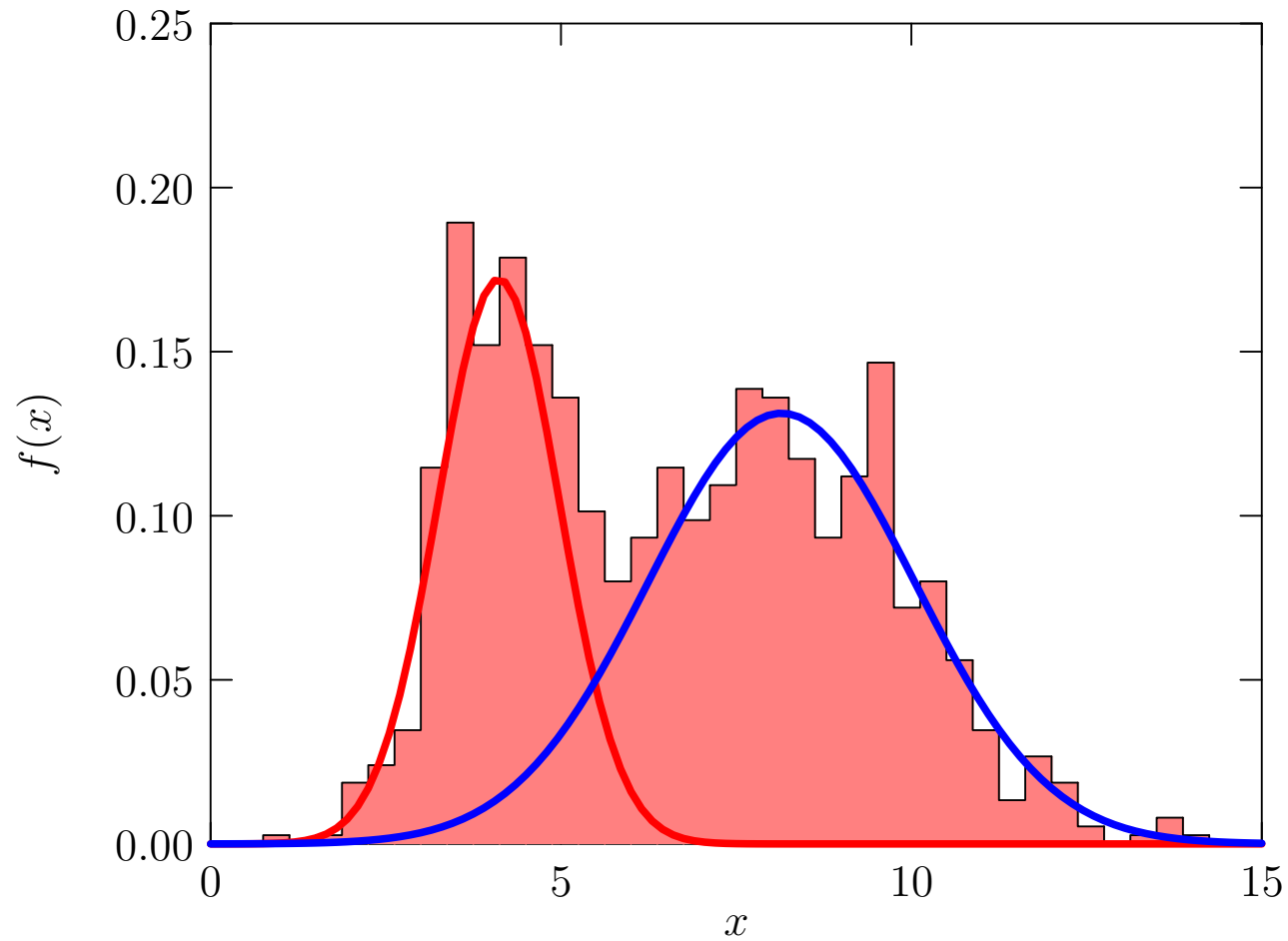
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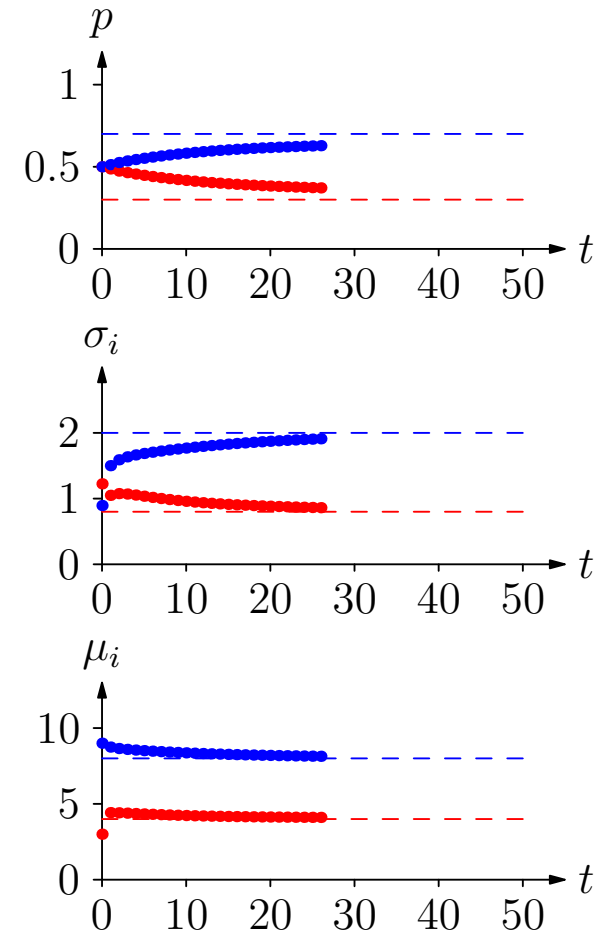
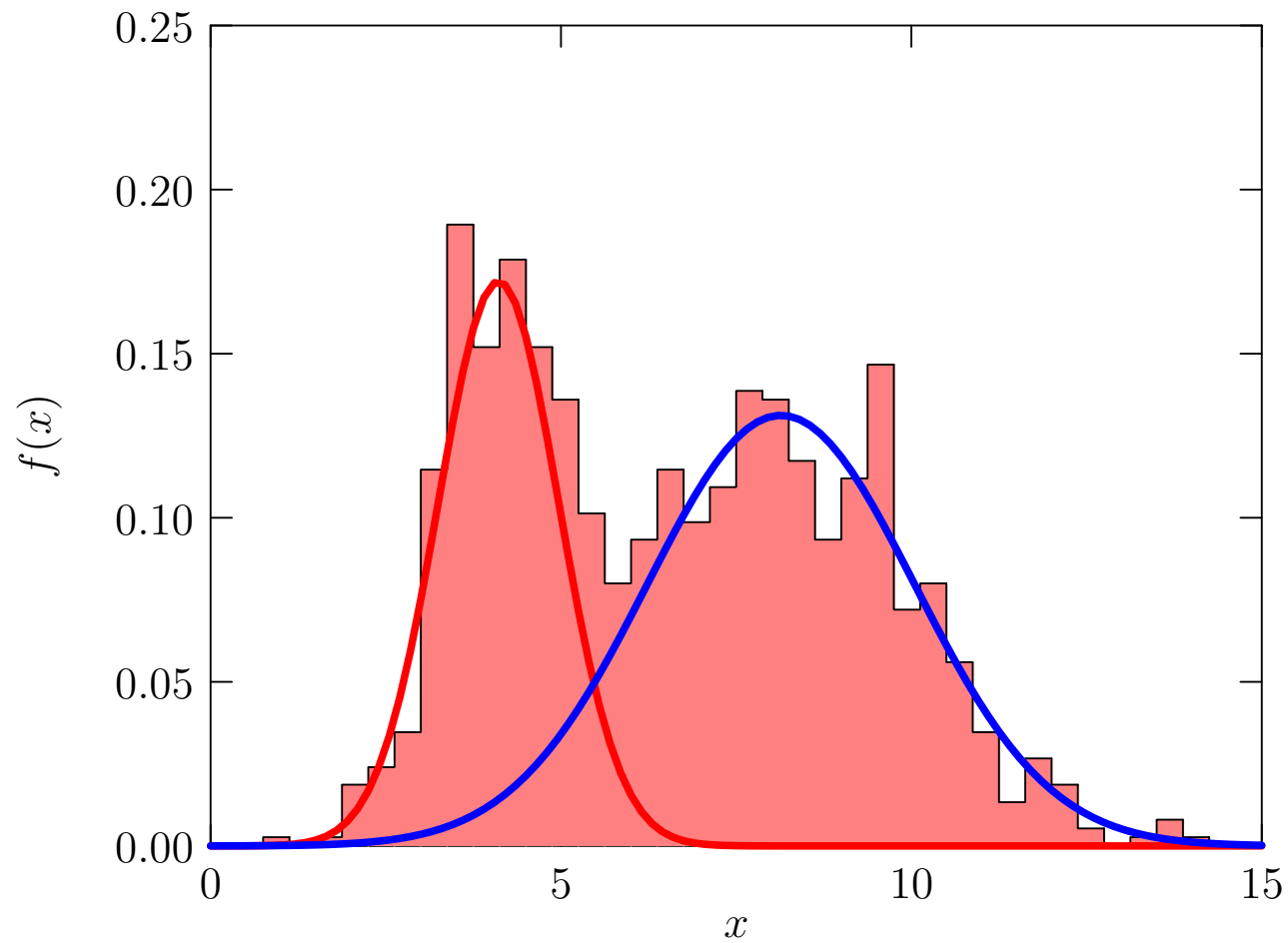
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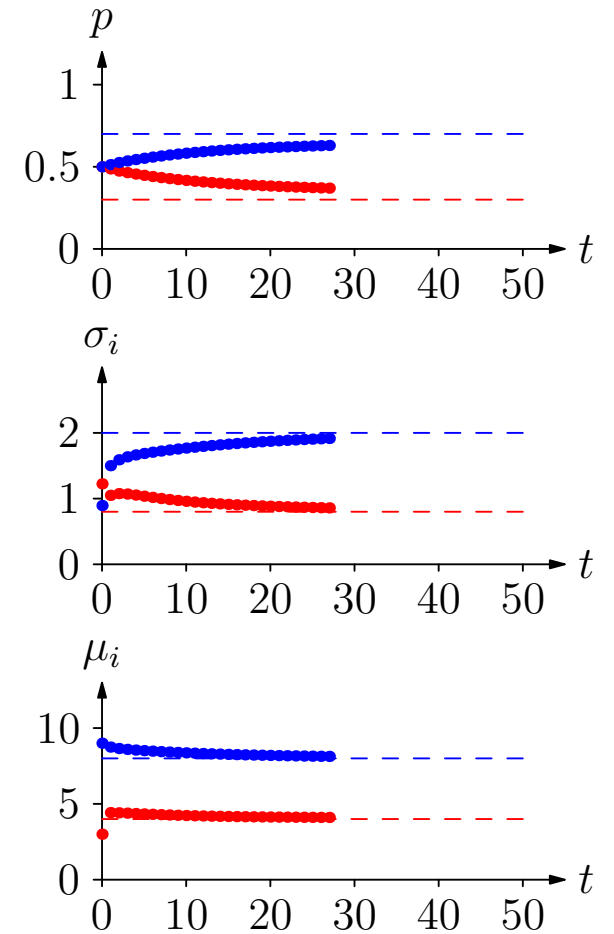
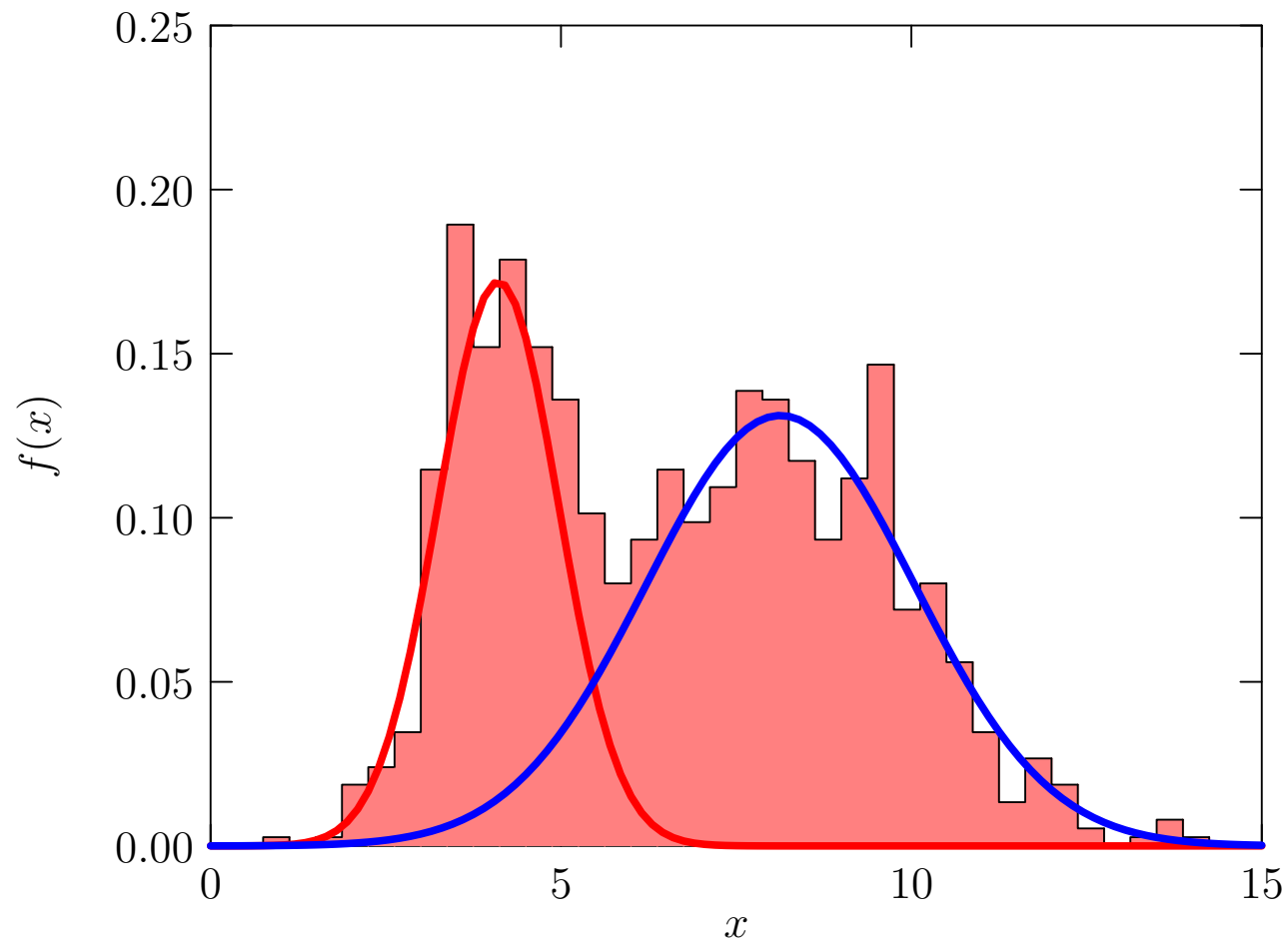
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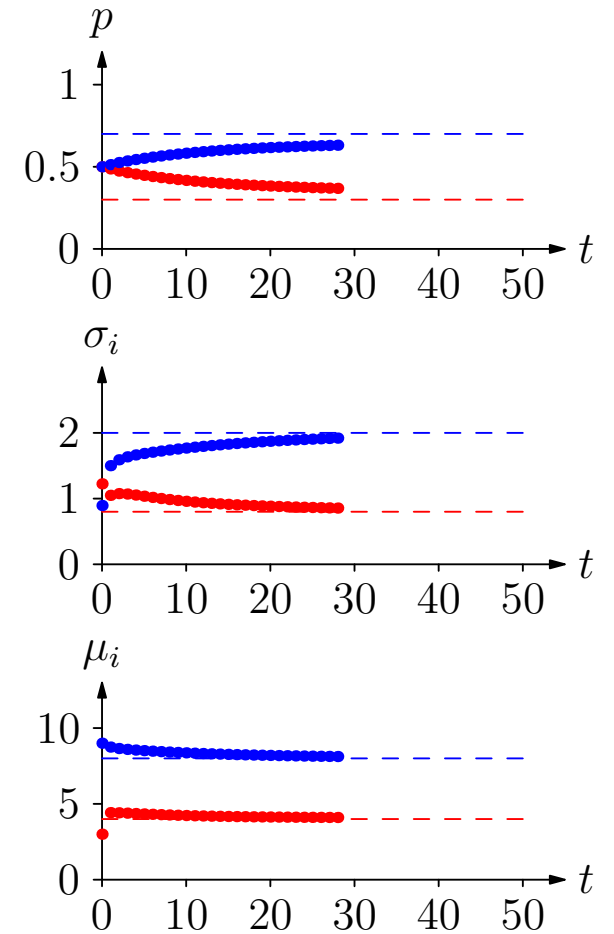
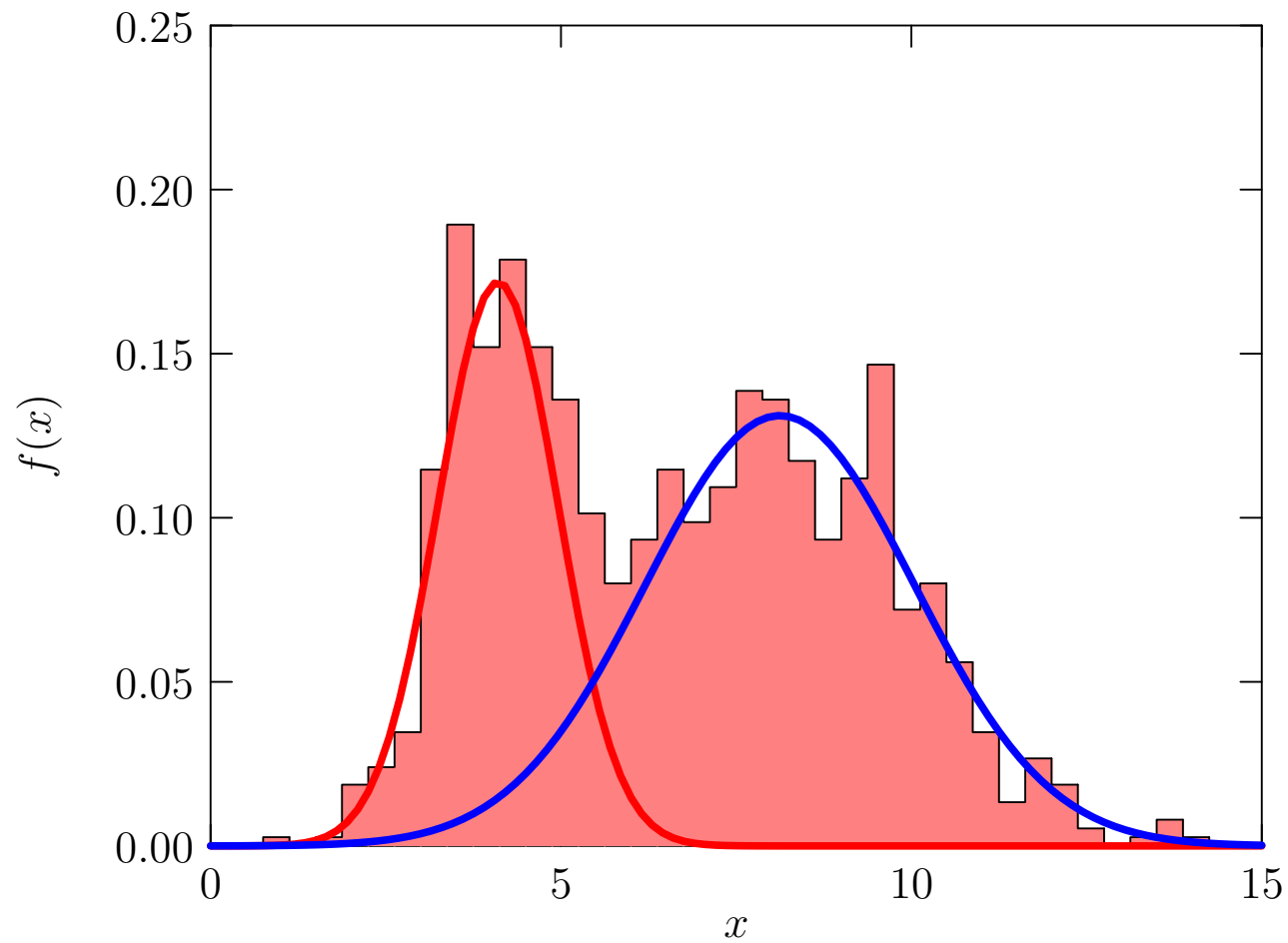
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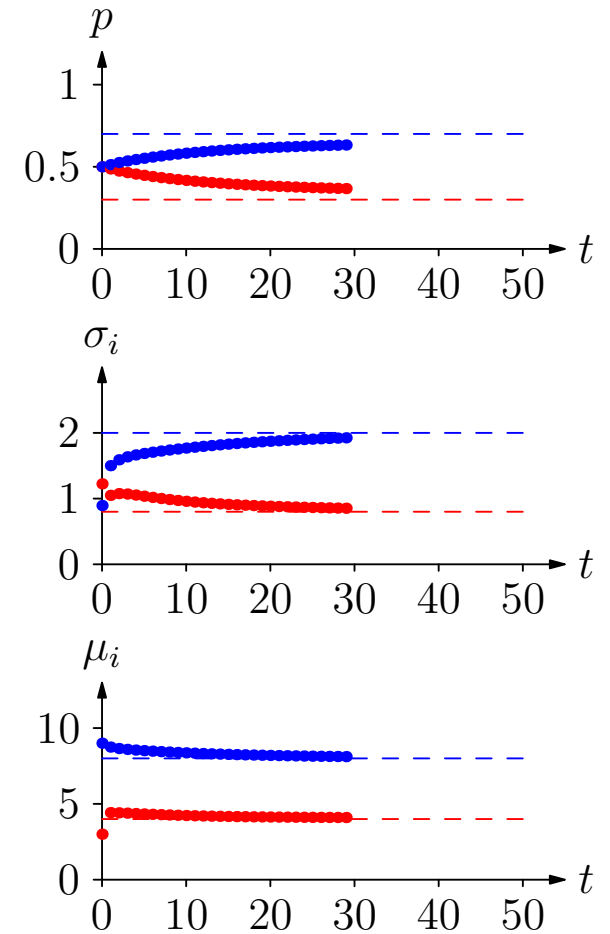
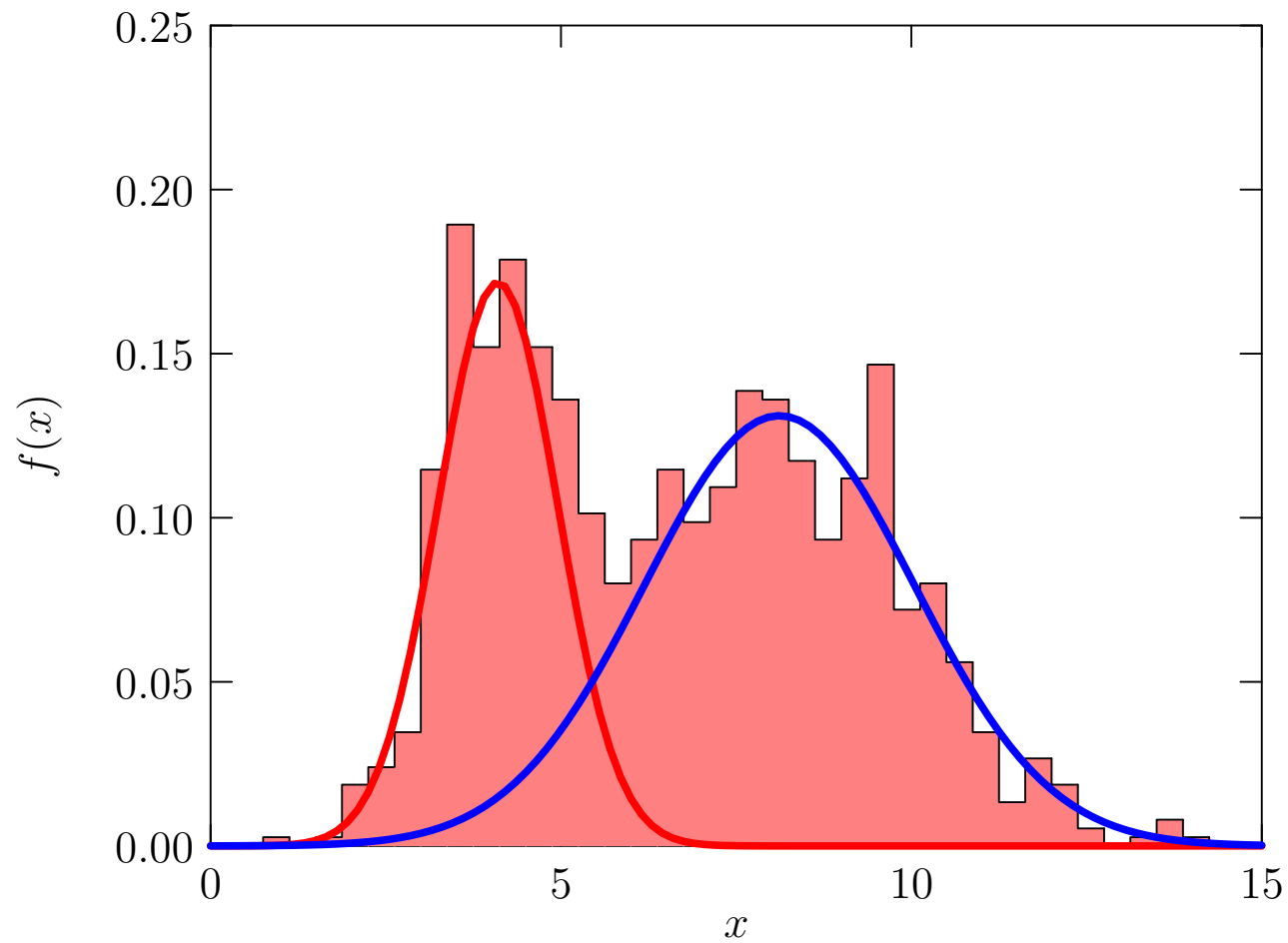
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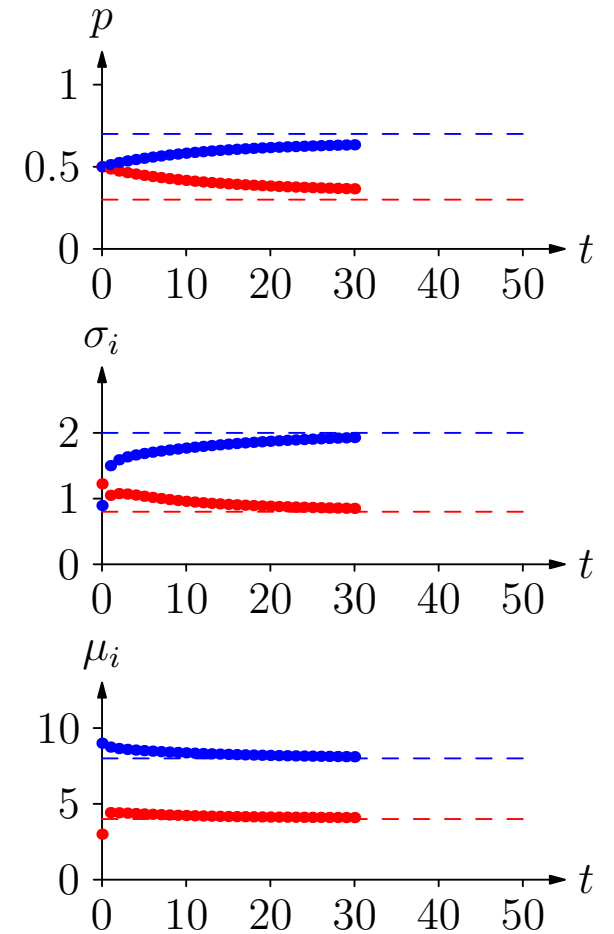
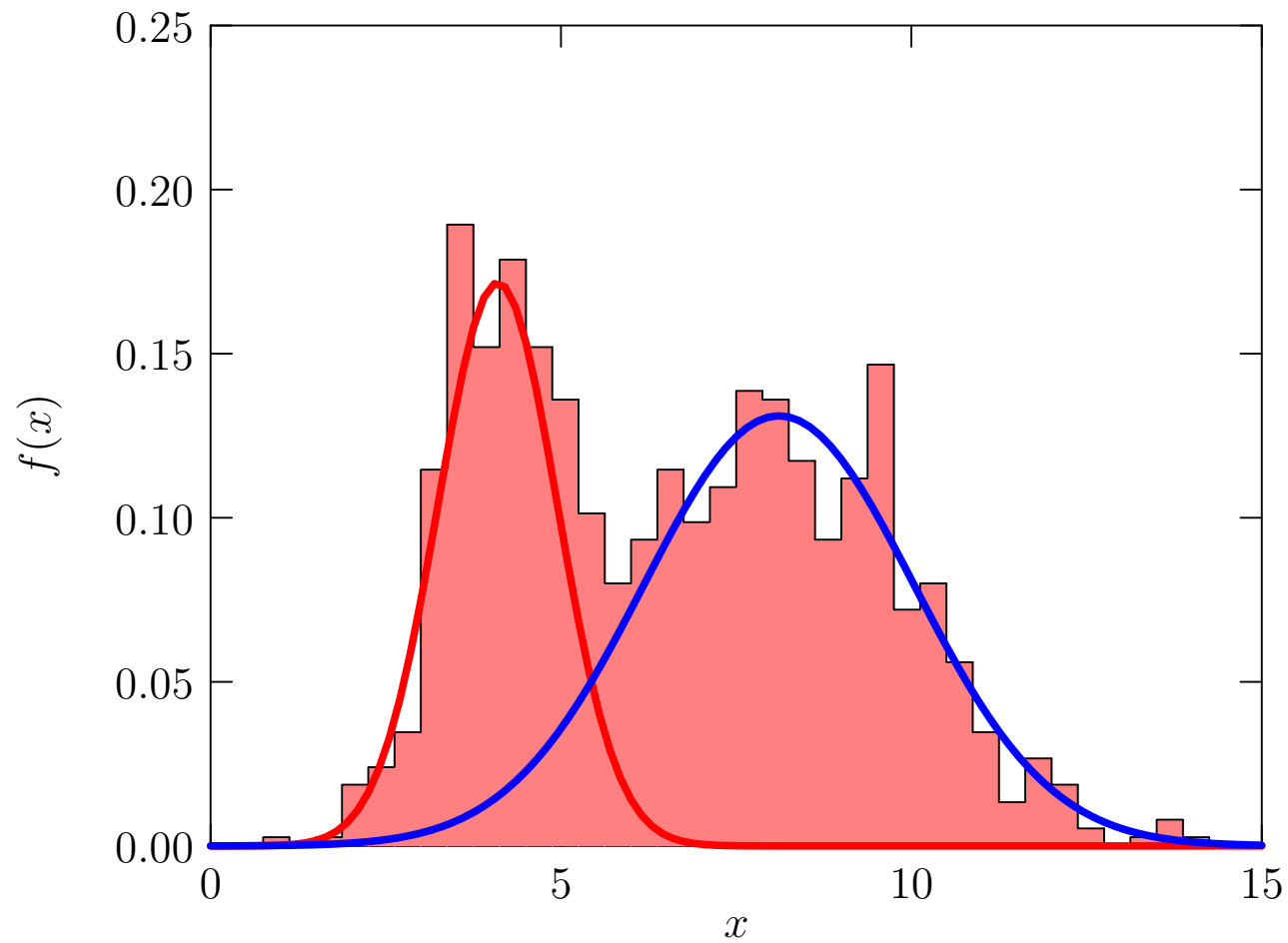
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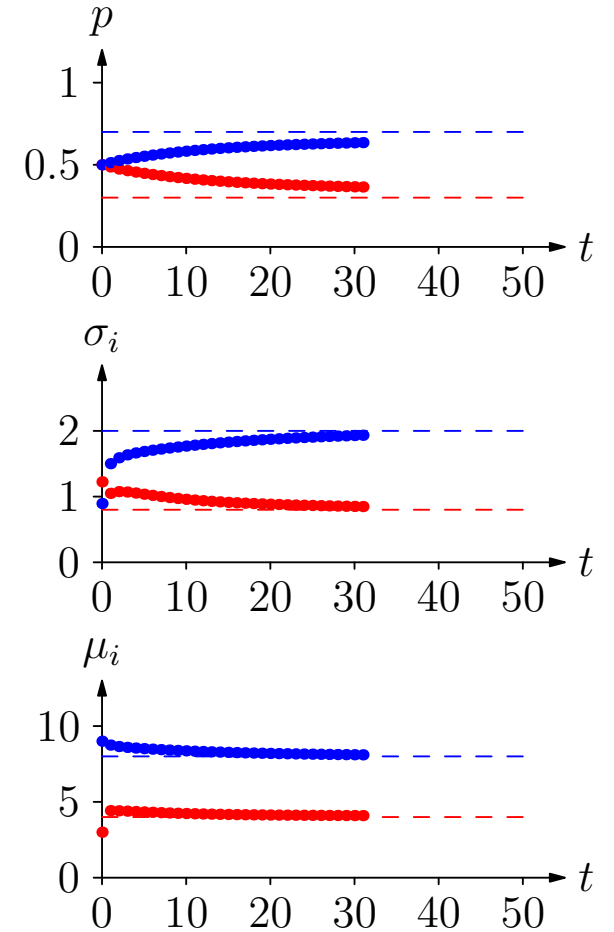
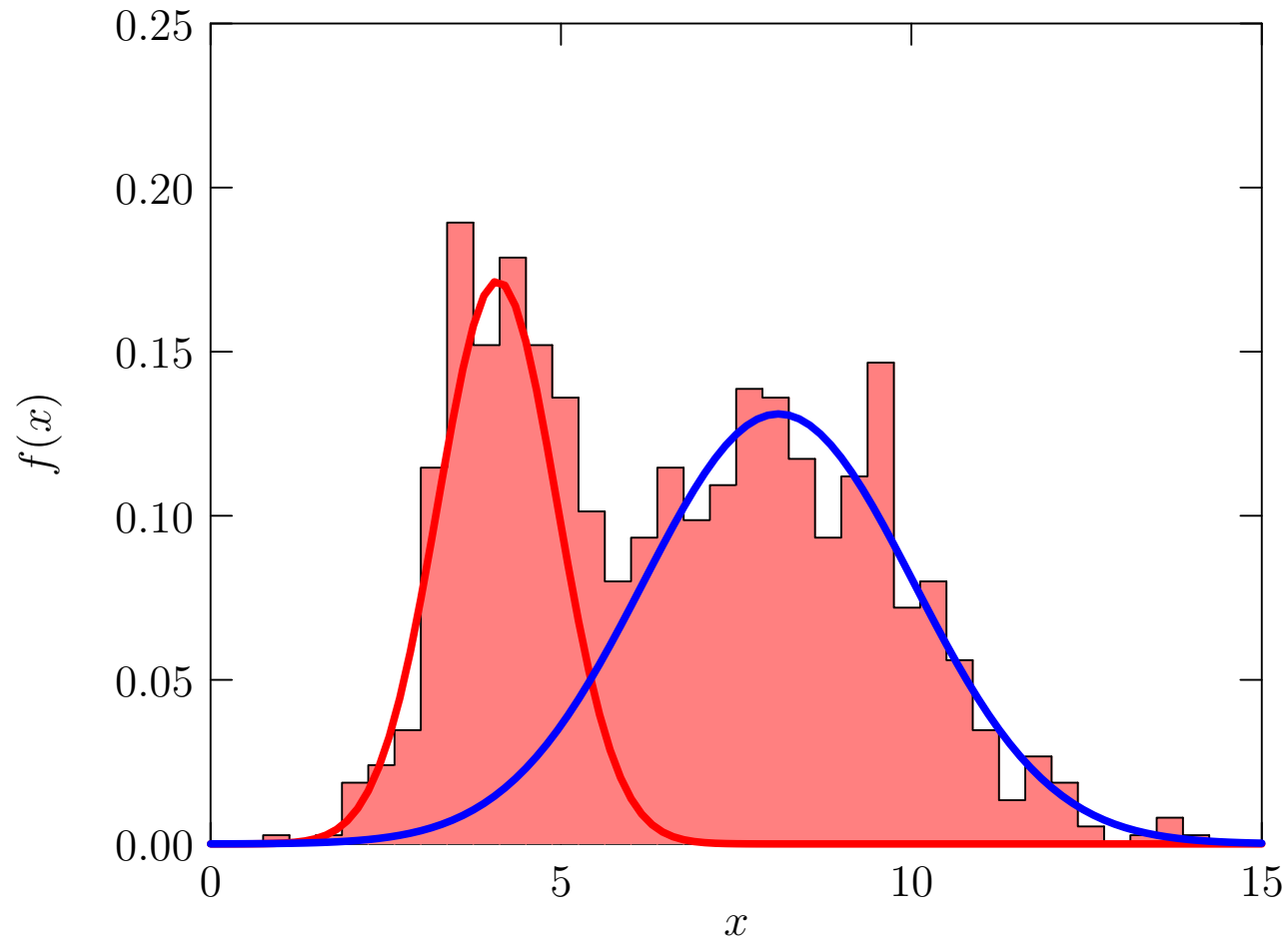
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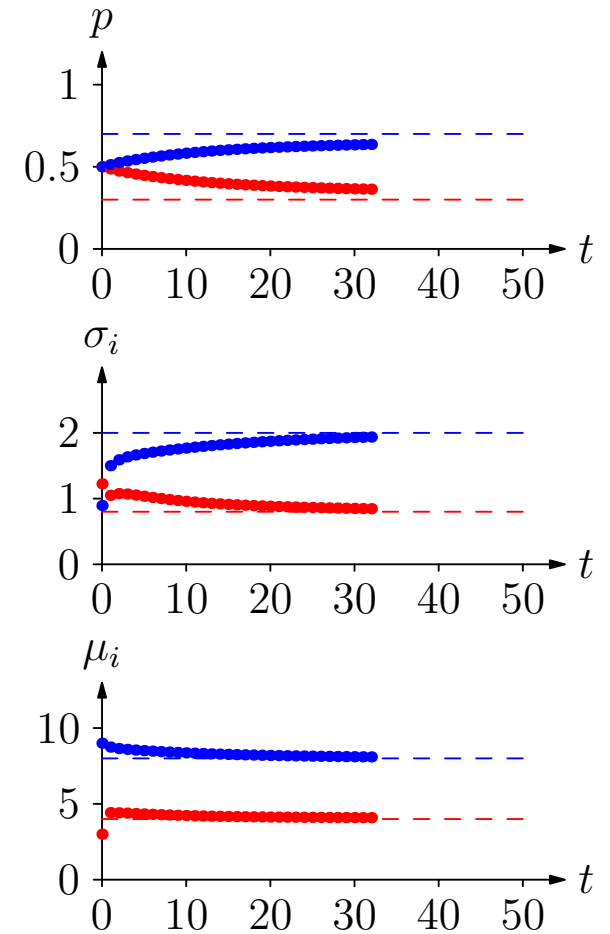
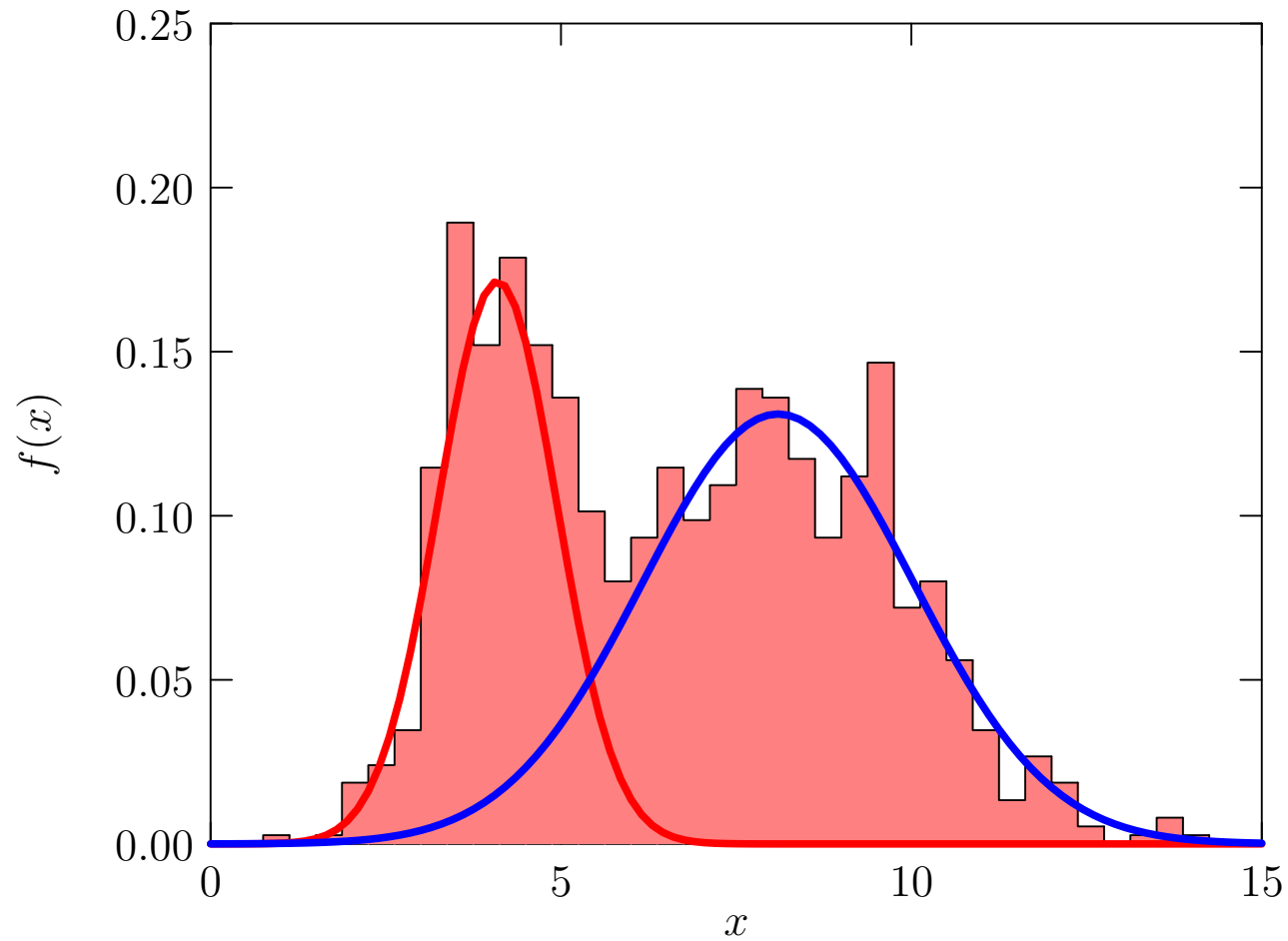
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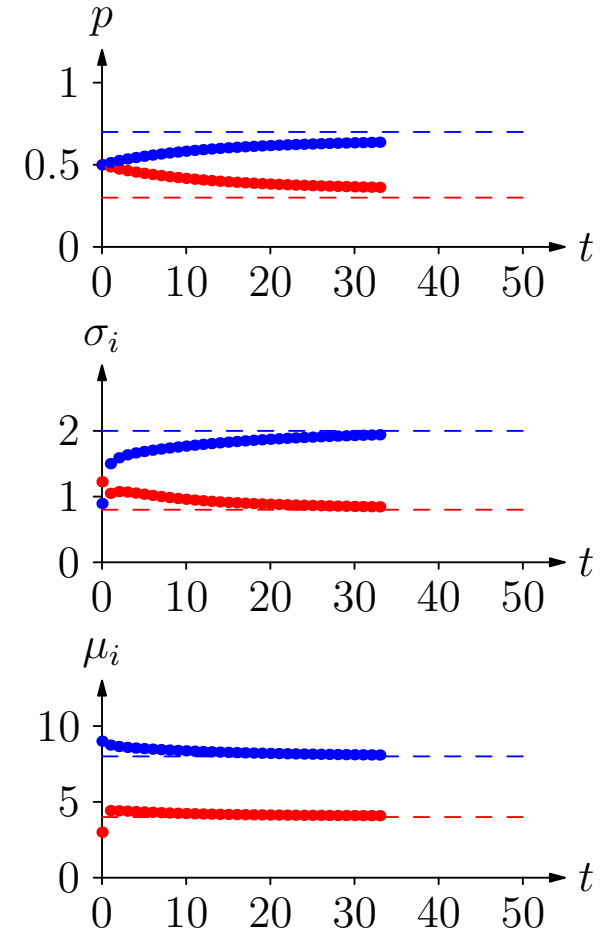
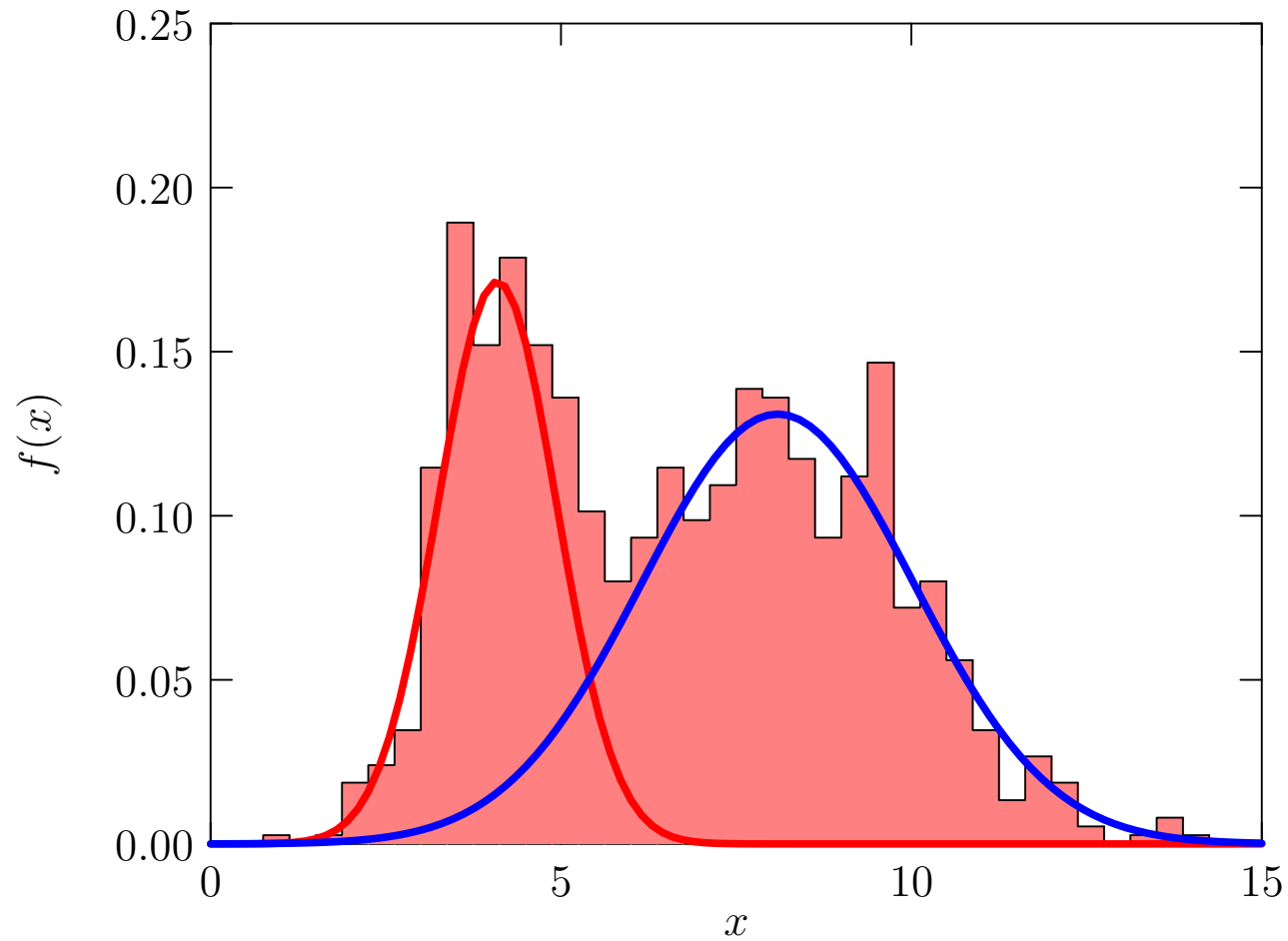
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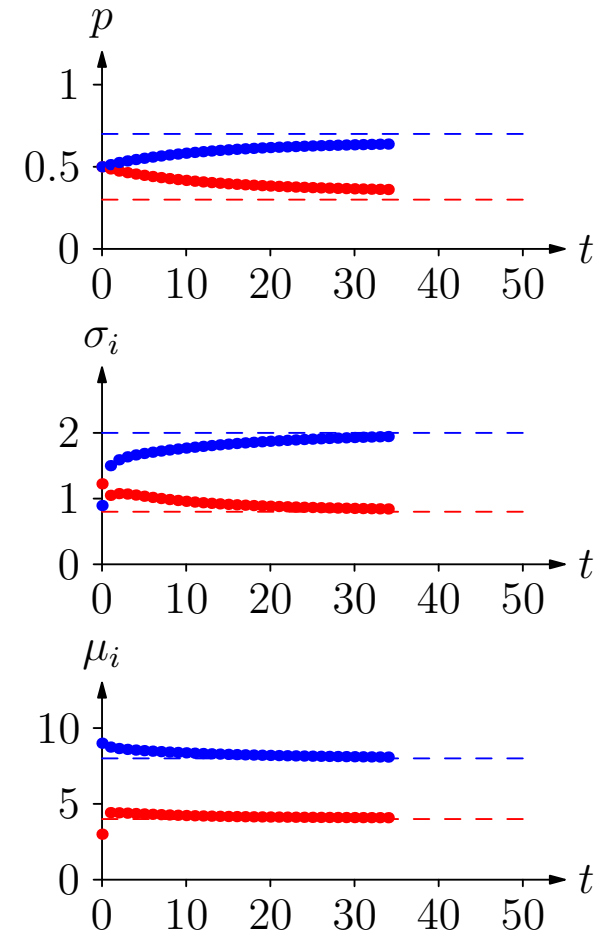
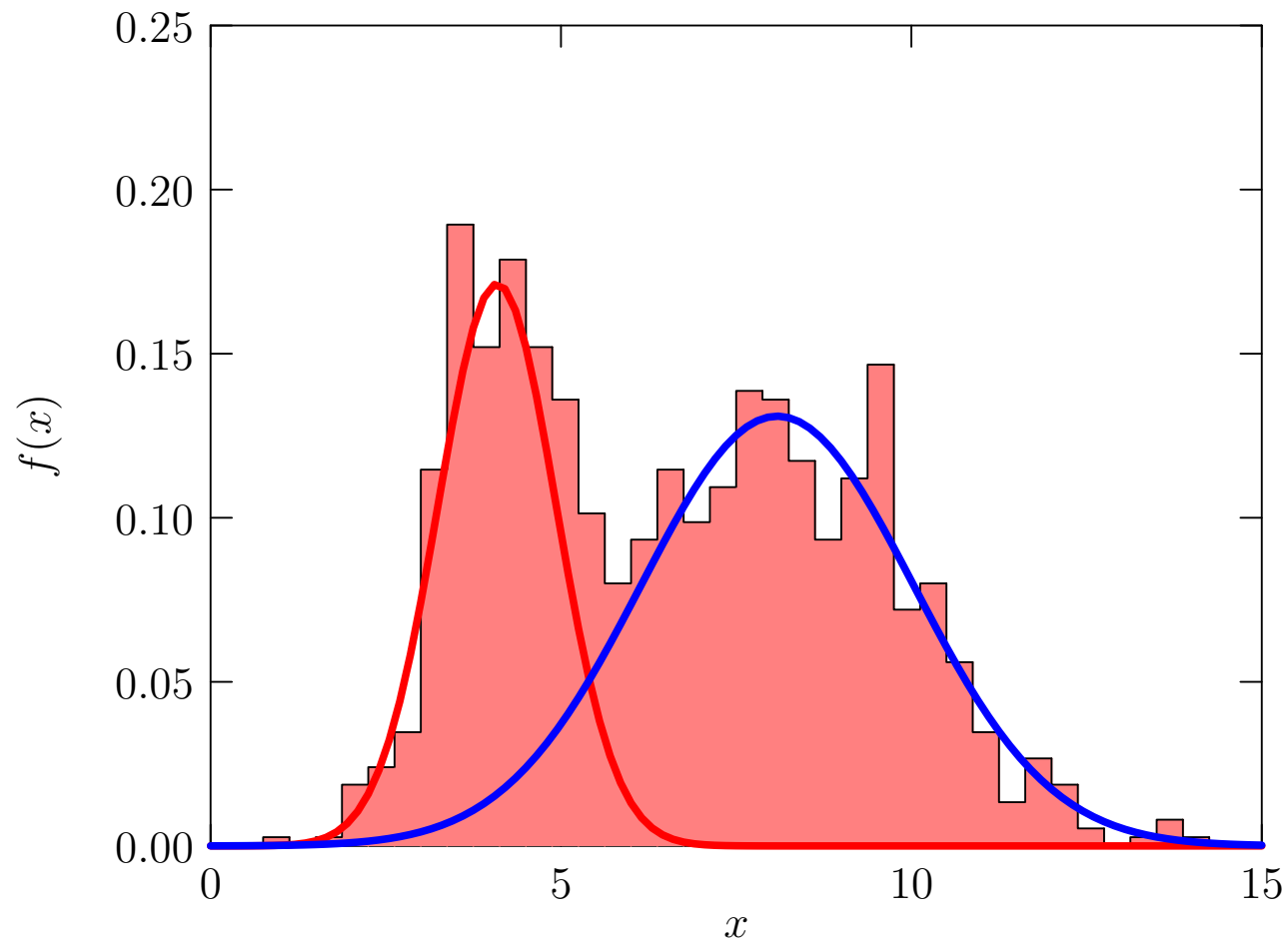
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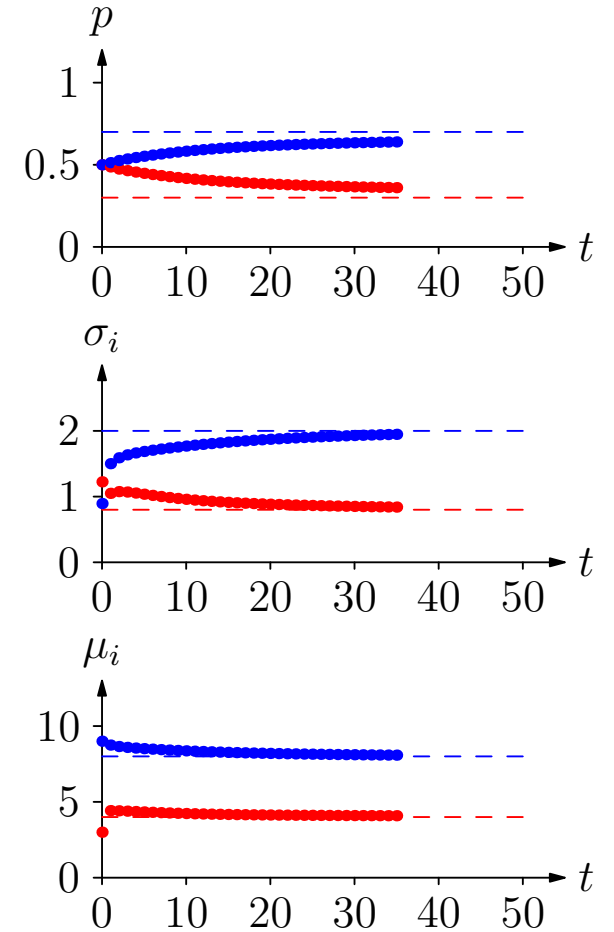
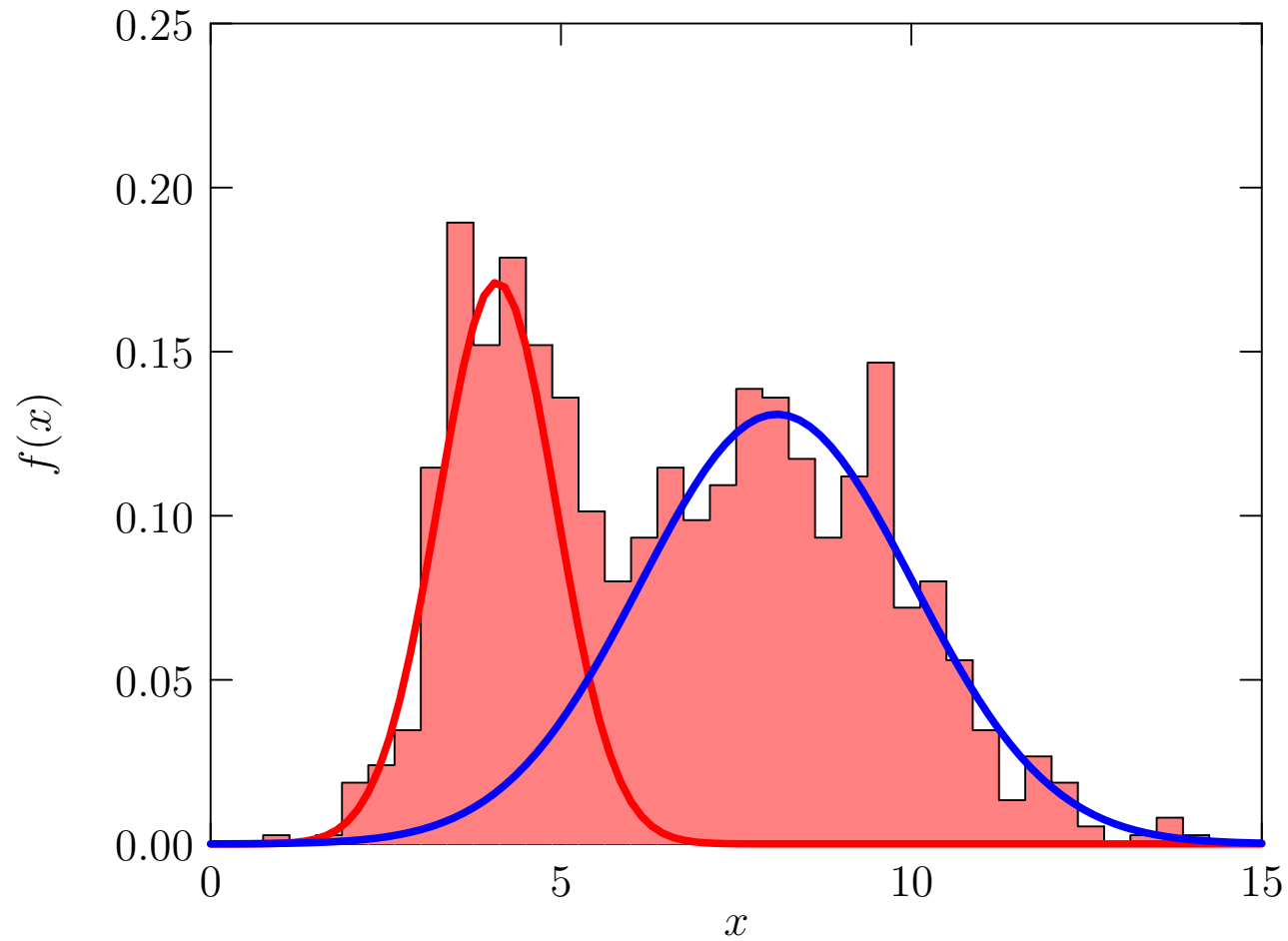
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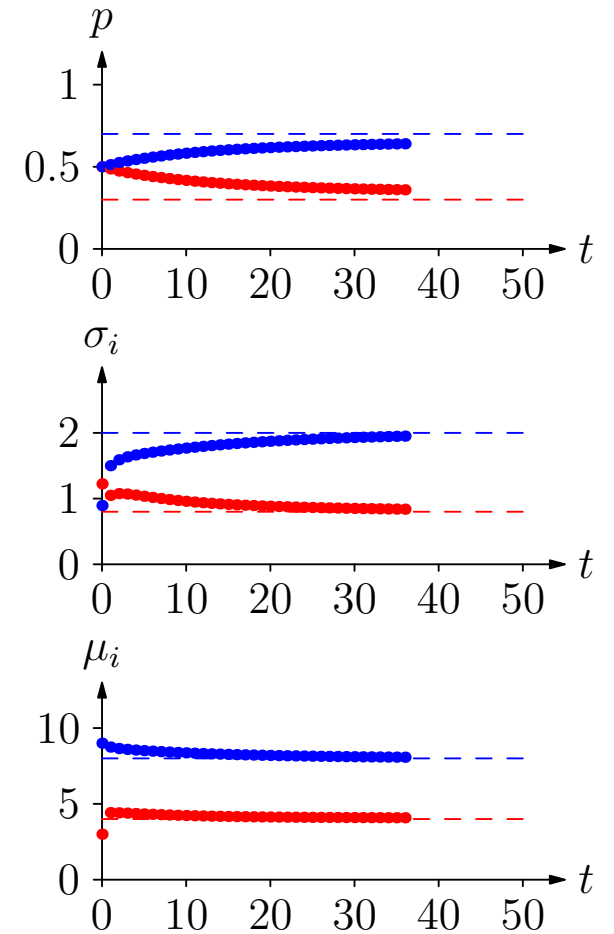
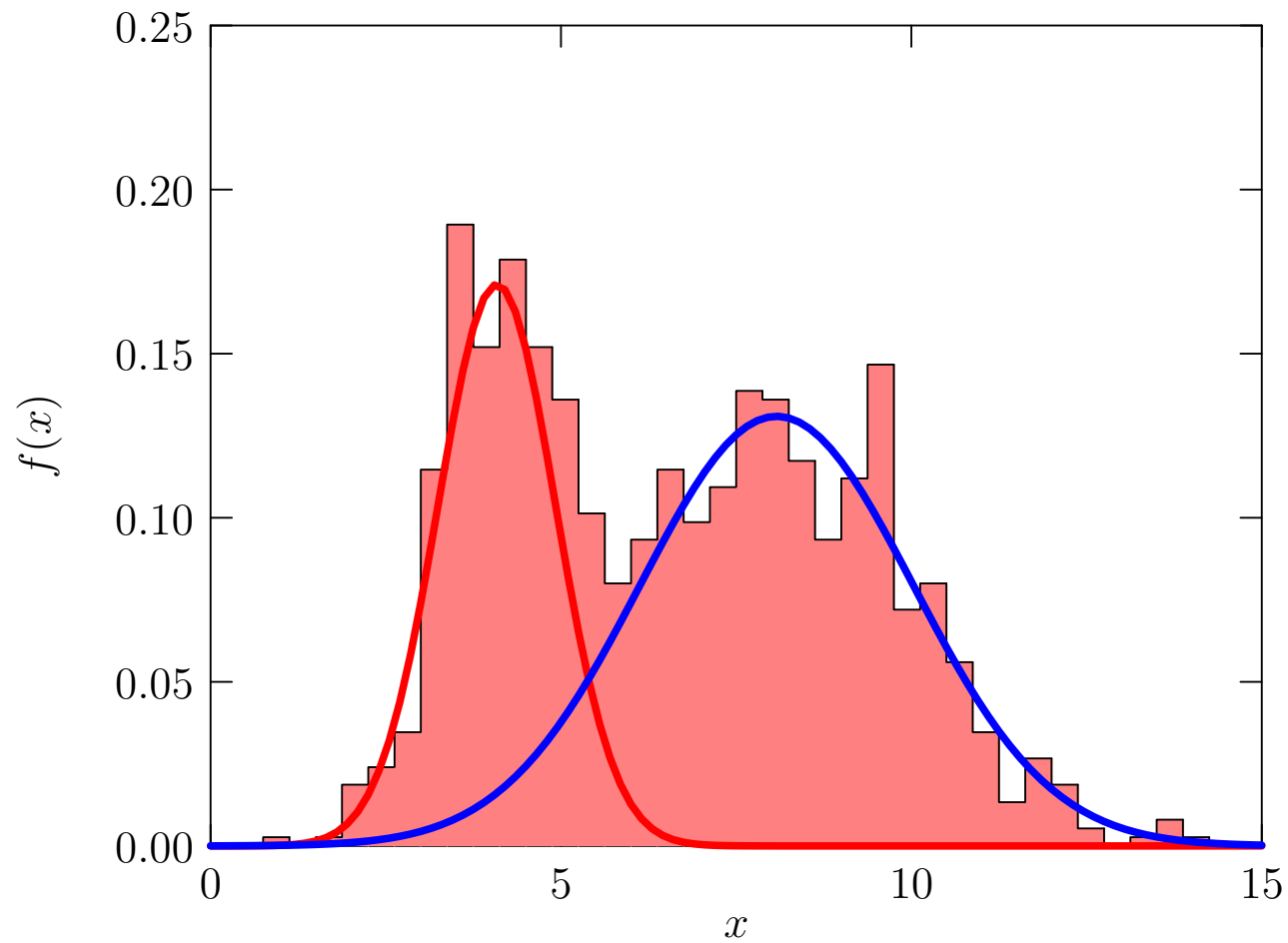
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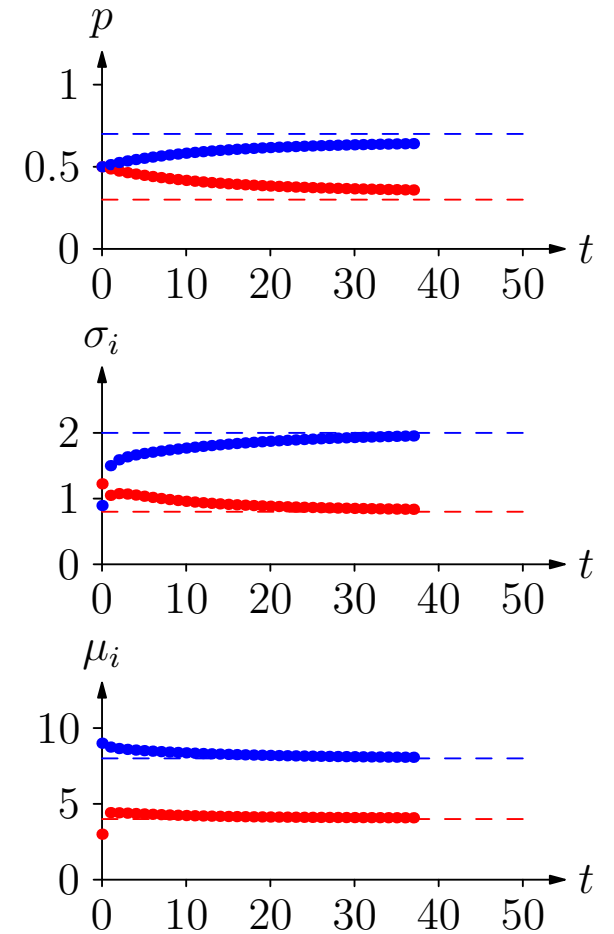
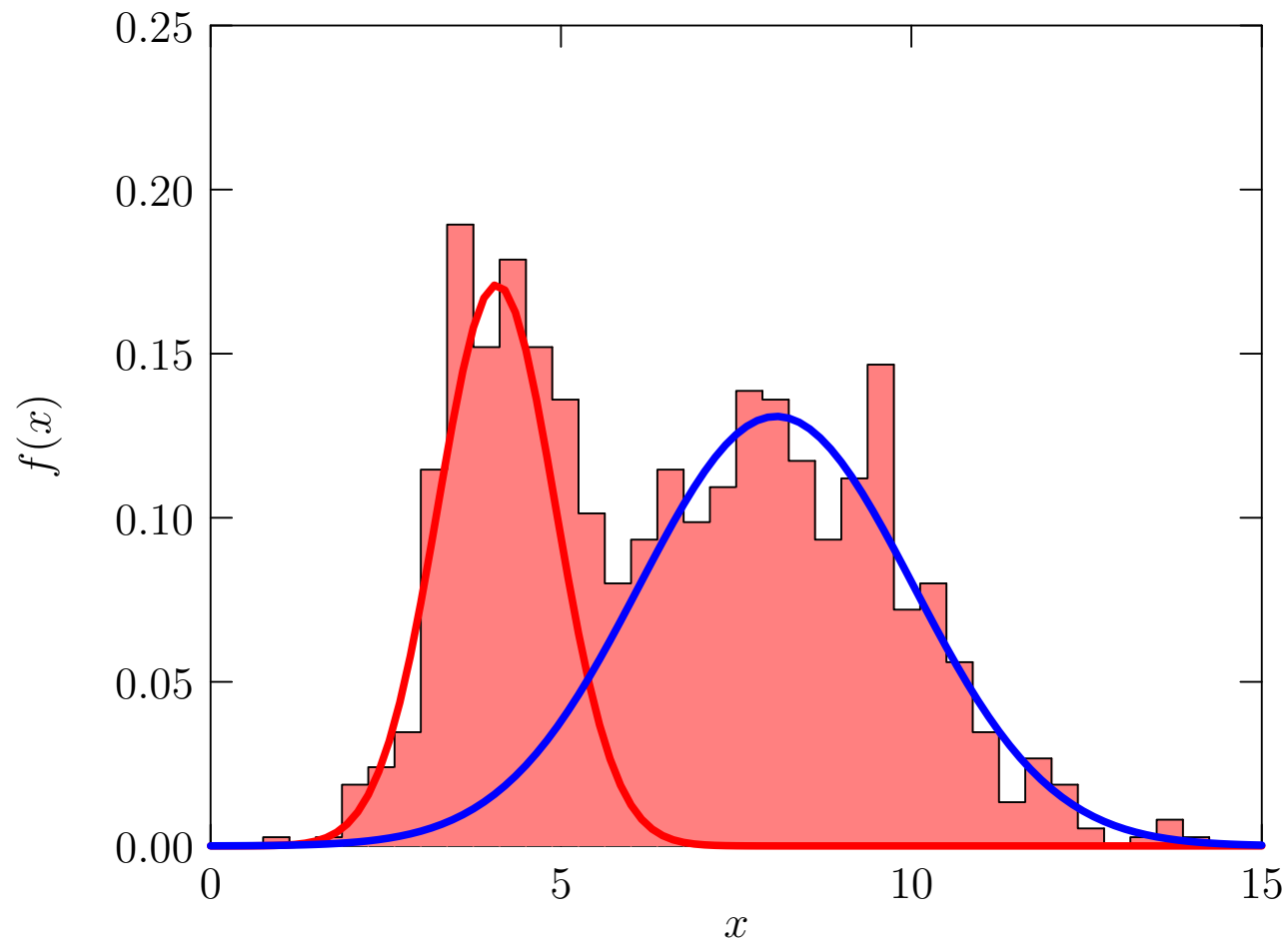
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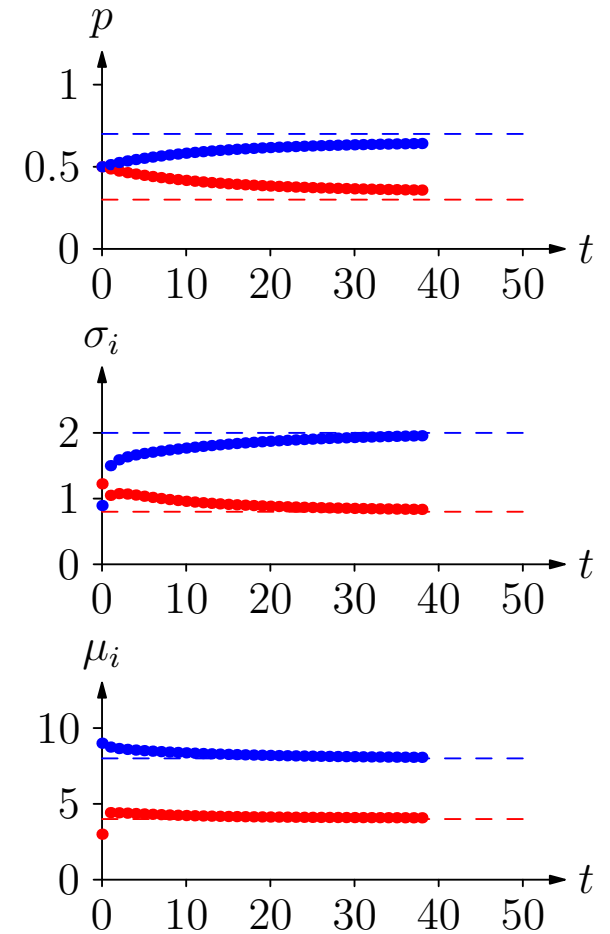
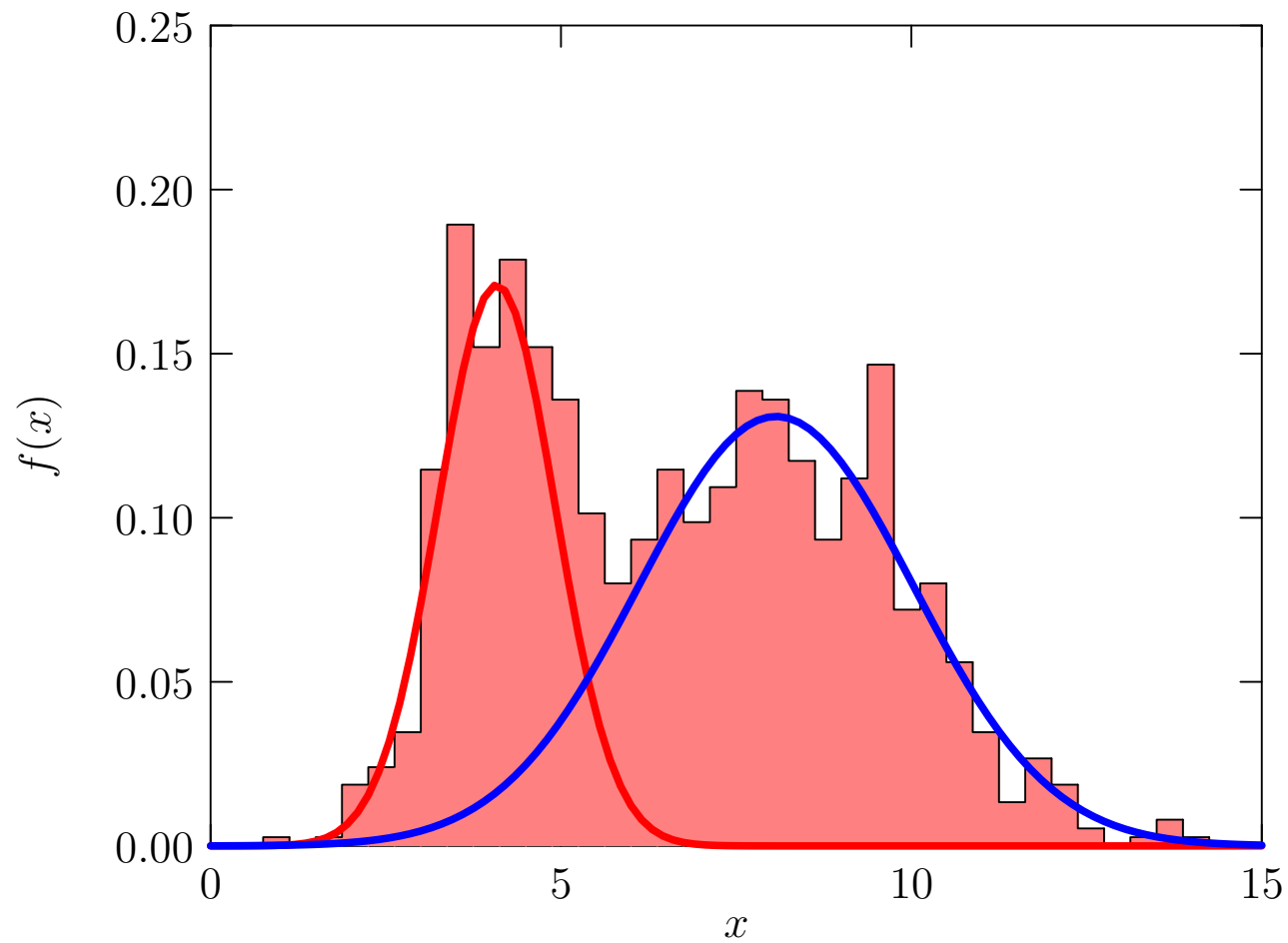
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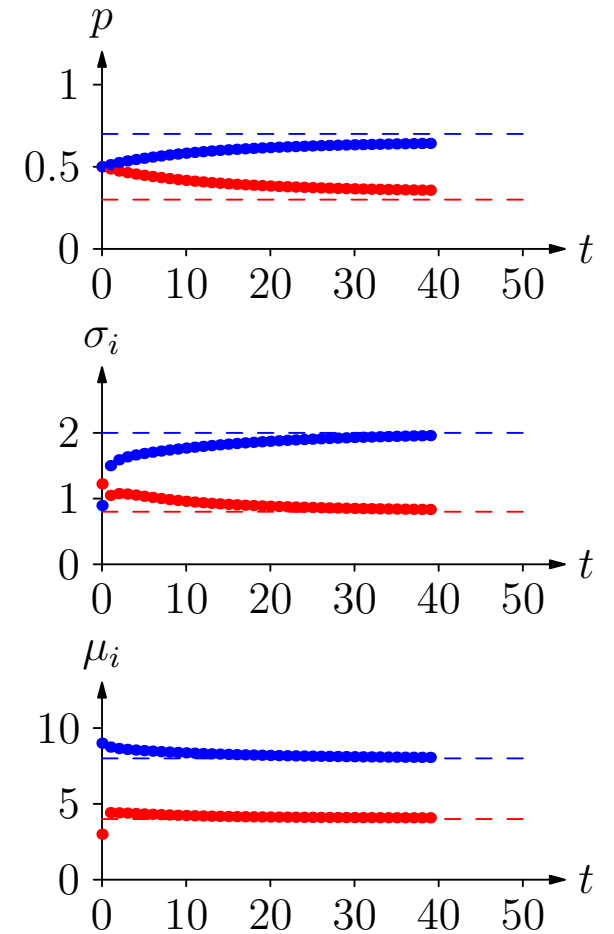
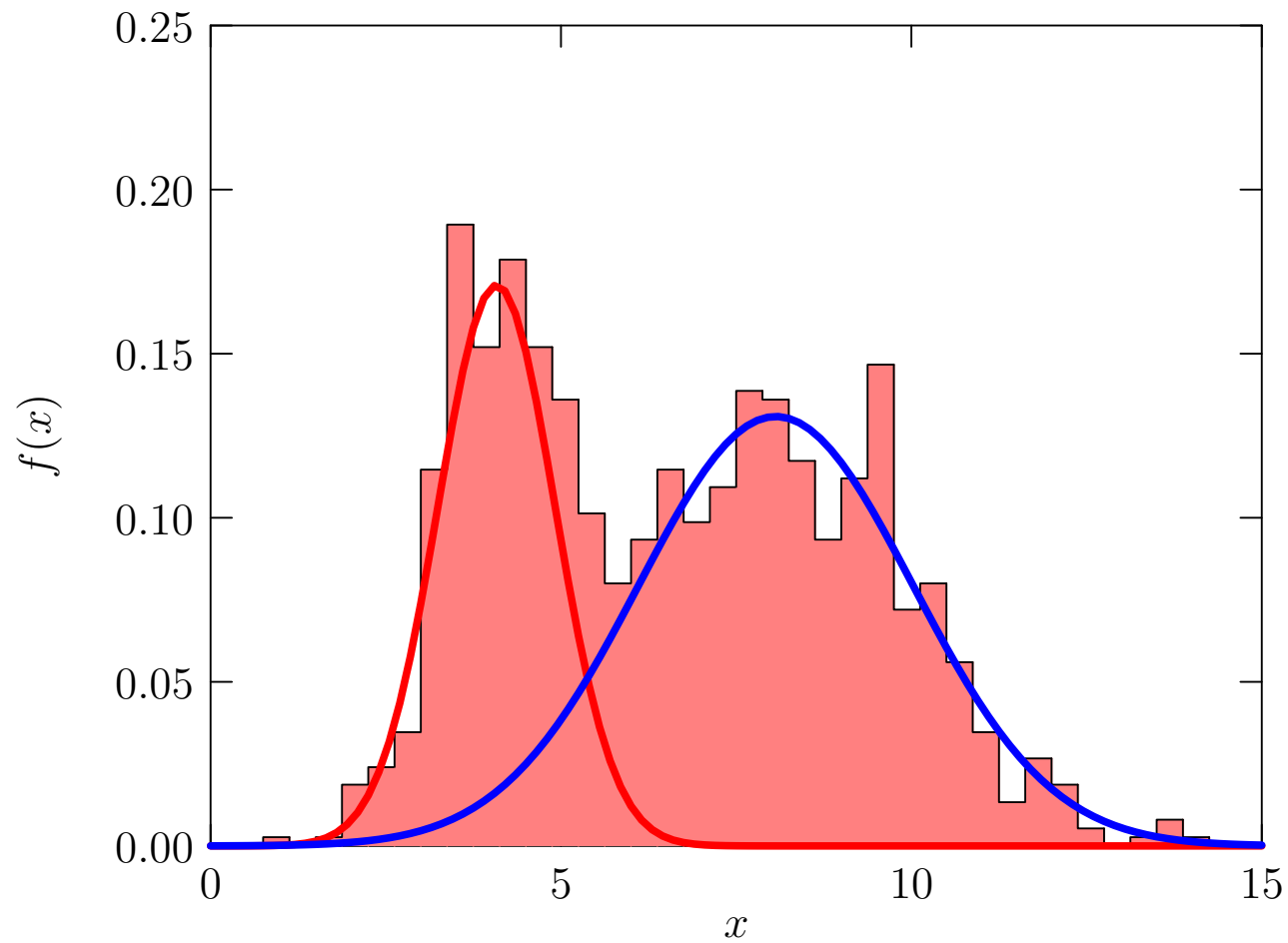
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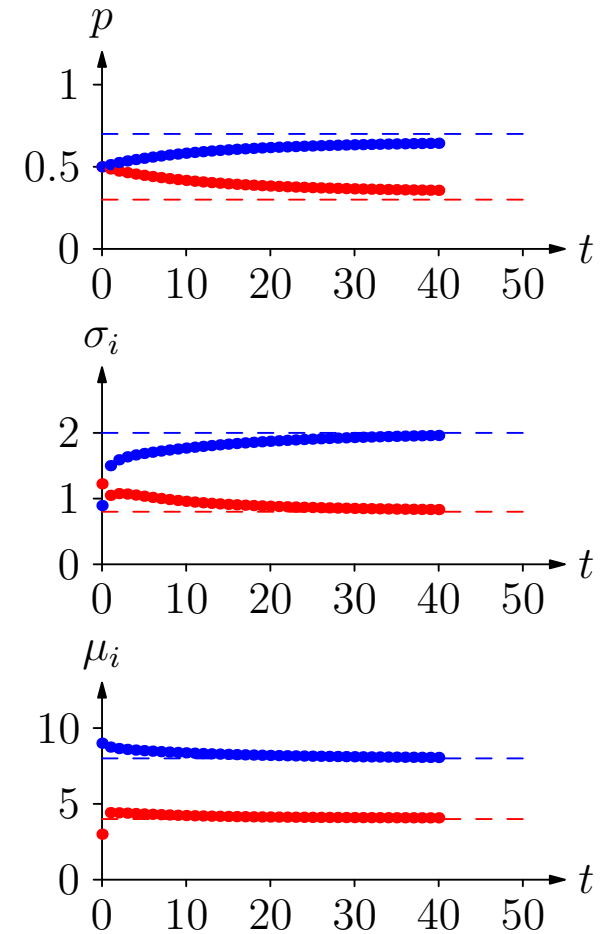
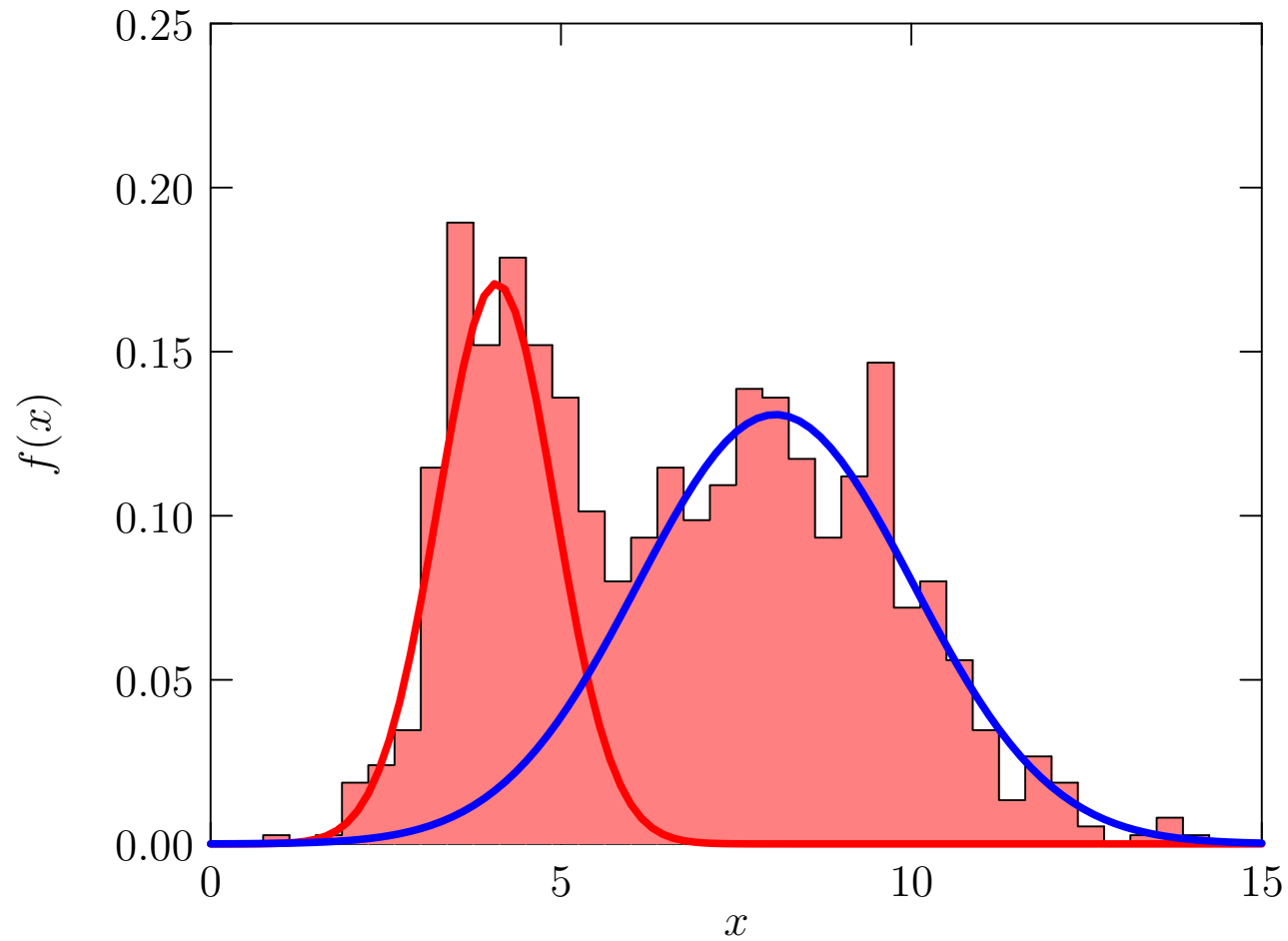
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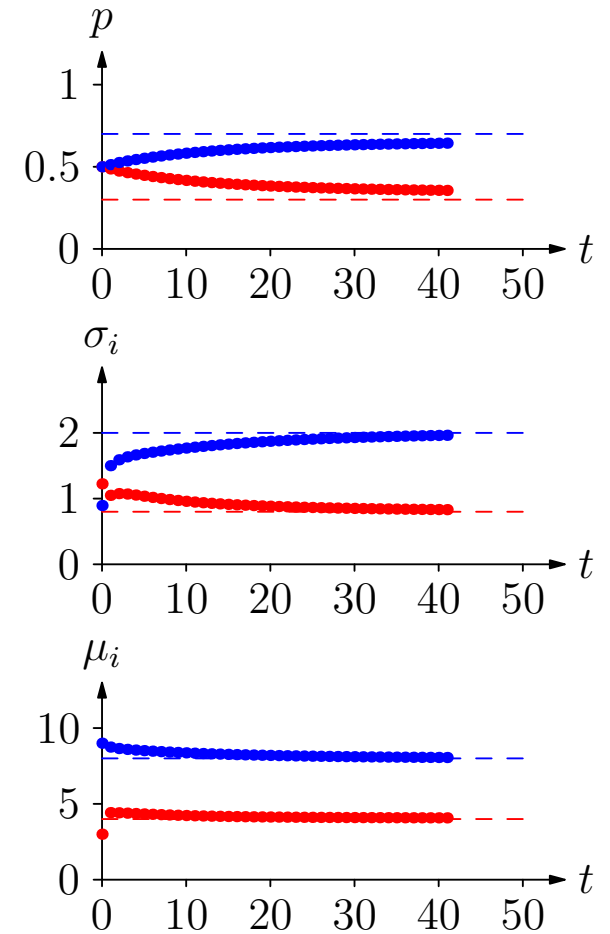
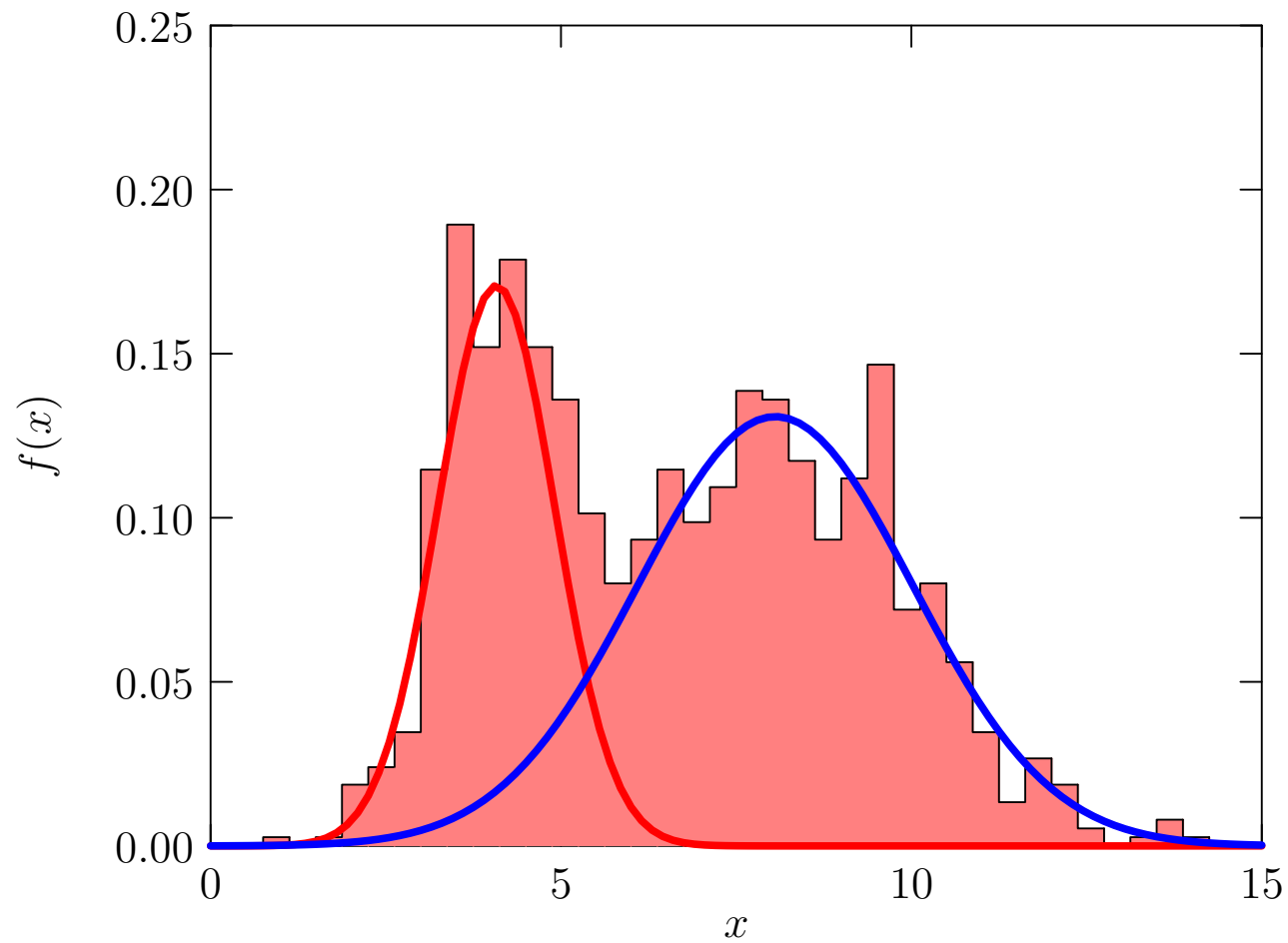
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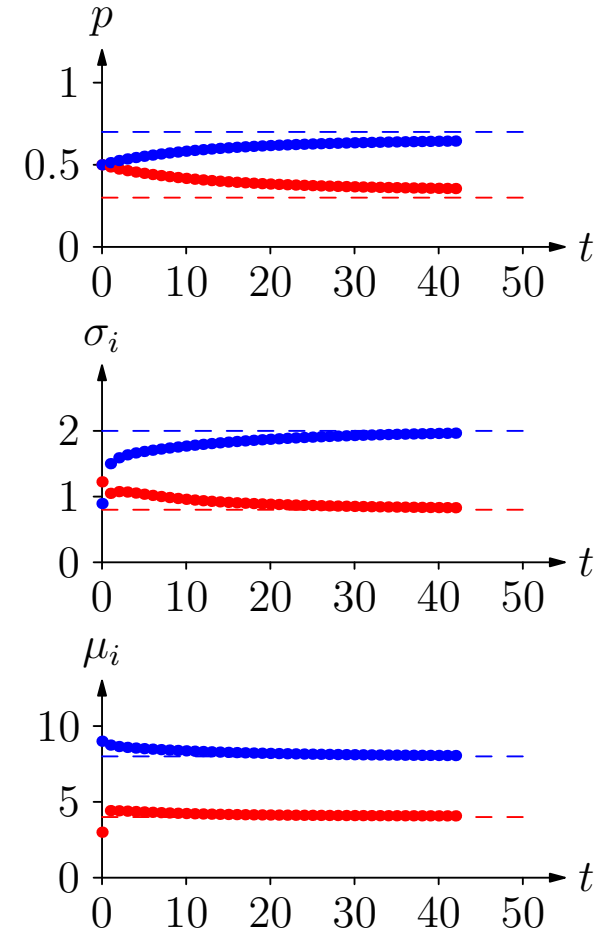
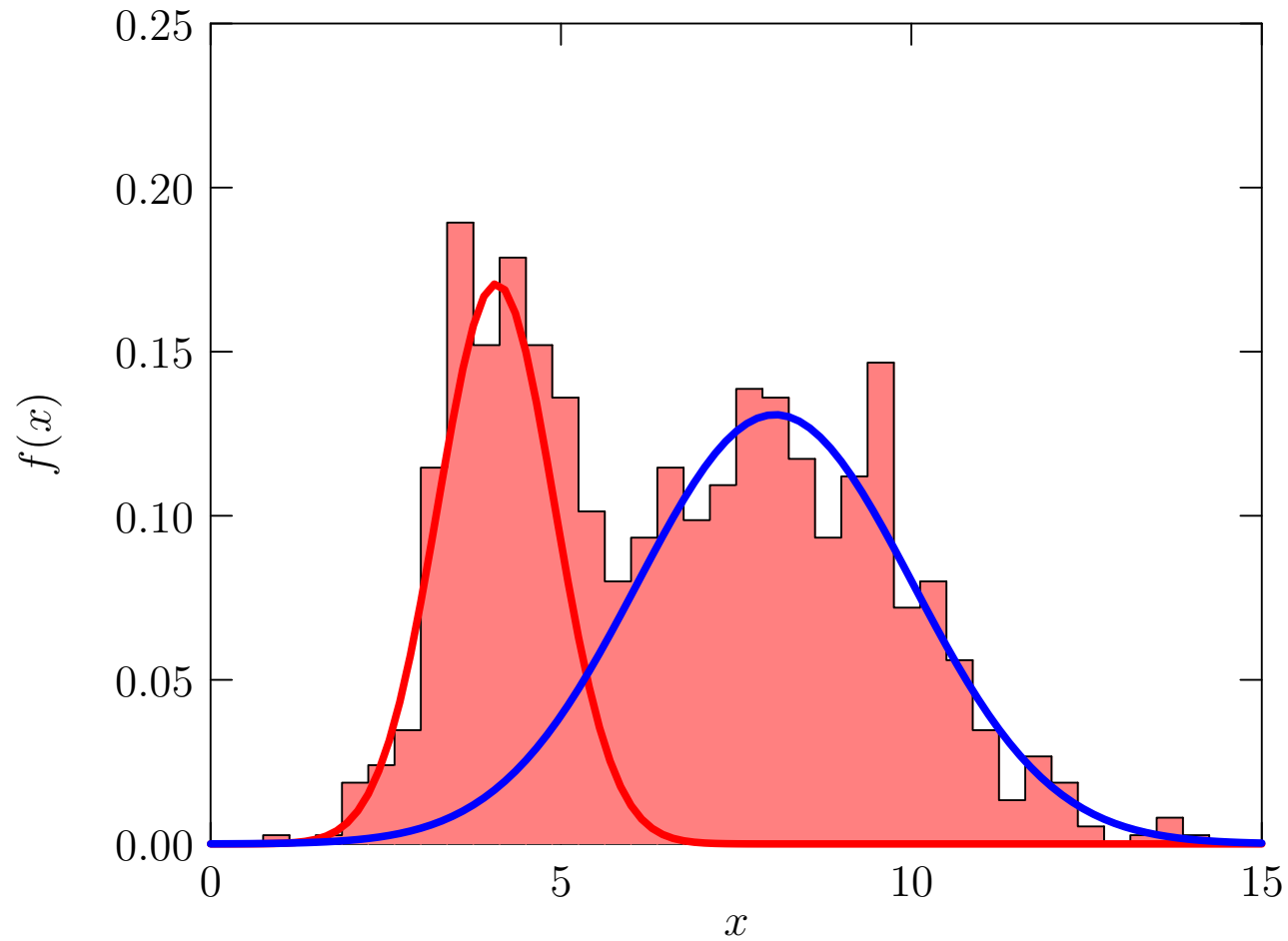
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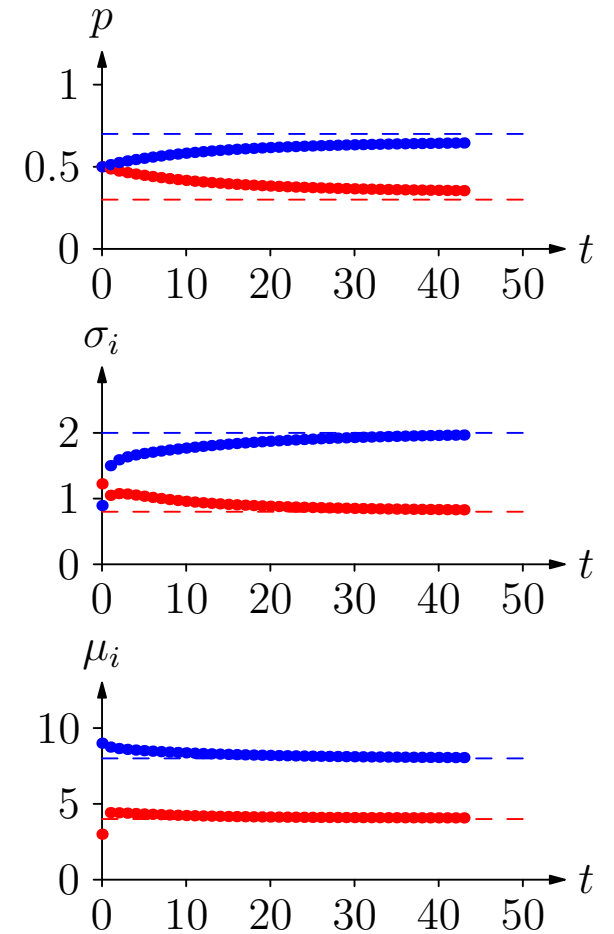
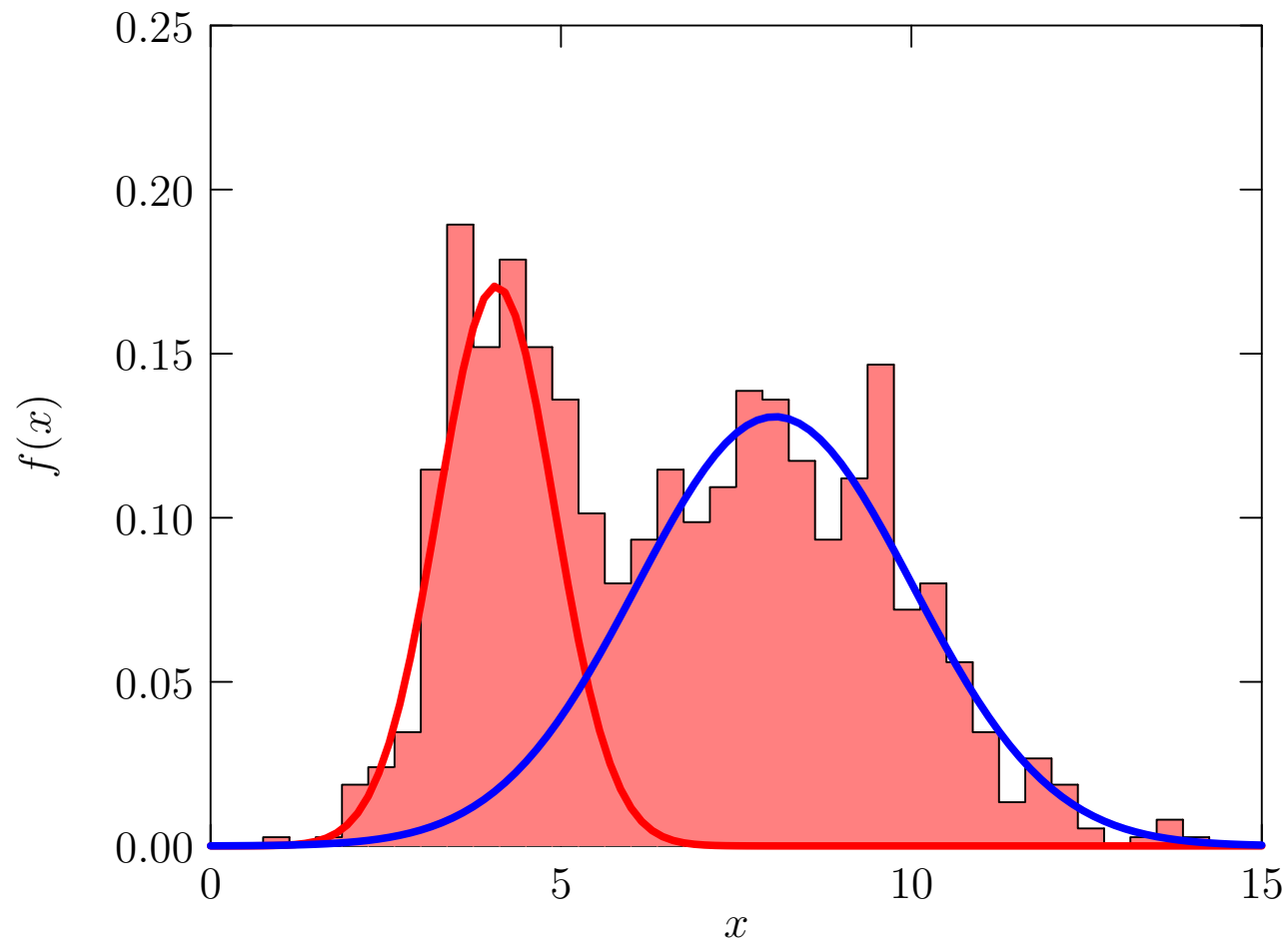
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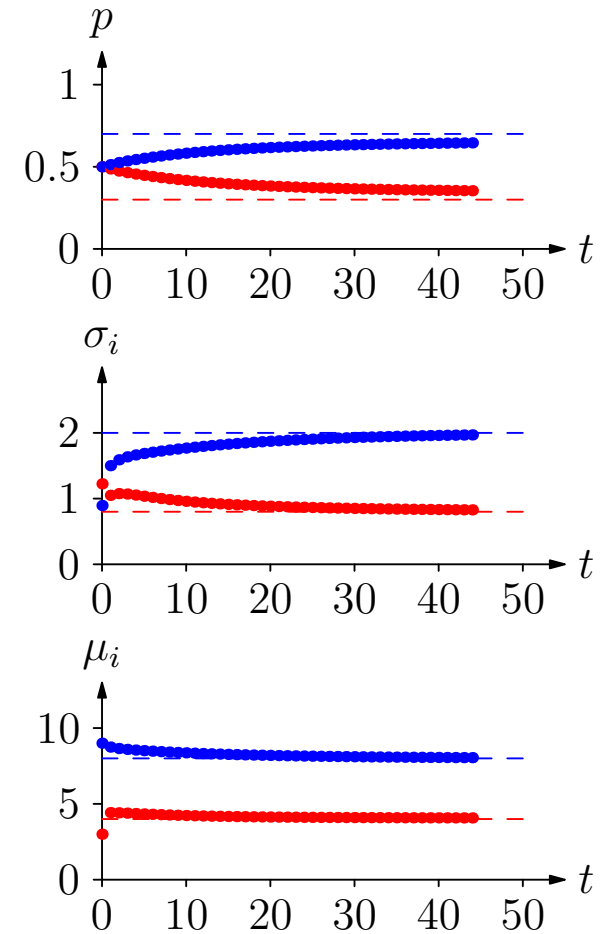
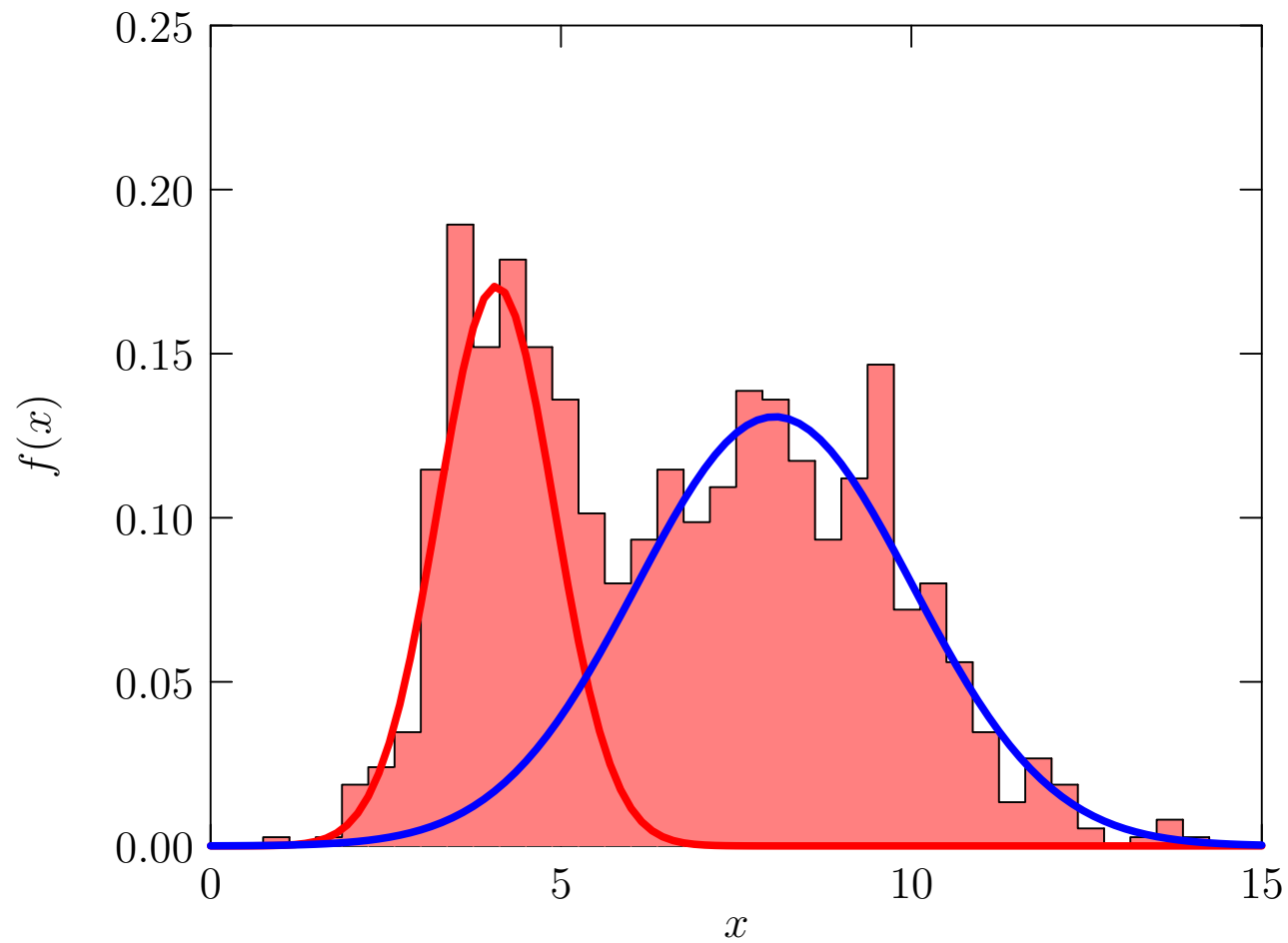
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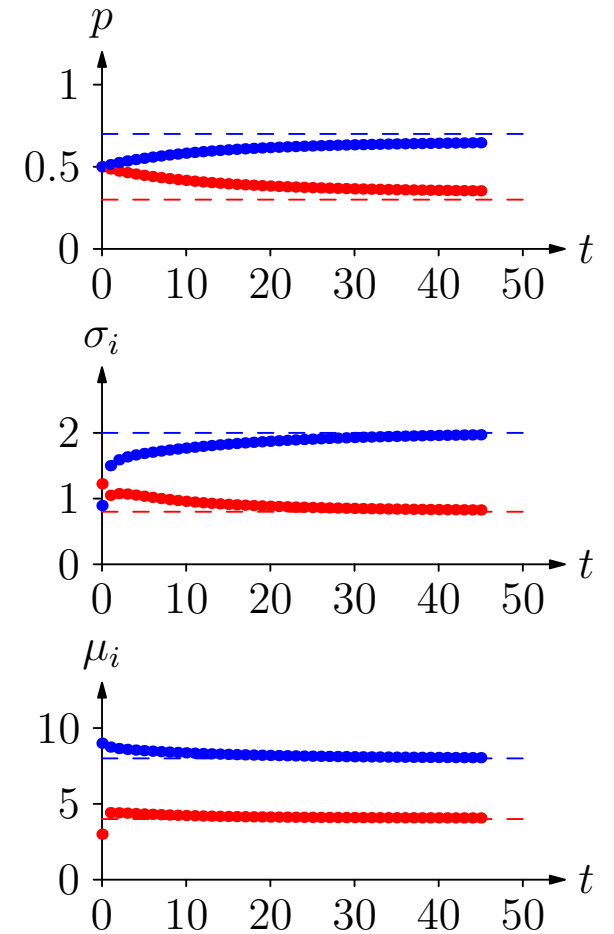
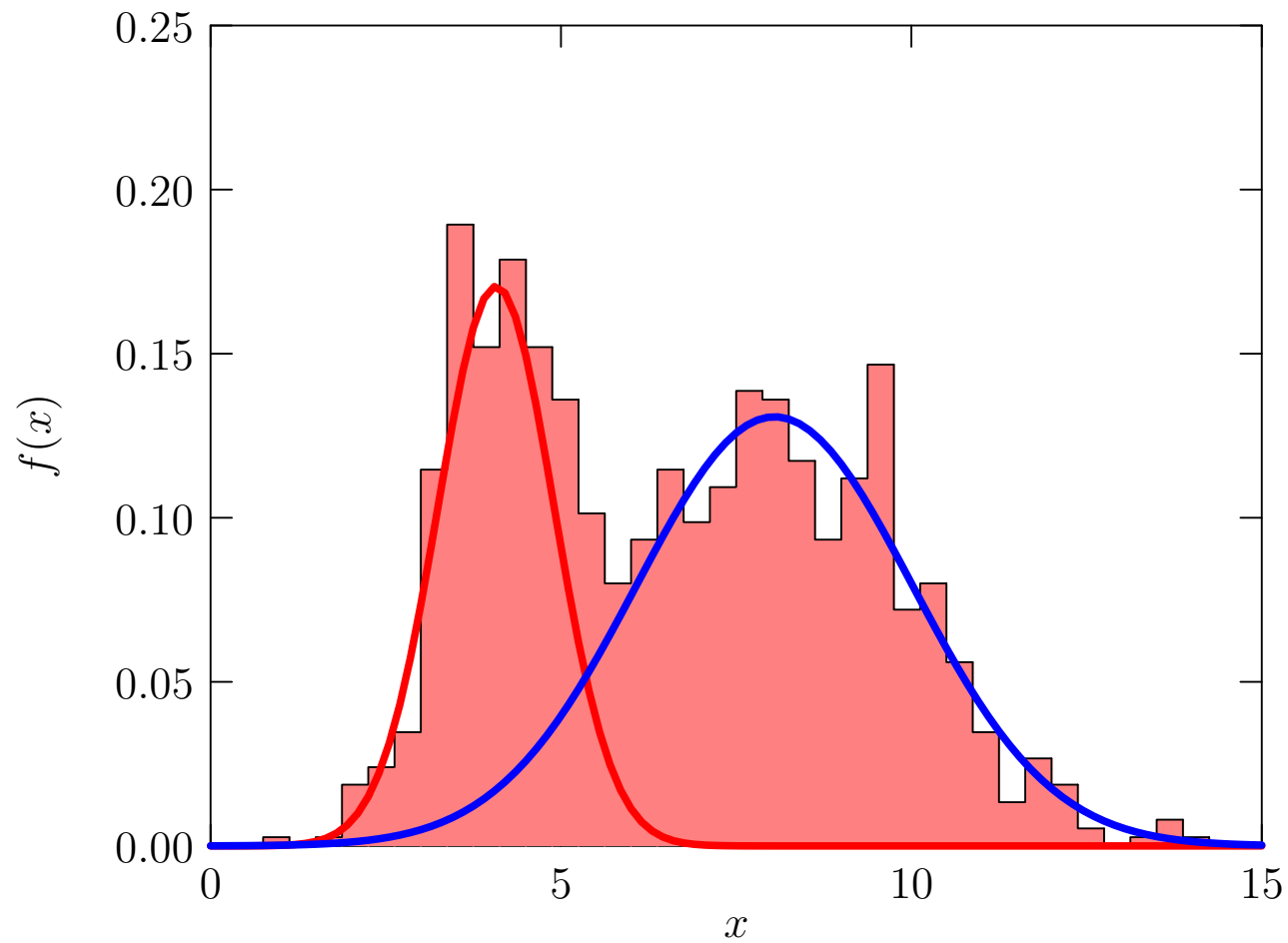
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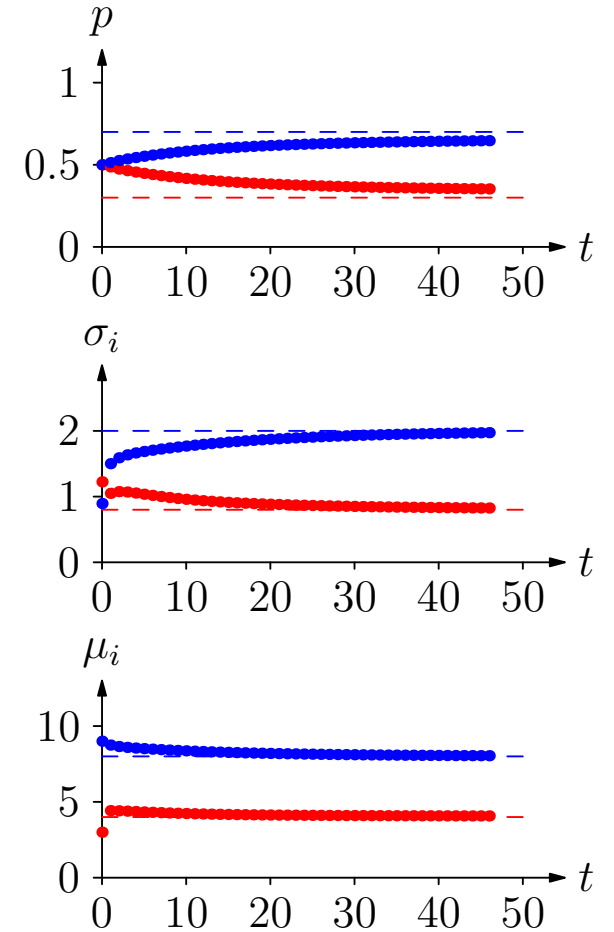
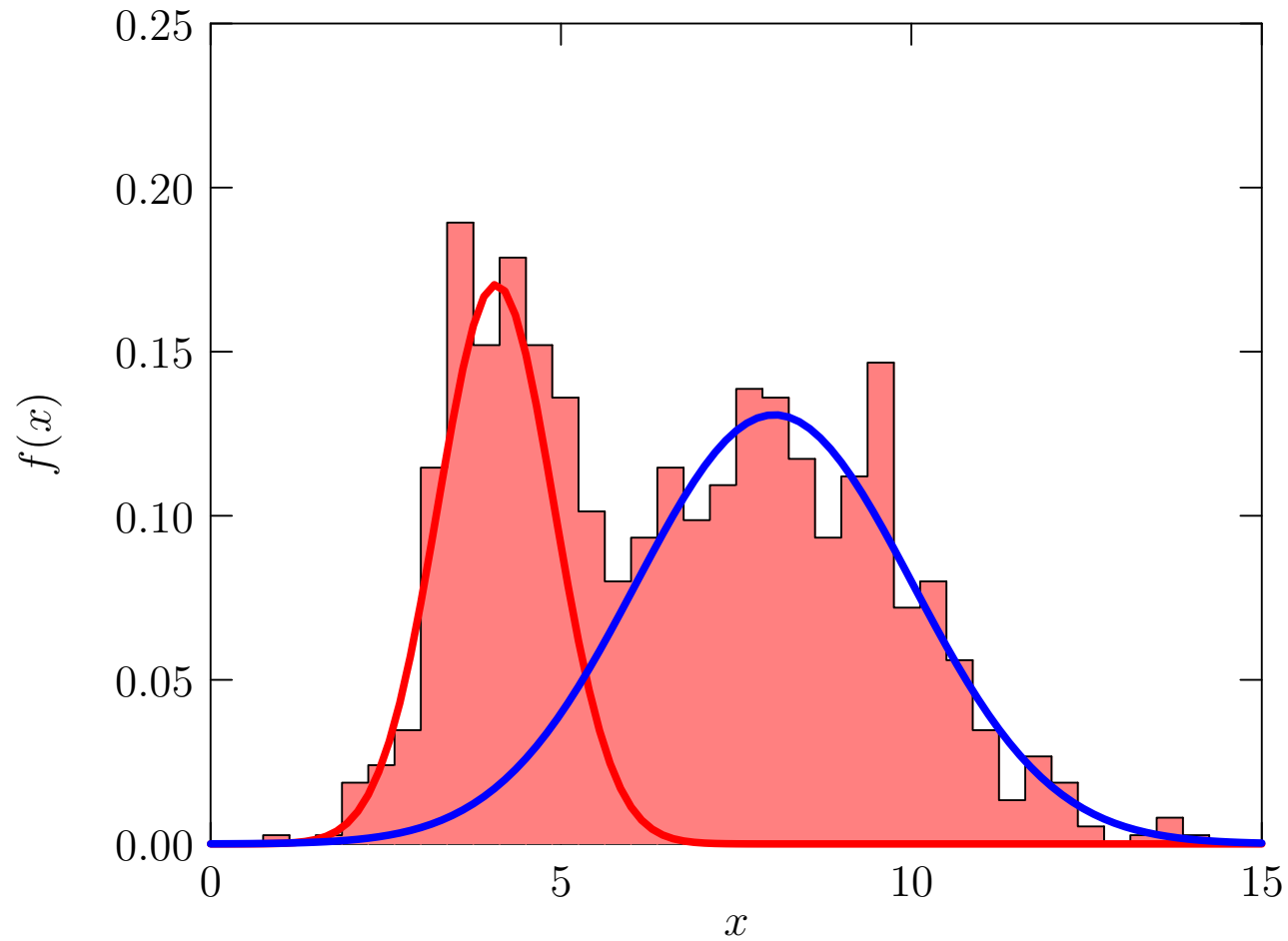
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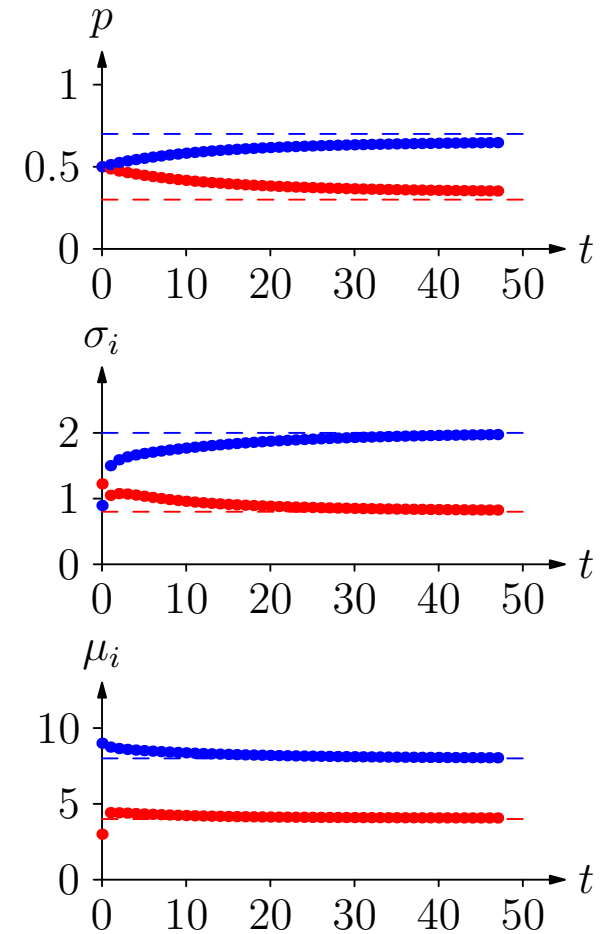
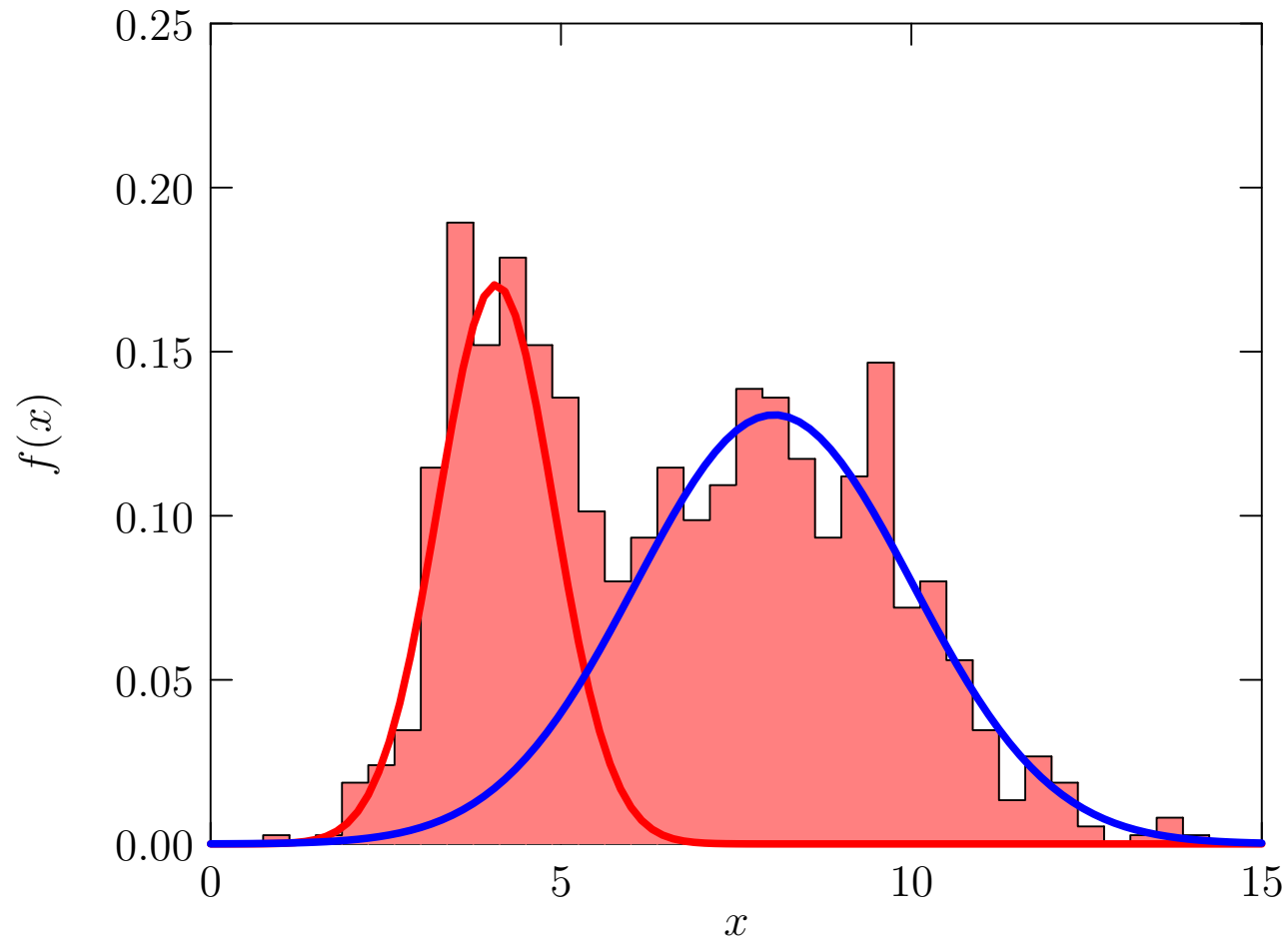
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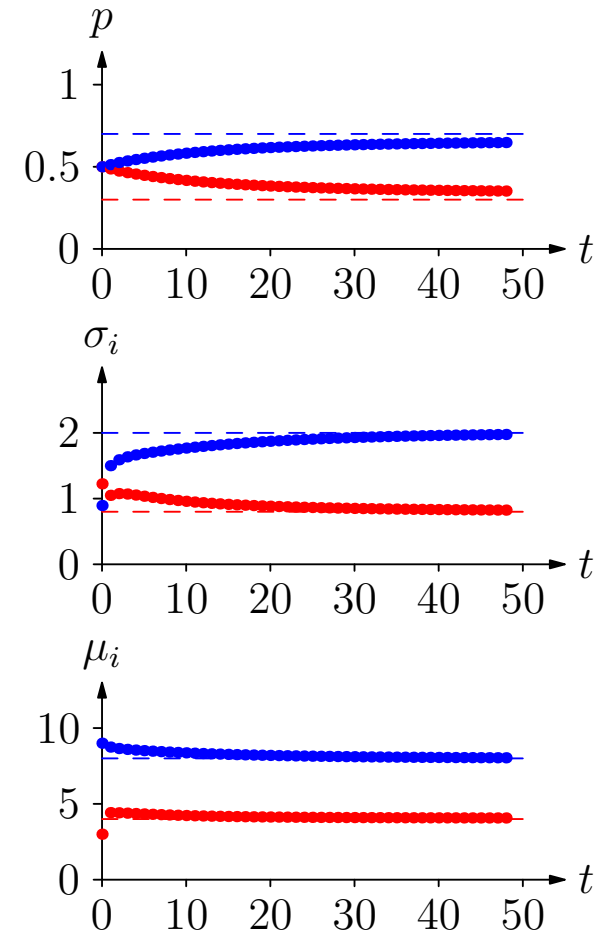
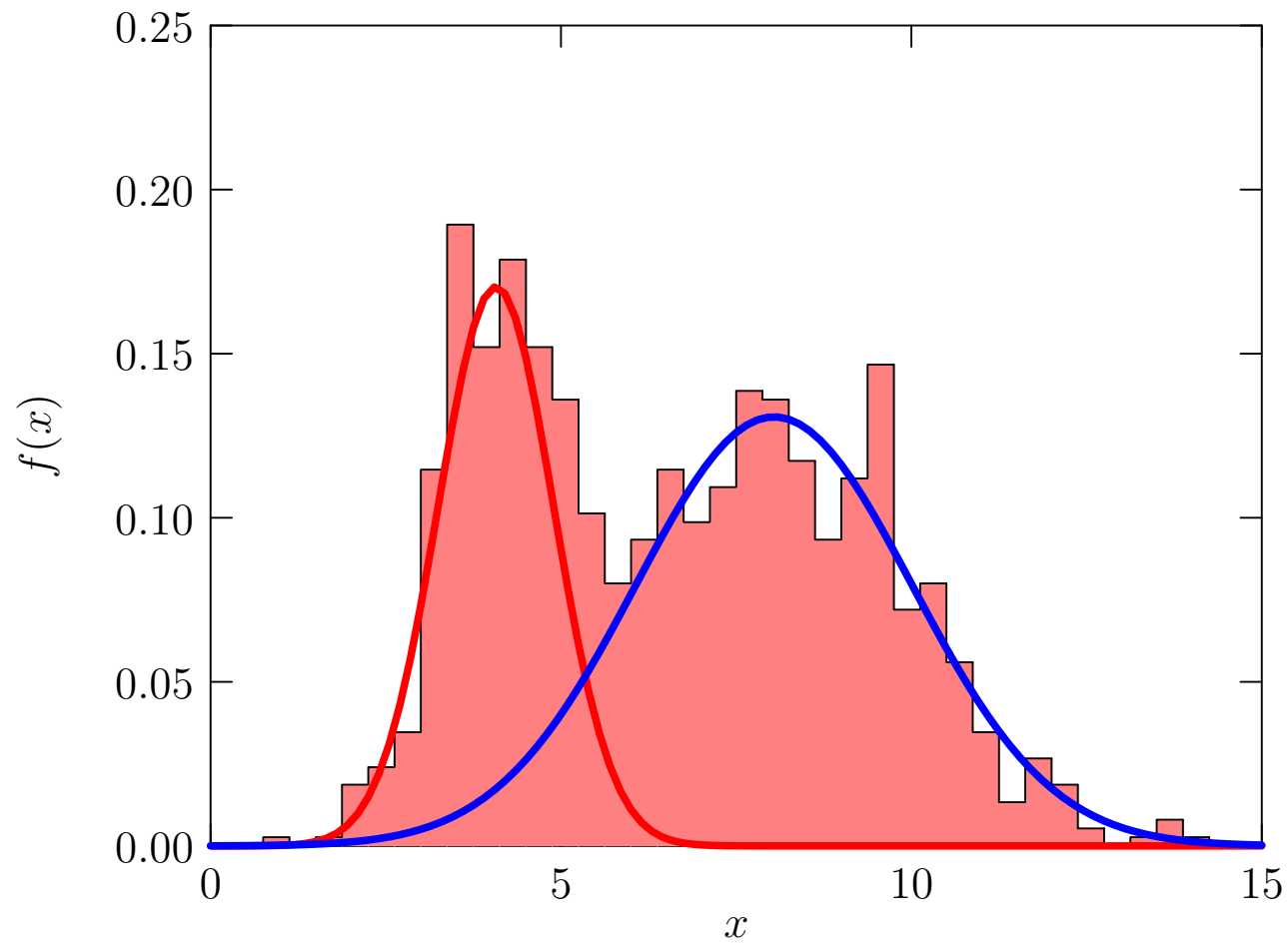
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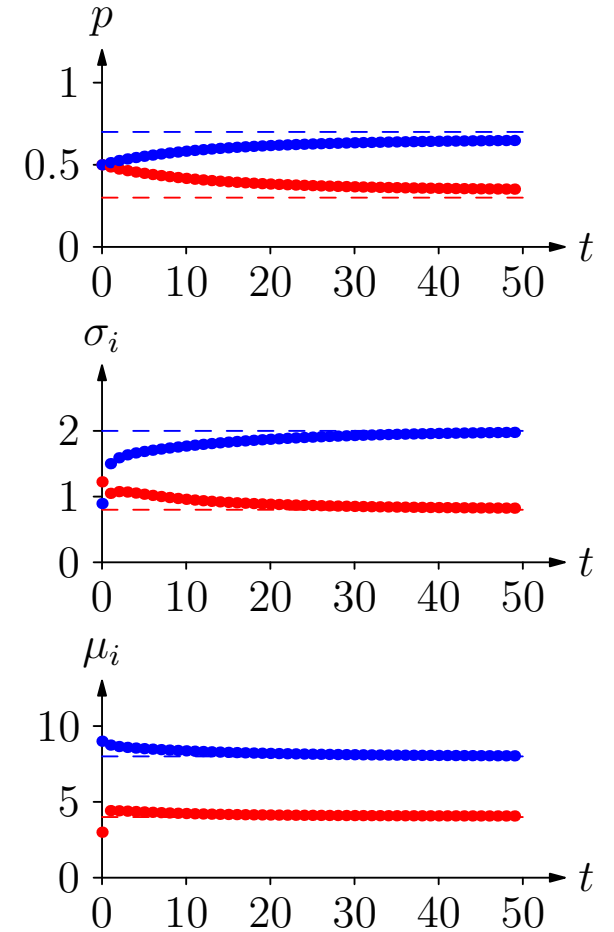
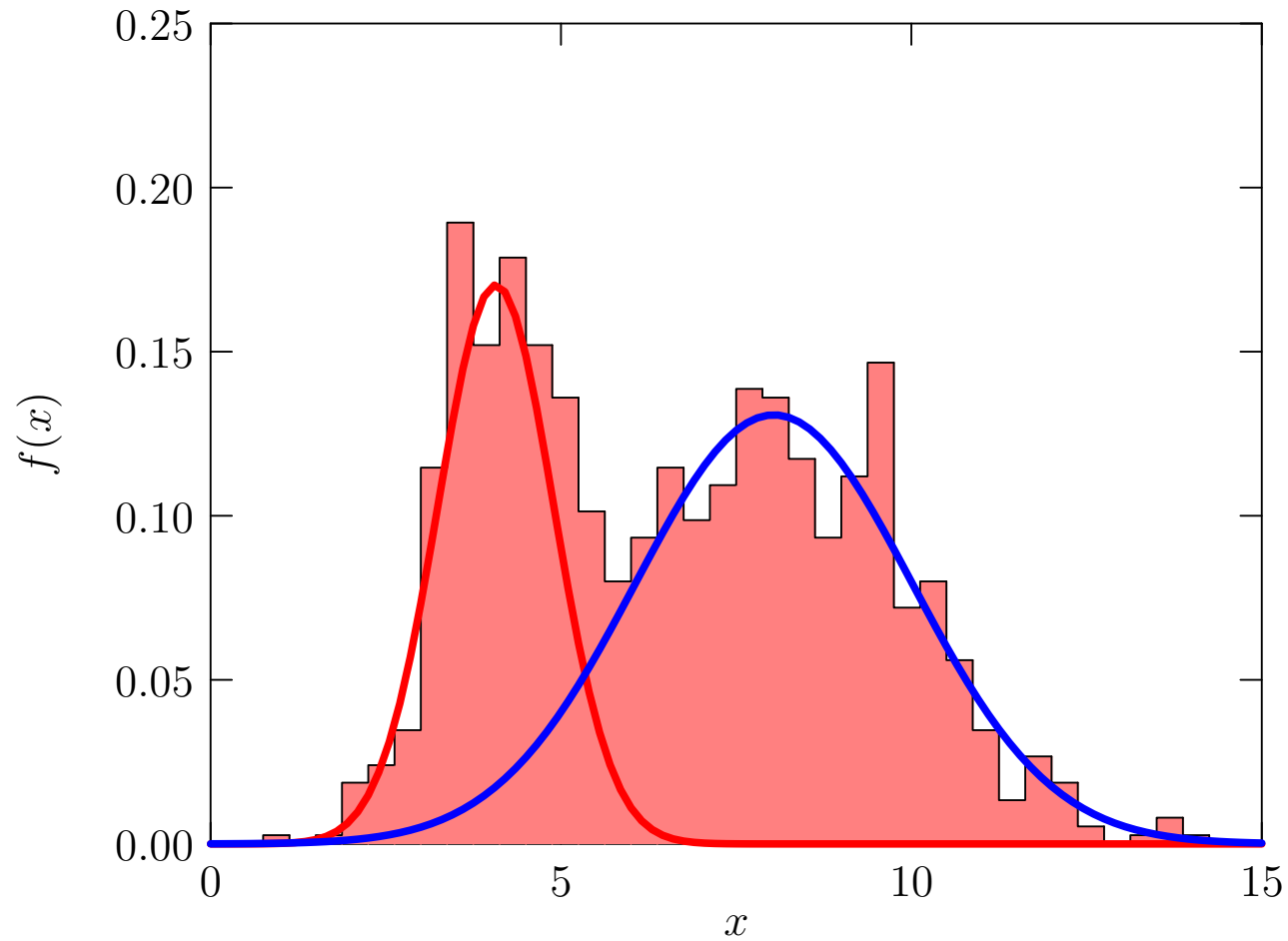
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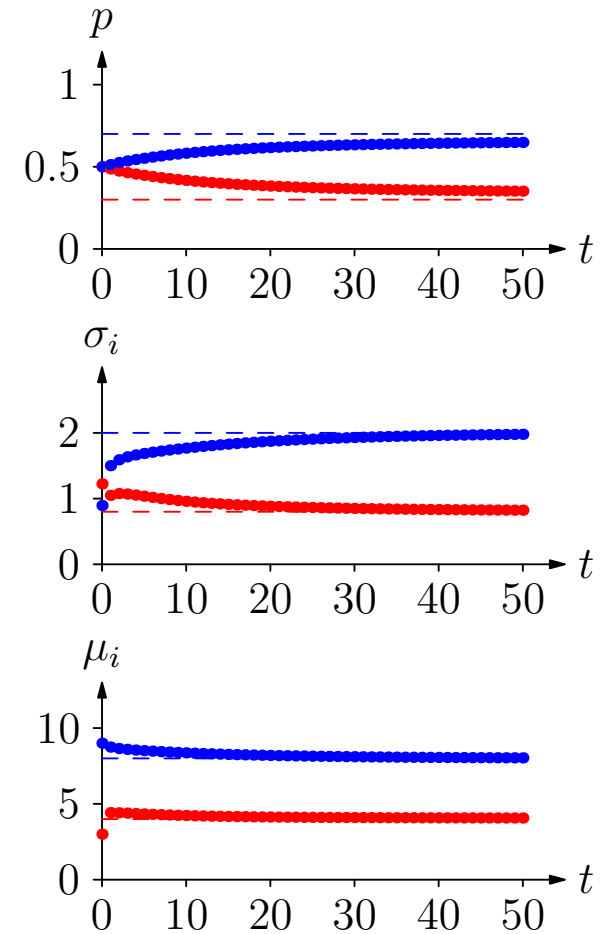
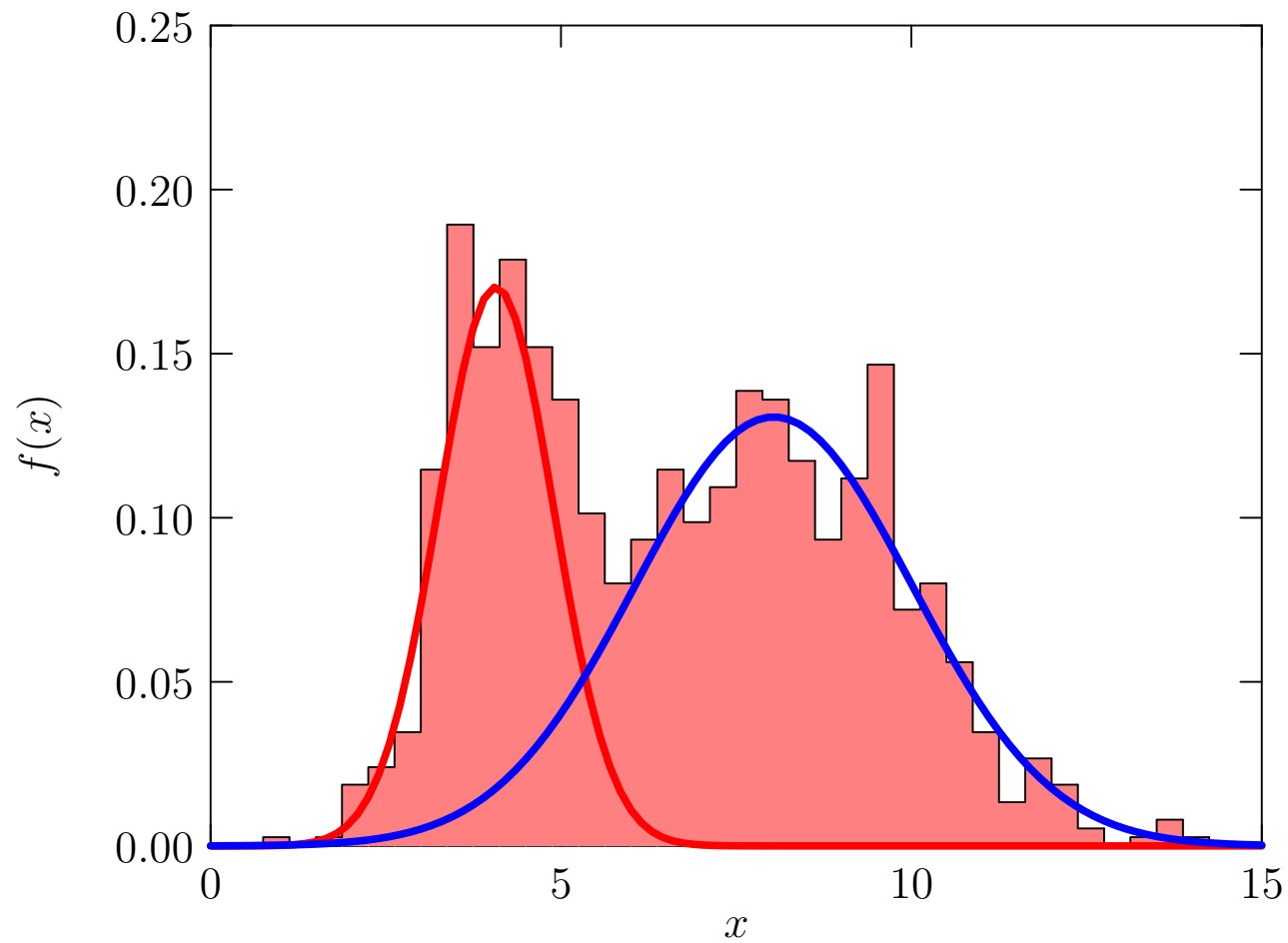
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Example



Example



Summary

- Building probabilistic models is an intricate process
- Identifying random variables that describe the system is the first step
- Often we need to introduce variables that we don't observe and need to be marginalised out
- The EM algorithm provide one approach to maximising likelihoods or MAP solutions when we have latent variables
- It often gives nice update equations, but convergence can be slow

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